Foreword

Education is not just knowledge transfer, but a spark for curiosity and critical thinking. In line with NEP 2020, NCF 2023, and SCF–AP, this Teacher Handbook empowers teachers in this transformative journey. Teachers today are facilitators of learning, shaping future-ready students with values, skills, and competencies. This handbook offers practical guidance for lesson planning, assessment, pedagogy, and reflection.

It includes curricular goals, learning outcomes, and innovative teaching strategies. Resources like TLMs, science labs, clubs, and joyful learning techniques are featured. It aligns with the Academic Calendar for structured year-round teaching. This is your professional companion encouraging creativity, collaboration, and growth. Let classrooms be inclusive, engaging, and empowering learning spaces. Teachers, lead with purpose and inspire lifelong learning in your students. May this handbook support meaningful classroom transformation?

The Department of School Education, Andhra Pradesh acknowledges the unwavering commitment of educators, academic experts, and resource persons who contributed to the development of this Teacher Handbook

Together, let's shape the future of education in Andhra Pradesh.

– Department of School Education, Andhra Pradesh

Government of Andhra Pradesh, Amaravati
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EXAMINATION SHEDULE									
	FA1	FA2	SA1	FA3	FA4	SA2	Pre Final 1 For X Class	Pre Final 2 For X Class	Public Exams For X Class
Syllabus	Jun,July	Aug,Sep	June-Oct	Nov,Dec	Jan				
Time Slot	Aug-25 4-7	Oct-25 13-16	Nov-25 10-19	Jan-26 5-8	FEB-26 9-12	April-26 6-15	FEB-26 18-27	March-26 2-12	16.03.26 to 02.04.26

No of Working Days

MONTH	JUNE	JULY	AUGUST	SEPTEMBER	OCTOBER	NOVEMBER	DECEMBER	JANUARY	FEBRUARY	MARCH	APRIL	TOTAL
No of working days For General	15	26	21	18	23	24	25	19	23	21	18	233
No of Holidays For General	4	5	10	12	8	6	6	12	5	10	5	83
No of working days For Minority	15	26	21	21	23	24	20	21	23	21	18	233
No of Holidays For Minority	4	5	10	9	8	6	11	10	5	10	5	83

For General Schools

Dasara Holidays	24.09.2025 to 02.10.2025
Pongal Holidays	10.01.2026 to 18.01.2026

For Minority Schools

Dasara Holidays	24.09.2025 to 02.10.2025
Pongal Holidays	10.01.2026 to 15.01.2026
Christmas Holidays	21.12.2025 to 28.12.2025

YEAR PLAN

S.NO	NAME OF THE CHAPTER	Month	No. of Instructional Periods Allocated	No. of Non-Instructional Periods Allocated	Total No of Periods	Page No.
1	Force and Pressure	Jun	10	3	13	3-19
2	Coal and Petroleum	July	15	8	23	20-31
3	Friction	Aug	8	7	15	32-41
4	Chemical effects of Electric Current	Sep	11	5	16	42-57
5	Sound	Nov	11	1	12	58-77
6	Combustion and Flame	Dec	11	7	18	78-94
7	Light	Jan	12	1	13	95-117
8	Some Natural Phenomena	Feb	10	6	16	118-135

June 2025

No bag Day

Cluster Complex

Teacher Resources

Sunday	Monday	Tuesday	Wednesday	Thursday	Friday	Saturday
1 Sunday	2	3	4	5	6	7 Bakr Eid
8 Sunday	9	10	11	NI 12 Reopening day	13 NI	14 Second Saturday
15 Sunday	16 1.1	17 1.2	18 1.3	19 1.4	20 1.5	21 International Yoga Day No Bag Day
22 Sunday	23 1.6	24 1.7	25 1.8	26 1.9	27 NI	28 No Bag Day Mega PTM
Sunday 29 National Statistics Day	30 1.10					

Week1:

Week 2:

Week 3: 1.1 – 1.5

Week 4: 1.6 – 1.9

Week 5: 1.10

↓ TEACHER'S NOTES ↓

Chapter 1: Force and Pressure



Learning Outcomes

1. Understand the effects of force.
2. Explain the concept of pressure.
3. Draw a flow chart on the types of forces.
4. Measure the net force acting on the system / object.
5. Conduct experiments to understand the pressure exerted by the liquids and gas

Prior Concepts

1. Students should have prior knowledge about moving objects.
2. Students should have prior knowledge about pushing, pulling, playing with a ball/magnet, muscles/ performing different types of work, cause of winds and cyclones, concepts of liquids/gases/atmosphere.

Teaching Learning Materials (TLMs)

- Spring balance, rubber balloons, syringes, wooden blocks, weights, droppers Teacher References

1. Link 1 (Class 8 physics textbook)
2. Link 3 (Important questions)
3. Link 4 (Solutions - Biju's)

Hook Activity: *Push or Pull — Everyday Forces!*

Objective:

To introduce students to the concept of force as a push or pull that changes an object's motion, shape, or position.

Procedure:

1. Show Real Objects like:
    - A ball
    - A chair
    - A door
    - A rubber band
2. Ask Students:
    - *What happens when you push a ball?*
    - *What happens when you pull a rubber band?*
    - *When you open a door, are you using a push or a pull?*
3. Explain:
   Every time you push or pull an object; you apply a force.
   Force changes the state of motion or shape of objects.

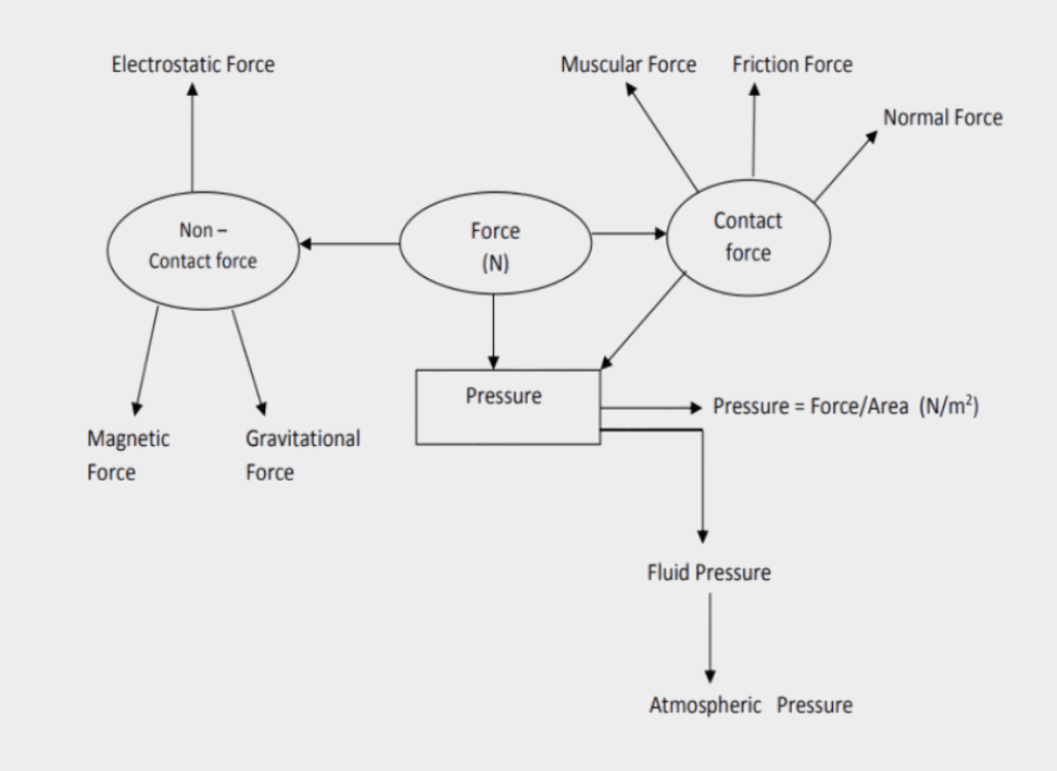
Quick Challenge:

Let them push and pull a few objects on their desk (erasers, books, pencil boxes) and write down:

- Which objects moved easily?
- Which were harder to move?
- Did the shape of anything change?

This hooks their curiosity and sets the base for discussing types and effects of force!

Concept Map



Synopsis

In this chapter, students explore the fundamental concepts of force and pressure. The chapter begins by introducing the concept of force as a push or pull that can change an object's state of rest or motion. It explains that forces have magnitude and direction and are measured in Newton (N) units. Various examples are given to help students understand different types of forces, such as muscular forces, friction, gravitational forces, and magnetic forces.

1. Muscular force: force that is caused by the action of the muscles in our body

2. Friction: the force responsible for changing the state of motion of objects.

3. Magnetic force: force exerted by a magnet.

4. Electrostatic force: The force exerted by a charged body on another charged body.

5. Gravitational force: Objects or things fall towards the earth because it pulls them.

The chapter then delves into pressure, explaining that conductors is the force applied per unit area. It introduces the formula for pressure: **Pressure = Force / Area.** Students learn about atmospheric pressure and pressure on fluids.

Period-wise Topics

Period No.	Concept/Topic	Learning Outcome	Remarks
1.1	Introduction 1.1 Force. a push or a pull	understand the concept of force	
1.2	1.2 Forces are due to an interaction	explain that force is due to interaction	
1.3	1.3 Exploring forces	understand different types of force	
1.4	1.4 a force can change the state of motion Activity 1.3 Activity 1.4	explain how a force can change its state of motion and its shape	
1.5	1.5 a force can change the shape of an object	concludes that force can change the shape of object	
1.6	1.6 contact forces, Muscular force, Friction	give examples for different contact forces	
1.7	1.7 non-contact forces Magnetic force, Electrostatic force	give examples for different non-contact forces	
1.8	1.8 pressure 1.9 pressure exerted by liquids and gases Activity 1.8	conduct experiments to understand the pressure exerted by liquids and gases	
1.9	Activity 1.9, Activity 1.10	explains that liquid exert pressure on the walls of container	
1.10	1.10 Atmospheric pressure. Activity 1.11	calculates the normal atmospheric pressure	

Period 1.1	**TOPIC:** Introduction; 1.1 Force - a push or a pull	**Activity**

Learning Outcome:
Students will be able to Understand the concept of force

Behavioural Expectations

Hook

- Teacher shows a video where force is applied.
  Video Link (*play and pause at 0:07)*
- Ask students what is happening in the pictures shown, collect some responses

Explicit Teaching/Teacher Modelling (I Do)	Group Work (We Do)	Independent Work (You Do)
1. Begin the lesson with the discussion of force. 2.  Explain that force is a push or pull using the **figures 1.1(a)(b)(c)** 3. Teachers can enact different forces in the class. 4. Give an overview of the chapter, topics that will be covered throughout the chapter.	1.Students push or pull objects like a chair or box to observe effects of Force 2. **Activity:** Group discussions **Instructions:** • Divide class into groups. • Ask students to discuss and fill textbook *"Activity1.1" page 4* • Ask each group to share their discussion one by one Activity 1.1	students make a list of activities where a push or pull or both applied by them when they are applying a force in various activities at their home.

Summary	Factual Question	CFU	Home Work
The action of push or pull is called -Force.	**Box Item Question in Page No.6**	1. What is force?	**Question : 1, 3 in Page No. 34 (Text Book)**

Period 1.2	TOPIC: 1.2 Forces are due to an interaction	Activity

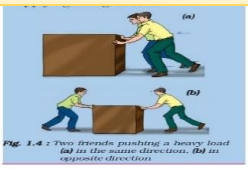
Learning Outcome: Students will be able to explain that force is due to interaction

Explicit Teaching/Teacher Modelling (I Do)	Group Work (We Do)	Independent Work (You Do)
	 *Fig 1.3 (b) : Who is pulling whom ?* of girls in Fig. 1.3 (b) are trying to pull each other. Similarly, the cow and the man in Fig. 1.3(c) appear to pull each  *Fig 1.3 (c) : Who is pulling whom?* Students will retrieve the nature of interaction between objects for force to exist. Based on the observations they will complete the following table.	Based on the examples of force given in the text book, Fig 1.2,1.3 students write in their own words it in their notebook and complete the following table. S.No. | No of objects/Persons | Direction of the force applying | | | | | | | | | | | | Total | Resultant force |

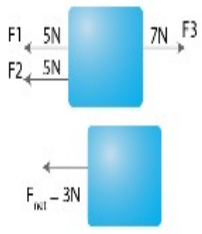
Summary	Factual Question	CFU	Home Work
When we push or pull an object then it will move or in rest	1)Do all objects require a force to be set in motion? Why or why not? 2)Is it possible for an object to be in motion without any force acting on it? Provide a reasoning.	 Who is pulling whom?	a) An object is pulled when it will move?

Period 1.3	TOPIC: 1.3 Exploring forces	Activity

Learning Outcome:
Students will be able to understand different types of force

Explicit Teaching/Teacher Modelling (I Do)	Group Work (We Do)	Independent Work (You Do)
Recall the tug of war activity in Fig1.5 Teacher explains the magnitude and direction of force activity Teacher explains the two or more forces acting in the same object simultaneously **Closing** • Discussion on the independent task. The teacher can collect a few responses. •Clear the doubt of the students. Closing: Forces arise due to interaction and can combine (same direction) or oppose each other (opposite direction).	 **Do the activity 1.3, Retrieving the** answers to the questions present in the text book. Activity**:** Fig. 1.4 – Two People Pushing a Heavy Load 1. Form pairs of students and give them a box or heavy object. 2. One student push alone; then both pushes together in the same direction. Discuss how the force increases. 3. Then, have them push from opposite directions and observe whether the box moves or stays in place.  **Do the activity, Retrieving the** answers to the questions present in the text book. **Activity:** Fig. 1.5 – Tug-of-War	Students record observations and answer: "What happens when forces are balanced or unbalanced?" Student can play the rope

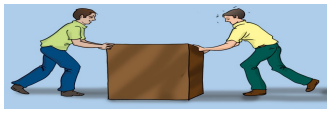
	1.Split the class into two teams. Conduct a small tug-of-war. 2.Observe when the rope moves and when it stays still: 3.The rope moves toward the team applying a greater force. 4.When both apply equal force, the rope stays in place. Change the number of students and again make the students to observe the changes.	activity. Student know the forces are acting on the rope in opposite directions. Student can add the magnitude of two or more forces when they are acting on the same direction. Students record observations and answer: "What happens when forces are balanced or unbalanced?"

Summary	Factual Question	CFU	Home Work
1.Forces applied in an object in the same direction, they add to one another. 2.Forces applied on an object in the opposite direction, they subtract from one another.	1. Box item Question in Page No 10 2. Box item Question in Page No 11	1.What happens if two friends pushing heavy load in the opposite direction? 2.6N, 5N forces are acting in West direction on a body. 20 N force is acting on the same body in the East direction. Calculate the net forces?	1.Calculate the net force when 6N, 3N forces are acting on a body in the opposite direction  2. Calculate the Net force

Period 1.4	Topic: 1.4 a force can change the state of motion	Activity 1.3, Activity 1.4

Learning Outcome:
Students will be able to explain how a force can change its state of motion and its shape

Explicit Teaching/Teacher Modelling (I Do)	Group Work (We Do)	Independent Work (You Do)
➢ Do a quick recap of the previous class how the force can change the state of motion of body using Activity 1.2 ➢ Explain the state of motion of an object using the activity 1.3 ➢ Explain that a force can change the state of motion ➢ Pose the questions to summarize the observations	**Teachers can also follow activity on *page 14 (activity 1.4)*** Choose a heavy object like a table or a box, which you can move only by pushing hard. Try to push it all by yourself. Can you move it? Now ask one of your friends to help you in pushing it in the same direction [Fig.1.4(a)]. Is it easier to move it now? Can you explain why? Now push the same object, but	Student records the observations observed in Activity 1.3, 1.4 in their note books Object on which the Force applied | Initial State | Final State | | | | | | | | | | | | | | | |

Closing Discussion on the independent task. The teacher can collect a few responses.	ask your friend to push it from the opposite side [Fig.1.4 (b)]. Does the object move? If it does, note the direction in which it moves. Can you guess which one of you is applying a larger force?  *Two friends pushing the load in the same direction*  *Two friends pushing a heavy load in opposite direction*	

Summary	Factual Question	CFU	Home Work
Force can change the state of motion	Why speed of a tyre increases whenever it is pushed. Why? (Fig 1.7)	- Can we see the forces acting on a body?	Q. No 2, 4 in page no 36 (text book)

Period 1.5	Topic: 1.5 a force can change the shape of an object	Activity

Learning Outcome:
Students will be able to conclude that force can change the shape of object

Explicit Teaching/Teacher Modelling (I Do)	Group Work (We Do)	Independent Work (You Do)
Explain that a Force can change the shape of an object.	**Activity: Paper Folding Challenge** Materials required: paper Instruction: • Give each group a sheet of paper (preferably square). • Instruct the groups to fold the paper in different ways to create various shapes, such as a paper aero plane, fan, or origami animals. • Have the students observe how applying force through folding changes the shape and size of the paper.	Students will apply the knowledge to complete the Table 1.2 present in the Activity 1.5

	• Encourage them to experiment with different folding techniques and discuss their finding	

Summary	Factual Question	CFU	Home Work
Force can change shape of the object	Do all materials require the same force to alter their shapes? If it is not, why?	Give two examples of situations in which applied force causes a change in the shape of an object.	**Project work regarding force can change the shape of the object. Q. No 6 in page no 36 in text book**

Period 1.6	Topic: 1.6 contact forces, Muscular force, Friction	Activity

Learning Outcome:
Students will be able to give examples for different contact forces

Explicit Teaching/Teacher Modelling (I Do)	Group Work (We Do)	Independent Work (You Do)
Do a quick revision of the previous class. • Introduce different types of force i.e. contact force. • Explain the definition of contact force and different types of contact force. 1. muscular force 2. friction • Show a video of contact force and ask students to explain their understanding from the same. Link 7 **Closing** Discussion on student's doubts.	***Contact Force:*** • Make students into two homogeneous groups. Assign first group to list out the various activities in day-to-day life in which muscular force will be observed. ***Frictional Force:*** Assign second group to list out the various activities in day-to-day life in which frictional force will be observed • Give students some time to discuss amongst themselves. • Ask students to present the list of activities.	Student writes the examples of advantages and disadvantages of frictional friction in day-to-day life.

Summary	Factual Question	CFU	Home Work
1.The force resulting due to the action of muscles is known as the muscular force. 2.Since muscular force can be applied only when it is in contact with an object, it is also	**Why the bikes will slip on wet surface roads after rain?**	1. What is contact force? 2. Write examples of contact forces.	1. Define Contact forces. 2. Write daily life examples of contact forces.

called a contact force. A 3.The force responsible for changing the state of motion of objects is called the friction 4.Frictional force acts always opposite to the direction of motion of the body.			

Period 1.7	**Topic: 1.7 non- contact forces Magnetic force, Electrostatic force, Gravitational force**	**Activity 1.6, 1.7**

Learning Outcome:
Students will be able to

- give examples for different non-contact forces

Explicit Teaching/Teacher Modelling (I Do)	**Group Work (We Do)**	**Independent Work (You Do)**
Do a quick recap of the previous class. • Introduce non-contact forces. • Explain different types of non-contact force. 1. Magnetic force 2. Electrostatic force 3. Gravitation force • Define magnetic force. • Demonstrate Activity 1.7 (Textbook, page no: 22) to the students to build clear understanding. • Demonstrate different actions to show the gravity around us.	***Student will do the activity 1.6* given on Textbook, page *number 20*** ***Activity: Understanding magnetic forces*** **Materials Needed:** • Magnets (bar magnets or magnetic wands) • Various objects made of different materials (paperclips, coins, aluminium foil, wooden sticks, plastic objects, etc.) **Instructions:** • Begin by providing a brief introduction to magnetism and magnetic force. • Explain that magnets have two poles, North and South, and like poles repel while opposite poles attract. Also, inform the participants that magnetic force can act through certain materials like paper or plastic. • Divide the participants into small groups. • Give each group a set of objects made of different materials. Ask them to predict which objects will be attracted to magnets and which ones will not. • Ask them to write in their notebooks. • Now, provide the groups with	Student writes daily examples in which he observes the ***gravitational force.***

	magnets and allow them to test each object to see if it's attracted to the magnet or not. As they test each object, have them record their observations and results in their notebook. • Gather the groups together and discuss their findings. ***Students will do the activity 1.7 (page No 22 in text book)*** ***Activity: Understanding Electrostatic force and Gravitational force.*** **Materials Needed:** - Balloons - Small pieces of paper or confetti - Plastic cups - Plastic straws - Paper clips - A smooth tabletop or floor **Instructions** • Begin the activity with a brief introduction to electrostatic force and gravitational force. • Explain that both are examples of non-contact forces and have distinct properties. Mention that electrostatic force is the force between electrically charged objects, while gravitational force is the force of attraction between objects with mass. • Divide the class into small groups. **Electrostatic Force** Provide each group with a balloon and ask them to rub it against their hair or clothing to create static electricity. Then, have students hold the balloons near small pieces of paper or confetti. They should observe how the paper is attracted to the balloon due to electrostatic force. Discuss why the paper sticks to the balloon.	

Summary	Factual Question	CFU	Home Work
1.The force exerted by a magnet on a piece of iron is also a noncontact force. 2.The force exerted by a charged body on another charged or uncharged body is known as electrostatic force. 3.Objects or things fall towards the earth because it pulls them. This force is called the force of gravity, or just gravity	**Why like poles repel and opposite poles attract each other of two bar magnets?**	1. What are the examples of non-contact forces? 2. Have you ever played with magnets? How do they attract or repel each other? Can you think of something that uses magnets in your daily life?	Define magnetic force. What is electrostatic force, and how does it differ from other types of forces? Define gravitational force and its role in the universe. Question No 4, 7 in text book Page no 36

Period 1.8	TOPIC: 1.8 pressure 1.9 pressure exerted by liquids and gases Activity 1.8	Activity 1.8

Learning Outcome: Students will be able to

Hook Activity: Bed of Nails — The Pressure Trick!

□ Concept:

This activity shows how pressure is distributed over an area. When many nails are used, the force of your body spreads out, so, no single nail presses hard enough to puncture your skin.



□□ Materials Needed:

- A wooden board
- Dozens or hundreds of nails (all hammered in, heads sticking up, evenly spaced)
- Protective supervision for kids!

⚡How to Do It:

1. **Prepare the Nail Board:**
   Hammer the nails into the wooden base, leaving the pointed ends upward but spaced evenly (the more nails, the better the pressure distribution).
2. **Test the Concept:**
   First, press your hand gently — you'll feel the nails, but it won't hurt, because your hand's pressure is spread out.
3. **Stand or Sit:**
   Carefully stand or sit on the bed of nails (preferably with adult supervision or as a teacher-led demonstration). Because the body weight is spread across all the nail points, it doesn't hurt or cause injury.

□ Safety Note:

- Only attempt this with proper setup and adult supervision.

- Nails should be closely packed.
- Never jump onto the board — lower yourself gently.

Explicit Teaching/Teacher Modelling (I Do)	Group Work (We Do)	Independent Work (You Do)
• Recap of the previous concepts. • Teacher can begin the concept of pressure by posing questions to class like: 1. Have you ever wondered why it's easier to push a pin into a soft surface like clay compared to a hard surface like a wall? *(collect few responses)* • Define pressure. • Teachers can show the below video to the students for better understanding of the concepts. *(play video from 2:11-5:35)* • Link 8 *(Force and pressure)* • Introduce the formula of pressure. For better understanding, teachers can use some videos. One link is given below. • Link 9 *(What is pressure - YouTube)* **Closing** • What is pressure? • What is the formula of pressure?	**Activity:** To understand the effect of a force depends on the magnitude of the force and the area over which it acts. Material required: Transparent Plastic Pipe with 25 cm long and 5-7.5 cm diameter, rubber sheet, balloon, water. Instructions: Student do the activity 1.8 and repeat the activity 4 to 5 times by pouring different amount of water in the pipe and observe the relation between bulge in the rubber sheet and the height of the water column in the pipe. Alternate Activity: • Divide the class into groups. • Give each group a balloon and some bulletin board pins. • Ask students to blow the balloon and keep it gently on pins. • Ask them to create a little pressure with the help of a book on it. • Ask them to write their observations and discuss the results. Instruct the students to experiment with different levels of pressure and observe the behaviour of the water stream from the hole. • Facilitate a discussion where students share their observations and reflect on how the pressure exerted by squeezing the bottle affects the water flow from the hole.	Student records the observations observed in the activity 1.8 in his note book. S.no. | Height of the water column | Diameter of the bulged part of the rubber sheet | | | | | | | | | | | | | | | | | | | |

Summary	Factual Question	CFU	Home Work
Summary The force acting on unit surface area is called Pressure	Why Shoulder bags are provided with broad straps and not thin strap? Why the tools meant for cutting and piercing	• What is pressure? • What is the formula of pressure? • When you shake a carbonated beverage	• What is pressure? Question No 10 in Page No 36 in text book

	always have sharp edges?	and then open it, why does the liquid spray out forcefully?	

Period 1.9	**TOPIC: Activity 1.9, Activity 1.10**	**Activity** 1.9,1.10

Learning Outcome:
Students will be able to explain that liquid exert pressure on the walls of container

Explicit Teaching/Teacher Modelling (I Do)	**Group Work (We Do)**	**Independent Work (You Do)**
• Recap of the previous class. • Pose questions or present examples to capture the student's attention and set the context for discussing pressure exerted by liquids and gas. ask the question • Why does a balloon inflate when we blow air into it?" or "Have you ever wondered why water squirts out of a small hole in a water bottle when we squeeze it?" molecules collide with the walls of the container. Emphasize that an increase in the number of gas molecules or an increase in their speed results in higher gas pressure • Define pressure exerted by liquids and gases • Focus on pressure exerted by liquids. Explain that the pressure exerted by a liquid depends on its depth and density. • Explain that the pressure exerted by a gas depends on the force with which the gas	Students do the activity 1.10 Student write the answers for the questions like 1.Do the different streams of water coming out of the holes fall at the same distance from the bottle? 2.What does this indicate? **Alternate Activity:** Water bottle fountain Materials required: plastic bottle, bulletin board pins, water. Instructions: • Divide the students into small groups of 3-4 members. • Provide each group with a plastic water bottle, water, bulletin board pin. • Instruct each group to fill their water bottle about three-fourths full with water. • Ask the groups to tightly seal the water bottle with its cap. • Instruct the students to carefully poke a small hole near the bottom of the water bottle using the bulletin board. • Encourage the groups to observe what happens when the water bottle is squeezed or squeezed and released. • Instruct the students to experiment with different levels of pressure and observe the behaviour of the water stream from the hole. • Facilitate a discussion where students share their observations and reflect on how the pressure exerted by squeezing the bottle affects the water flow from the hole.	Student will note down the observations and writes the conclusion in their own words in their note book.

Summary	**Factual Question**	**CFU**	**Home Work**
Liquids exert pressure on the walls of container		1. Imagine you're diving deep into a swimming pool. How	Ask students to write some examples of pressure exerted by

		does the pressure of the water increase as you go deeper? 2. When you shake a carbonated beverage and then open it, why does the liquid spray out forcefully?	liquids and gases in their notebooks.

Period 1.10	**TOPIC: 1.10 Atmospheric pressure. Activity 1.11**	**Activity**

Learning Outcome:
Students will be able to calculates the normal atmospheric pressure

Explicit Teaching/Teacher Modelling (I Do)	**Group Work (We Do)**	**Independent Work (You Do)**
• Begin the lesson by asking students about their experiences while going in high altitudes or using straw while drinking. Encourage them to share their experiences or observations. • Explain that atmospheric pressure is the force exerted by the weight of the air in the Earth's atmosphere on objects on the surface of the Earth. It is the pressure caused by the air molecules in the atmosphere pressing down on everything around us. Discuss the unit of atmospheric pressure, instrument used to measure it. • summarize the main points covered in the lesson, emphasizing the definition of atmospheric pressure and its effects.	Teachers can ask the students to do the ***activity 1.11* given on *page number 30 in groups***. Material required: Rubber sucker.	Student do the Extended Learning-Activities and Projects in page no 38 in text book.

Summary	**Factual Question**	**CFU**	**Home Work**
The pressure exerted by air around us is known as atmospheric pressure.	If the you're your head were 15x15 cm, how much force, air will exert on your head.	What is atmospheric pressure? a) The force exerted by air molecules on objects on Earth's surface b) The force exerted by gravity on objects in the atmosphere c) The force exerted by the Sun's rays on the Earth's surface d) The force exerted by wind on buildings and structures What is the unit of measurement for atmospheric pressure? a) Celsius b) Kilograms per cubic meter c) Newtons d) Pascals	Students do the research on Otto Von Guericke and write a paragraph about it in their notebook.

Teacher's Diary

Name of the Teacher:		Name of the Month:	
Name of the Lesson:		Class:	

Period No	**Name of the Concept to be taught**	**Date**	**Activities Conducted during the teaching**	**TLM Used**	Remarks
1.1	Introduction 1.1 Force. a push or a pull				
1.2	1.2 Forces are due to an interaction				
1. 3	1.3 Exploring forces				
1. 4	1.4 a force can change the state of motion Activity 1.3 Activity 1.4				
1. 5	1.5 a force can change the shape of an object				
1. 6	1.6 contact forces, Muscular force, Friction				
1. 7	1.7 non-contact forces Magnetic force, Electrostatic force				
1. 8	1.8 pressure 1.9 pressure exerted by liquids and gases Activity 1.8				
1. 9	Activity 1.9, Activity 1.10				
1. 10	1.10 Atmospheric pressure. Activity 1.11				

1	What were some of the specific strategies that I used to encourage participation? How effective were they? What will I do differently next time?
2.	Were there any concepts or activities that students found particularly difficult? How will I adapt my approach to address these difficulties in the next lesson?

3.	What additional resources or modifications could improve the effectiveness of this lesson in future implementations?
4.	How well did I adjust my teaching based on student reactions or unforeseen challenges?

Teacher's Signature Head Teacher's Signature

Head Teacher's Suggestions:

Teacher Notes:

July 2025

No bag Day

Cluster Complex

Teacher Resources

Sunday	Monday	Tuesday	Wednesday	Thursday	Friday	Saturday
		1 NI	2 2.1	3 2.1	4 NI	5 No Bag Day
6 Sunday Muharram	7 NI	8 2.2	9 2.2	10 2.2	11 NI	12 Second Saturday
13 Sunday	14 2.3	15 2.3	16 2.3	17 2.3	18 NI	19 No Bag Day Cluster meeting
20 Sunday	21 2.4	22 2.4	23 2.4	24 NI	24 2.5	26 No Bag Day SMC Meeting
27 Sunday	28 2.5	29 2.5	30 NI	31 NI		

Week1: 2.1

Week 2: 2.2

Week 3: 2.3

Week 4: 2.4 – 2.5

Week 5: 2.5

↓ TEACHER'S NOTES ↓

Chapter 2: Coal and Petroleum

Learning Outcomes:

1. differentiate exhaustible and inexhaustible natural resources
2. Understand the formation of coal in the earth
3. give examples of by products from coal
4. explain the process of petroleum refining
5. give the examples of some useful natural resources



Prior Concepts/Skills:

1. Basic understanding of natural fuels and their resources
2. Understanding of physical processes such as boiling point and distillation etc.

Teaching Learning Materials (TLMs)

1.Get outline map of India. Mark the places in the map where India. Mark and natural gas are found.

2. Images of renewable and non-renewable resources

Teacher References

www. energyarchive.ca.gov

web.ccsd.k12.wy.us

web.pcra.org

Hook Activity

Hook Activity: "Fossil Fuel Formation Simulation"

Objective:
Help students visualize how coal and petroleum are formed over millions of years from dead plants and animals under heat, pressure, and layers of earth.

Materials Needed:

- Slices of bread (representing layers of earth)
- Small leaves, twigs, or soft sponges (representing plant material)
- Small plastic animals or paper cut-outs (representing marine organisms)
- Heavy books or weights
- Clear plastic wrap

Activity Steps:

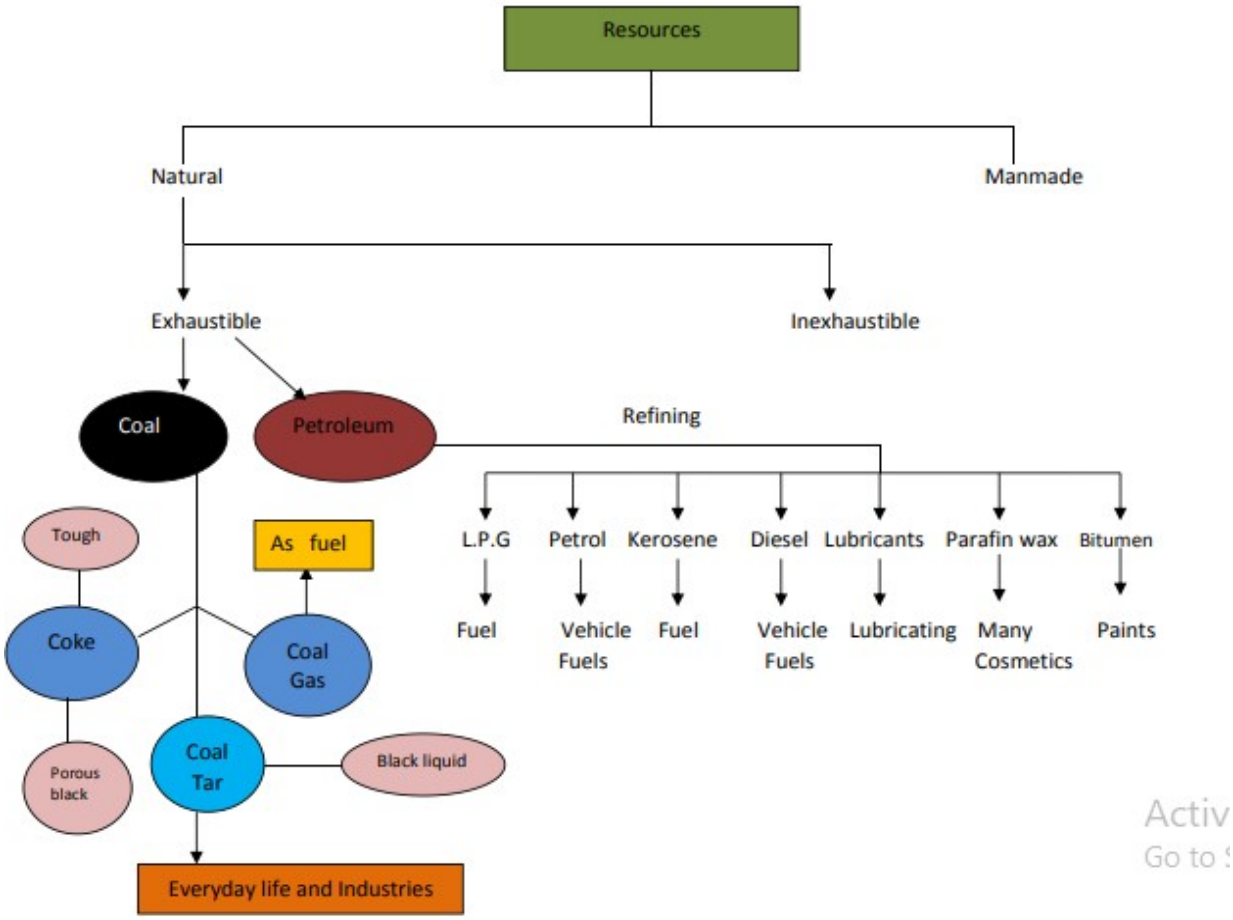
1. **Layering:**
   Students create layers using bread (soil layers) and leaves/twigs (plants) or animal cut-outs (marine creatures) to mimic nature's fossil-forming setup.
2. **Compression:**
   Wrap the stack tightly in plastic wrap and place heavy books on top to simulate pressure.

3. **Time Travel Talk:**
   While waiting, discuss how natural processes take **millions of years** to convert organic matter into **coal** (from plants) and **petroleum** (from marine life) under heat and pressure.

4. **Reveal & Relate:**
   After some time, students observe how the layers flatten and compress — a mini version of fossil fuel formation!

Discussion Prompt:
"Imagine this process happening deep underground over millions of years — this is how the coal and petroleum we use today were formed! Why do you think it's important to use them wisely?"

Concept Map



Synopsis

- **Natural resources:** The sources which are obtained from nature are called natural resources.
- **Types of natural resources**
  - Inexhaustible natural resources: Sunlight, air, etc.
  - Exhaustible natural resources: Forests, coal, petroleum. minerals, wild life, natural gas, etc.
- **Carbonisation:** The slow process of conversion of dead vegetation into coal is called carbonisation.
- **Destructive distillation of coal:** The main products obtained by the destructive distillation of coal are as follows:
  (1) Coke (2) Coal tar (3) Coal gas
- **Petroleum:** It is dark brownish to green coloured viscous liquid fossil fuel. It has a strong foul smell due to the **presence** of sulphur containing compounds in it. It is commonly called crude oil.

The economy of a nation depends to a great extent on petroleum wealth, that's why petroleum is called the black gold.

- **Uses of petroleum**
  (1) Petroleum products are used as fuels.
  (2) Lubricating oils, and Vaseline are used as lubricants.
  (3) Paraffin wax is used for manufacturing candles, polishes, waxed paper, water proofing, etc.
  (4) Some of the by-products of petroleum after purification are used in the preparation of medicines, ointments, face creams and cosmetics**.**

- **Natural gas:** Natural gas was formed millions of years ago along with petroleum when microscopic sea plants & animals died & got buried under the sand & mud. These plants & animals under anaerobic conditions changed to gas.
- **Composition:** methane (about 85%), ethane (about 10%) propane (about 3%) and butane
- **Uses of natural gas:**
  - As a fuel
  - As a source of hydrogen & carbon
- **Conservation of fossil fuels:** In India, the Petroleum Conservation Research Association (PCRA) offers the following tips to conserve petrol & diesel while driving.
  (1) Drive at a constant & moderate speed as far as possible. Driving at a high speed or slow speed wastes a lot of fuel.
  (2) Switch off the engine if you have to wait at traffic lights or for any other reason.
  (3) Check the tire pressure regularly.
  (4) Make sure that you send your vehicle to the garage for regular maintenance.

Period wise Topics

Period No.	Concepts/Topic	Learning Outcomes	Remarks	Signature of the Teacher
2.1	Introduction to exhaustible and inexhaustible natural resources Activity 2.1	differentiate exhaustible and inexhaustible natural resources		
2.2	Activity 2.2 Coal, Story of coal	Understand the formation of coal in the earth		
2.3	Coke, Coal tar, Coal gas	give examples of by products from coal		
2.4	2.2 petroleum, refining of petroleum	explain the process of petroleum refining		
2.5	2.3 natural gas 2.4 some natural resources are limited	give the examples of some useful natural resources		

Period 2.1	Topic: Introduction to exhaustible and inexhaustible natural resources	Activity 2.1

Learning Outcome: Students will be able to

- differentiate exhaustible and inexhaustible natural resources

Behavioural Expectation

1. We will listen attentively when others are speaking
2. We will actively participate in classroom discussions and activities
3. We will bring the required material to class every day
4. We will complete our tasks (classroom and homework) on time.

Hook Teacher shows the images of renewable and non-renewable resources, and asks the following questions

- Ask students to identify different resources in the pictures
- Ask students where they have seen these resources
- Can they explain the meaning of these terms -renewable and non-renewable?

Explicit Teaching/Teacher Modelling (I Do)	Group Work (We Do)	Independent Work (You Do)
• **Ask the whole class:** How would you all define natural resources? • What are man-made resources? • Call any student to answer the question in their own way. • Give positive feedback even if they go wrong. • Define natural resources and man- made resources. ◦ The resources which are obtained from nature are called natural resources. ◦ The resources made by humans using the natural resources, are called man-made resources. • Explain the two categories of the natural resources – ◦ Inexhaustible (Renewable) Resources ◦ Exhaustible (Non-renewable) Resources • Explain the categories along with examples	• ***Group Activity:*** Divide the class into groups of 4-5 students, ask them to perform **Activity number 2.1.** • Create discussion around it and ask students to present it in the class.	**Activity:** • Ask students to do ***Activity 2.1.*** • Make a list of various materials used by us in daily life and classify them as natural and man-made. **Natural** | **Man-made** | | | | | | | | | |

Summary	Factual Question	CFU	Home Work
1. Sunlight and air are Inexhaustible natural	1. Differentiate between renewable and non-	1) What is an exhaustible natural	1. Define natural resources.

resources 2. coal, petroleum, natural gas are exhaustible natural resources.	renewable resources. 2. Classify the following as exhaustible and inexhaustible natural resources- air, sunlight, water, forests, wildlife, coal.	resource? 2) How are natural resources different from man- made resources?	2. Can we use natural resources forever? Give answers in your own words.

Period 2.2	**Topic: Activity 2.2 Coal, Story of coal**	**Activity**

Learning Outcome: Students will be able to understand the formation of coal on Earth

Hook: Teacher will ask- *""I'm thinking of a black rock that has been used for centuries to produce heat and energy. It's buried deep underground and is a crucial part of electricity generation."*

Ask students -

- What do you think this energy source could be?
- Have you heard of any natural resources that match these descriptions?
- After collecting a few guesses, reveal that you're talking about coal. Share some basic information about coal, its colour, where it's found, and its historical significance as an energy source.

Explicit Teaching/Teacher Modelling (I Do)	**Group Work (We Do)**	**Independent Work (You Do)**
• Do a quick recap of the previous class. • Introduce the topic of coal • Define coal. • Show the video to students given in group activity • Introduce the concept of coal formation in the natural environment.	• Show the video to the students and follow the instructions given below. Link 1 Divide the class into small groups. • Ask students to write the uses of coal and discuss the below questions in their respective groups. **Questions:** • What did you like about the video? • What new did you learn from this video? • Two new words you learnt from this video. • What is coal? • Have you seen coal? What is its colour? and where are they used by humans? • What is decomposed in the formation of coal? What percent of energy consumption comes from coal?	***Activity:*** Students can create a poster of different uses of coal and show it in the class

Summary	**Factual Question**	**CFU**	**Home Work**
1.Coal, petroleum and	1. What is coal?	- Explain the process	- What are the uses of

natural gas are fossil fuels 2. Fossil fuels are exhaustible resources 3. Fossil fuels are formed from the dead remains of living organisms millions of years.	2. List some uses of coal. **3.** How is coal formed in the natural environment? 4. What is carbonization? Why is coal called fossil fuel?	of formation of fossil fuels. - What are the effects of burning coal on the environment? - Describe characteristics of coke - Write a short note on coke.	coal? - Earlier, Coal was used to cook food, but not used now. Why? - Why do industries use coal? - Write a paragraph on the formation of coal - Which state is popular for coal? - Where are the biggest coal mines in India?

Period 2.3	**Topic: Coke, Coal tar, Coal gas**	**Activity**

Learning Outcome:

Students will be able to give examples of the by-products of coal and understand the types of coal and various products of coal

Explicit Teaching/Teacher Modelling (I Do)	**Group Work (We Do)**	**Independent Work (You Do)**
• Do a quick recap of the previous class. • Ask students about the coal studied in the previous period. • Introduce the topic of various types of coal. ◦ Anthracite, ◦ Bituminous, Lignite. • Introduce and explain the topic: Various products of coal. ◦ Coke ◦ Coal Tar ◦ Coal Gas	**Activity: Group discussion and presentation.** **Instructions:** o Show the video to the students and ask the questions given below. Link 3 o Ask students to discuss the following in their respective group: o What did you learn from this video? o What is the carbon content in the anthracite coal? o Which type of coal is known as hard coal? o Name the process through which coal is formed. o As students share their answers to the questions given above, summarise it for everyone's benefit. o What is destructive	***Activity:*** Create a comparison chart that highlights the characteristics of different types of coal • Students can create a comparison chart that includes columns for each type of coal and rows for characteristics and uses. **Types of coal** | **Characteristics** | **Uses** | | | | | | | |

	distillation of coal? o What are the different products of destructive distillation of coal?		

Summary	Factual Question	CFU	Home Work
Coke, coal tar and coal gas are the products of coal	1. Name the products obtained and their uses when coal is processed in industry. 2. What is the reason that a particular type of coal gives the maximum amount of heat and energy? 3. Coal tar is a mixture of various substances (T/F) 4. Coke is almost pure form of carbon. (T/F	1. Name the different types of coal present in the environment? **2.** What are the different products of coal? 3. Differentiate between coal tar and coal gas. 4. What would happen if we use coal excessively?	• Which type of coal is used in industries and why • Write in brief about a) Coke b) Coal tar and c) Coal gas

Period 2.4	TOPIC: Petroleum, Refining of petroleum	Activity

Learning Outcome:

Students will be able to explain the process of some useful natural resources

Explicit Teaching/Teacher Modelling (I Do)	Group Work (We Do)	Independent Work (You Do)
• Do a quick recap of the previous concepts. • Introduce the topic petroleum to the students briefly. • For better understanding, show the video to the students. Link 5 • Ask students a few questions from the above video. • What are the new terms you have learnt in this video? • How do we get oil from the natural environment? • What is an oil well? • Introduce the topic of petroleum refining. • Explain the steps of petroleum refining through	**Activity:** Group Discussion. **Instructions:** • Discuss India's energy needs and future challenges related to petroleum consumption. • Ask students to brainstorm ideas for reducing petroleum dependency and increasing the use of some renewable energy sources. • Create discussion space in the classroom **Activity: Role play** **Instructions:** • Divide the class into small groups of 3-4 students each. • Assign roles within each group, such as "Crude oil Group" "Paraffin wax	**Activity:** Ask students to draw **Figure 3.4 on** a chart paper and produce it in the class. Get an outline map of India. Mark the places in the map where petroleum is found. Show the places where petroleum refineries are situated

the fractional distillation process. ● Start the class by asking questions about the topic of petroleum refining taught in the previous class. ● Do a quick revision of the topic studied in the last period. ● Ask students if they have some doubts about topics studied in the previous class? ● Show students a video- Link 6	Group," "Lubricating oil Group," and "Asphalt Group." "Kerosene Group" Petrol group" ● Each group will define them ● Characteristic features, appearance ● the temperature at which they get extracted during the refinery process. ● Their different uses ● As students share their responses through role play, summarise it for everyone's benefit.	

Summary	Factual Question	CFU	Home Work
1.petroleum gas, petrol, diesel, kerosene, paraffin wax, lubricant oil are obtained by refining petroleum	**1.**Draw a labelled diagram showing the fractional distillation of petroleum. Give the uses of any three fractions. 2.Identify the group of **constituents** of petroleum. Diesel petrol, rayon Lubricating oil, Paraffin wax, bitumen Diesel, Bitumen, caustic soda Petrol, rayon, paraffin wax 3.What are the four different uses of petroleum products?	1.What are 'Petrochemicals'? Give the uses of petrochemicals obtained from petroleum. 2.Why is petroleum also known as 'black gold'? 3.How is petroleum formed? 4.What is refining?	1.Explain the process of formation of petroleum. 2.Imagine a world without petroleum-derived products. How would our daily lives be different? 3.Research about the history of oil in our country. 4.Read it before the class one by one in your next period. 5.Write a paragraph on the effect of petroleum production on aquatic organisms. Read it one by one in the class. 6.Write down various constituents of petroleum. 7.What are the different uses of petroleum? Name the petroleum product used for surfacing roads.

Period 2.5	**TOPIC: Natural Gas & some natural resources are limited**	**Activity**

Learning Outcome:

Students will be able to give the examples of some useful natural resources

Explicit Teaching/Teacher Modelling (I Do)	Group Work (We Do)	Independent Work (You Do)
1.Give a summary of the topic taught in the last class. 2.Introduce and explain the formation and use of natural gas. 3.Discuss about natural gas found in different parts of India. 4.Explain to students the importance of saving fossil fuels 5.Discuss about natural resources found in India, their extensive use and its consequences; ● fossil fuels ● forests ● minerals	**Activity: Role play** **Instructions:** Divide the class into small groups of 3-4 students each. Assign roles within each group, such as "Gas Explorer," "Gas Transporter," "Homeowner," and "Environment Observer." Explain to the groups that they will be acting out a scenario that depicts the journey of natural gas from the well to a home. Each group performs a brief role-play based on the following scenario: Gas Explorer: Start by pretending to drill for natural gas and extract it from the ground. Gas Transporter: Act out the transportation of natural gas through pipelines to various locations. Homeowner: Receive the natural gas and demonstrate its use for heating and cooking in a home. Environment Observer: Comment on the positive and negative aspects of natural gas usage, considering environmental impacts. **Presentations:** After each group's role-play, ask them to briefly explain their part in the scenario and how it contributes to the overall journey of natural gas. **Class Discussion:** Facilitate a class discussion to highlight the stages of the journey, the importance of natural gas, and the potential environmental considerations.	**Activity:** In an outline map of India, mark the places in the map where natural gases are found.

Summary	Factual Question	CFU	Home Work
coal and petroleum resources are limited. We should use them judiciously.	**1.** Define fossil fuels. 2.Can coal, petroleum and natural gas be prepared in the laboratory from dead organisms?	1.Why is CNG better than any other fuels? 2.Which govt organization advises people about saving petrol/diesel while driving? 3.Why is CNG better than any other fuels? 4.How is use of fossil fuel associated with global warming?	1.Write down ways to save fossil fuels. 2.Can coal, petroleum and natural gas be prepared in the laboratory? **Activity:** Choose any five families in your neighbourhood. Enquire whether their energy consumption (coal, gas, electricity, petrol, kerosene) has increased or decreased in the last five years. Enquire also about the measures they adopt to conserve energy. 3.What are the advantages of using CNG and LPG as fuels? 4.Fill in the blanks. (a) Fossil fuels are ……., and ……. (b) Process of separation of different constituents from petroleum is called ……. (c) Least polluting fuel for vehicles is…….

Teacher's Diary

Name of the Teacher:				Name of the Month:	
Name of the Lesson:				Class:	
Period No	**Name of the Concept to be taught**	**Date**	**Activities Conducted during the teaching**	**TLM Used**	Remarks
2.1	Introduction to exhaustible and inexhaustible natural resources Activity 2.1				
2.2	Activity 2.2 Coal, Story of coal				

2.3	Coke, Coal tar, Coal gas				
2.4	2.2 petroleum, refining of petroleum				
2.5	2.3 natural gas 2.4 some natural resources are limited				

1.	What were some of the specific strategies that I used to encourage participation? How effective were they? What will I do differently next time?
2.	Were there any concepts or activities that students found particularly difficult? How will I adapt my approach to address these difficulties in the next lesson?
3.	What additional resources or modifications could improve the effectiveness of this lesson in future implementations?
4.	How well did I adjust my teaching based on student reactions or unforeseen challenges?

Teacher's Signature Head Teacher's Signature

Head Teacher's Suggestions:

Teacher Notes:

August 2025

No bag Day

Cluster Complex

Teacher Resources

Sunday	Monday	Tuesday	Wednesday	Thursday	Friday	Saturday
					1 NI	2 No Bag Day
3 Sunday	4 FA - 1	5 FA - 1	6 FA - 1	7 FA - 1	8 Vara Lakshmi Vratham NI	9 Second Saturday
10 Sunday	11 3.1	12 3.2	13 3.2	14 3.3	15 Independence day	16 Sri Krishna Ashtami
17 Sunday	18 3.4	19 NI	20 3.5	21 NI	22 3.6	23 No Bag Day Cluster meeting
24 Sunday	25 3.7	26 NI	27 Sri Vinayaka Chavithi	28 NI	29 National Sports day NI	30 No Bag Day
31 Sunday						

Week1: FA - 1

Week 2: 3.1 – 3.3

Week 3: 3.4 – 3.6

Week 4: 3.7

TEACHER'S NOTES

Chapter 3: Friction

U2B8A2

Learning Outcomes

1. Identify the necessity of friction in day-to-day life.
2. Explain about the factors affecting friction.
3. Classify different types of frictions.
4. Give examples to understand fluid friction.
5. Suggest different ways to increase and reduce friction

Prior Concepts/Skills

1. Students should understand the concept of force and its impact on objects.
2. Students should predict the direction of an object when a force is applied to it.
3. Students should predict the magnitude of force to be applied on any object required for its displacement

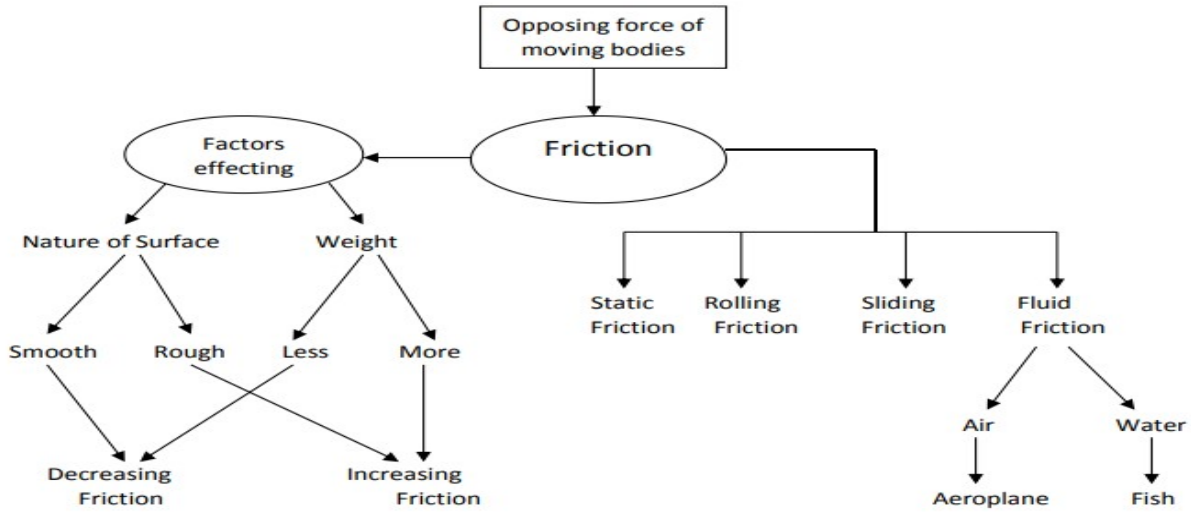
Teaching Learning Materials (TLMs)

Toy describing friction, Friction Toy - SCERT Textbook
Spring balance, Nylon thread, Brick, Plank, Different objects (small size)
Book, Round pencils – 6, Iron Nail, Hammer
Match box, Caram Board, Lubricant, Trolly suitcase, Ball Baring

Teacher References

- E:\Textbook\12 Rationalized Textbook\0854-Science\1 Source Files\Chapter 12\Chapter
- https://www.teachstarter.com/us/teaching-resource/all-about-friction-mini-book-us/
- https://guides.itsi.concord.org/friction.html
- Friction (Science Readers: Content and Literacy) by Suzanne Barchers

Concept Map



Synopsis

1. **Friction**- The resistance offered by the surfaces in contact when they move past each other.
2. **Force of Friction**- A force that resists motion when the surface of one object comes in contact with the surface of another.

3. **Factors affecting Friction**- Friction depends on three external factors that are, On the nature of the two surfaces that are in contact, the state of the smoothness of these two surfaces and the force that is acting on these two surfaces.
4. **Static Friction**- The frictional force that acts between the surfaces when they are at rest with respect to each other is called Static Friction. Example- Book kept at the table, wearing watches or rings, cards in a deck etc.
5. **Sliding Friction**- It comes into play when an object slides over another. Example – Sliding books on a table, rubbing hands together, sliding bricks on the ground, writing with a pen or pencil on paper etc.
6. **Rolling Friction**- It comes into play when an object is rolling over another. Examples – rolling of football, pulley in a well, skateboard tires etc.
7. **Fluid Friction**- It is defined as the friction between the fluid's layers when they are moving relative to each other. Examples – swimming in the water, sailing in the submarine, stirring coffee with a spoon, flow of ink in a ballpoint pen.

Period wise Topics

Period No.	Concepts/Topic	Learning Outcomes	QR Code	Any Specific Information
3.1	Introduction to friction and 3.1 force of friction Activity 3.1	understand different types of friction		
3.2	3.2 Factors affecting friction Activity 3.2	explain the factors affecting friction		
3.3	Activity 3.3	understand the friction on sloping surface		
3.4	3.3 Friction: A necessary evil	identify necessity of friction		
3.5	3.4 Increasing and reducing friction	suggest different ways to reduce friction		
3.6	3.5 Wheels reduce friction Activity 3.4	explain the role of wheels in reducing friction		
3.7	3.6 Fluid friction	give examples to understand fluid friction		

Period 3.1	**TOPIC: Introduction to friction and 3.1 force of friction Activity 3.1**	Activity

Learning Outcome:
Students will be able to

- understand force of friction

Behavioural Expectations

1. We will listen attentively when others are speaking
2. We will actively participate in classroom discussions and activities
3. We will bring the required material to class every day
4. We will complete our tasks (classroom and homework) on time.

Hook

- Teacher shows a video where force is applied. Video Link (*play and pause at 0:07)*

Explicit Teaching/Teacher Modelling (I Do)	Group Work (We Do)	Independent Work (You Do)
Teacher will discuss a few examples from daily life where frictional force comes into play and elaborate on the concept of frictional force. • Why would the carrom coins stop after moving when it is struck lightly? • Why a book, when it is pushed, stops after sliding for a few cm? She will demonstrate the book by sliding it on the table. • Why a pen stops after a while when rolled on the ground? The teacher will demonstrate the same. • Teacher will explain the concept and direction of friction force in the above activities.	Teacher will make groups of 4 or 5 students and ask them to perform the below activity and record their observations. Align the textbook with the help of a geometry box or any brick to make an inclined plane. Now try to roll different things from the inclined plane and observe which material will roll from the inclined plane and cover the farthest distance. 3. A.1.b] While sharing observations, students will be asked to share the direction of the frictional force.	Write the observations made by you from the activity Which object moves faster and covers more distance? Give the reason why some objects moves faster and some slow?

Summary	Factual Question	CFU	Home Work
The force acting on the object opposing its motion is called the force of friction.	What is friction? Where do you find more friction?	1. Which force acts on the carrom coins, book and pen in the above activity? 2. Why would the carrom coins stop after moving when struck lightly?	1. If we apply force on an object from the right, friction will act on that object from which direction? 2. Why does the textbook stop after sliding for a few CMS?

Period 3.2	**TOPIC: 3.2 Factors affecting friction Activity 3.2**	Activity

Learning Outcome:
Students will be able to

- explain the factors affecting friction

Explicit Teaching/Teacher Modelling (I Do)	Group Work (We Do)	Independent Work (You Do)
Nature of surfaces in contact Teacher will continue the previous discussion by highlighting the difference between the nature of the material of the cemented floor, plastic and cloth and how it was easier to slide when we use plastic but difficult in case of cloth. State of smoothness	Students will do the same activity in subgroups of 4 or 5. Set up an inclined plane in each group. Take some wooden blocks and roll it from the plane together.	Students will roll down a toy car from the below materials and arrange the friction force acting on the car in decreasing order: • Newspaper spread

Teacher will demonstrate the below activity with the help of an inclined plane.
At first, she will roll down a few objects (like the previous day's activity), like a pencil, pen, scale, rubber, sharpener, and notebook and ask students which object rolled down and went far.
Then a cloth will be spread on the inclined plane, and the same activity will be repeated.
Then a little bit of oil is spread on the surface, or it could be covered with an aluminium foil, and the same and same activity will be repeated.
The teacher will highlight the difference in the sliding of objects in all three materials.
Both nature of materials and the state of smoothness of materials affect friction.
Friction is caused by the irregularities on the two surfaces in contact. The higher the irregularities, the higher friction will be between them, which is why cloth provides higher friction than aluminium foil.

How hard the materials are pressed together
While connecting with students' experiences in the engineer activity teacher will take their observations and conclude that the harder the materials are pressed together, friction will be higher. Therefore, while dragging each other, students with less weight could slide away more quickly than heavy students.



Students will be asked to carefully observe the base of the blocks, which are moving slowly down the inclined plane and are going fast.
After this, students will use rectangular metal blocks and repeat the same activity. After this, two three blocks are tied together and are compared together with the single blocks.
Students will record observations of all three activities and share them with other groups according to the below questions:

on floor
• Dry marble floor
• Wet marble floor
• Cloth spread on floor
• Aluminium foil spread on floor
Spread a mat on the floor and make a student sit on it. Another student will drag the student sitting on the mat by pulling the mat. Now ask one more student to sit on the mat and drag the mat.

Q - State why it was difficult to drag the mat with two students compared to that of one.

Summary	**Factual Question**	**CFU**	**Home Work**
• Friction depends on the nature of surfaces in contact. • For a given pair of surfaces friction depends upon the state of smoothness of those surfaces. • Friction depends on how hard the two surfaces press together	Write the factors influencing the friction? Why heavier object has more friction?	1. What are the factors which affect the force of friction? 2. Why does a heavier object cause more friction than a lighter object?	1. Friction depends on the _____ of surfaces. 2 Sprinkling of powder on the carrom board _____ friction. 3. Why are metal blocks facing less friction than wooden blocks? 4. Why, when two or three blocks are tied together, go slower from the inclined plane than the single blocks? 5. What will happen if we smoothen out the base of the wooden blocks? Will they be faster or slower?

Period 3.3	**Activity 3.3**	Activity
Learning Outcome: Students will be able to understand the friction on sloping surface		

Explicit Teaching/Teacher Modelling (I Do)	**Group Work (We Do)**	**Independent Work (You Do)**
Sliding Friction The same wooden block, when pushed sideways, will slide on the table. Now sliding friction is coming into play. We can observe that sliding friction is smaller than static friction.	In sub-groups of 5, students will do the same activity with the help of wooden blocks or flat plates and record their observations. • Then slide it a bit (Sliding friction)	Categorise the pictures below according to the friction acting on them. List some commonly used home lubricants.

Summary	**Factual Question**	**CFU**	**Home Work**
When one body rolls over another body, rolling friction comes into play. Rolling friction is smaller than sliding friction.	**What is rolling friction?** **Where do you find rolling friction?** **Give an example helps rolling friction moving the objects.**	1. Which type of friction is experienced by a person walking on an icy sidewalk? a. Static friction b. Sliding friction 2. Which of the following scenarios best represents an example of sliding friction? a. Pushing a heavy box across a floor b. Holding a book stationary on a table c. A ball rolling down a hill d. A person walking on a sidewalk	1.Give two examples of sliding friction. 2. Sliding friction is _____ than the static friction.

Period 3.4	**TOPIC:3.3 Friction: A necessary evil**	Activity
Learning Outcome: Students will be able to identify necessity of friction		

Explicit Teaching/Teacher Modelling (I Do)	**Group Work (We Do)**	**Independent Work (You Do)**
Teacher will connect the energiser activity with this session with the help of statements like if we rub our hands for longer, we might feel discomfort or when we run very fast and fall down, we get bruises on our hands and legs if we get dragged down on the road. This shows that friction can be very harmful to us. Then he/she will demonstrate by	In subgroups of 5, some plastic bottles will be given to students, and they will be asked to open the cap and record the time taken to open the cap. Then, everyone will be asked to close and tighten the bottle cap. Then apply some oil to your hands, try to open the bottle cap, record the duration, and answer the questions	Q -Categorise the actions given in the worksheet into where friction is favourable and where it is not favourable. Q -List some commonly used home lubricants.

rubbing the oil in her/his hands and holding a knife to cut something. This will be difficult as oil reduces the friction between hands and knife, making our hands difficult to hold it. Here oil is acting as a lubricant. Substances which reduce friction are known as lubricants. Therefore, the teacher will explain both the favourable and unfavourable sides of friction. When two objects are rubbed together, heat is produced. This heat is sometimes useful for us, but sometimes harmful for us.	below: ➢ How much time did it take to open the bottle cap in the first case? ➢ How much time it took to open the bottle after putting oil in the hands? ➢ Friction will be higher in the first or second case? ➢ What is the role of oil in this activity? Is it increasing or reducing friction?	

Summary	**Factual Question**	**CFU**	**Home Work**
- Friction is a friend and foe. - The sole of the shoes and the tires of the vehicles are corrugated to increase friction. - . Friction is sometimes undesirable. - Friction can be reduced by using lubricants	Why friction is friend? Why friction is evil? Where friction is undesirable? Where friction is desirable?	1. Oil is used to increase or decrease the friction? 2. When the two objects are rubbed together, is the heat produced useful or harmful for us? What do you think?	1. While playing kabaddi, players are feeling that their hands are slipping away while catching each other. What do you think they should put on their hands to have a better grip? Choose from below: a. Soil b. Oil c. Powder d. Moisturising cream 2. In some machines, using oil as a lubricant may not be advisable. Why?

Period 3.5	**TOPIC: 3.4 Increasing and reducing friction**	Activity

Learning Outcome:
Students will be able to suggest different ways to reduce friction

Explicit Teaching/Teacher Modelling (I Do)	**Group Work (We Do)**	**Independent Work (You Do)**
Teacher will connect the energizer activity which is session playing with carom board. When we move striker on the carom board it will strike on the board intermittently. But if you spray powder on the board, it Move easily with less friction. Then he/she will demonstrate by moving with the rubber shoes on the normal floor stand floor tiles floor and on marble floor. Then teacher will explain the conditions to increase friction and reduce friction.	Make the students in sub groups of 5, some chappals made up of different materials are to be given and walk on different surfaces/flours. Then every student will be asked-concept of friction ● How do you feel? ● Where is the friction high? ● Where is the friction less? ● While riding a bicycle, after 3 years do you change the tires and chain?	Categories the actions given in the Worksheet into where the friction increases and decreases

To reduce friction in the machine parts we add oils or graphite powder-as a lubricant.	Why? Name some lubricants which are used in daily life- machine oil, engine oil.	

Summary	Factual Question	CFU	Home Work
The sole of the shoes and the tires of the vehicles are corrugated to increase friction. Lubricant is a material that reduces friction.	1. Why football players use shoes with spikes? 2.How lubricants help reduce friction?	1.Why do sportsmen shoes have spikes? 2.Why do we slip or slide on wet marble floors? 3.What is a lubricant?	1.Why are soles of shoes rugged? 2.What is a lubricant?

Period 3.6	**TOPIC: 3.5 Wheels reduce friction Activity 3.4**	Activity

Learning Outcome:
Students will be able to explain the role of wheels in reducing friction

Explicit Teaching/Teacher Modelling (I Do)	**Group Work (We Do)**	**Independent Work (You Do)**
Rolling Friction Now, place two pencils at the bottom of the wooden block, and We will see that block will easily roll on the table's surface. Here rolling friction is coming into play. Rolling friction is smaller than sliding friction hence cartwheels are used to transport heavier objects from one place to another.	Critically analyse the above activities	Collect the daily life situations where rolling friction helps reduce friction.

Summary	Factual Question	CFU	Home Work
When one body rolls over the surface of another body, the resistance to its motion is called rolling friction. Rolling reduces friction.	Why friction reduces motion of an object? Give an example of rolling friction	1. What type of friction is acting between the tyre and the surface? a. Rolling Friction b. Sliding Friction c. Static Friction d. Fluid Friction  2. Explain why objects moving	1. Which type of friction is acting on the slide and your body? a. Rolling Friction b. Sliding Friction c. Static Friction d. Fluid Friction  2.Why is sliding friction is

		in fluids must have unique shapes.	higher than rolling friction?

Period 3.7	**TOPIC: 3.6 Fluid friction**	Activity

Learning Outcome:
Students will be able to give examples to understand fluid friction

Explicit Teaching/Teacher Modelling (I Do)	**Group Work (We Do)**	**Independent Work (You Do)**
Fluid Friction Now tie a thread with the same block and dip it into a transparent small tub filled with water, and with the help of a moving thread try to move the block inside the water. It can be observed that while the thread is moving very fast outside the tub, the block is moving very slowly inside the tub due to fluid friction.	- Ask the students in sub groups 5, the activity how ball floats and rolls on the water. - Another activity observes the video of a plane flying in the air? How it is flying in the air. - The alternative word of fluid friction is Drag. - How the ball bearings work.	**Make a paper flight in different shapes and through its air.** **Observe in which shape it moves faster and more distance.**

Summary	**Factual Question**	**CFU**	**Home Work**
In science, the common name of gases and liquids is fluids. So, we can say that fluids exert force of friction on objects in motion through them. The frictional force exerted by fluids is also called drag.	Define drag? What is fluid friction? Where do you observe fluid friction?	1. What is meant by drag? 2. How can birds fly in the air?	1.Air resistance is an example of: a. Static friction b. Sliding friction c. Rolling friction d. Fluid friction 2. Which type of friction can be reduced by using streamlined shapes? a. Static friction b. Sliding friction c. Rolling friction d. Fluid friction

Teacher's Diary

Name of the Teacher:		Name of the Month:	
Name of the Lesson:		Class:	

Period No	**Name of the Concept to be taught**	**Date**	**Activities Conducted during the teaching**	**TLM Used**	Remarks
3.1	Introduction to friction and 3.1 force				

	of friction Activity 3.1				
3.2	3.2 Factors affecting friction Activity 3.2				
3.3	Activity 3.3				
3.4	3.3 Friction: A necessary evil				
3.5	3.4 Increasing and reducing friction				
3.6	3.5 Wheels reduce friction Activity 3.4				
3.7	3.6 Fluid friction				

1.	What were some of the specific strategies that I used to encourage participation? How effective were they? What will I do differently next time?
2.	Were there any concepts or activities that students found particularly difficult? How will I adapt my approach to address these difficulties in the next lesson?
3.	What additional resources or modifications could improve the effectiveness of this lesson in future implementations?
4.	How well did I adjust my teaching based on student reactions or unforeseen challenges?

Teacher's Signature Head Teacher's Signature

Head Teacher's Suggestions:

Teacher Notes:

September 2025

Sunday	Monday	Tuesday	Wednesday	Thursday	Friday	Saturday
	1 4.1	2 4.2	3 4.3	4 NI	5 ~~Milad Un~~ -Nabi National Teacher's Day	6 No Bag Day
7 Sunday	8 International Literacy NI	9 4.4	10 4.4	11 4.5	12 4.6	13 Second Saturday
14 Sunday Hindi Diwas	15 International Day of Democracy NI	16 4.7	17 4.8	18 4.9	19 NI	20 No Bag Day Cluster and SMC meeting
21 Sunday	22 4.10	23 NI	24 Dussera Holiday	25 Dussera Holiday	26 Dussera Holiday	27
28 Sunday	29 Dussera Holiday	30 Durgaashtami				

E8W4U1 No bag Day

H6H1Q4 Cluster Complex

K5M8W3 Teacher Resources

Dasara Holidays 24.09.25 to 02.10.25
Dasara Holidays for Minority Institutions
27.09.25 to 02.10.25

Week1: **4.1 – 4.3**

Week 2: **4.4 – 4.6**

Week 3: **4.7 – 4.9**

Week 4: **4.10**

Week 5:

TEACHER'S NOTES

Chapter 4: Chemical Effects of Electric Current



Learning Outcomes:

1. Classify liquids as good conductors and poor conductors of electricity.
2. Explain the reasons for conduction of electricity through acids.
3. Conduct simple investigations to seek answers to queries like "Does pure water conduct electricity'?
4. Draw labelled diagram of passing of current through water.
5. Understand the uses of electroplating.

Prior Concepts:

Before starting this lesson, students should:

- Know the basics of **electric circuits** (closed path, current flow).
- Understand what are **conductors and insulators**.
- Have basic knowledge of **water, acids, and bases** from previous classes.
- Be familiar with the concept of **electrical energy** and its uses.

Teaching Learning Materials (TLM):

- Simple **circuit setup**: Battery, wires, switch, and bulb or LED.
- Different liquids: Tap water, saltwater, lemon juice, vinegar, distilled water, oil, sugar solution.
- Graphite electrodes or metal nails.
- Electroplating kit or copper sulphate solution.
- Charts showing the **electroplating process**.
- Diagrams and models of **electric circuits**.

Teacher References:

- NCERT Science Textbook Class 8.
- SCERT Andhra Pradesh Science Textbook Class 8.
- Teacher handbooks and lab manuals.

⚡Hook Activity (Engaging Introduction):

Activity Name: *"Liquid Detective: Conductor or Not?"*

Objective:
To arise curiosity about the electrical properties of different liquids.

Setup:
Create a simple circuit using a battery, wires, and a small bulb. Keep several liquids ready (water, lemon juice, saltwater, oil, vinegar).

Steps:

1. Show the class an **empty beaker** and ask: *"Do you think this liquid can light the bulb?"*

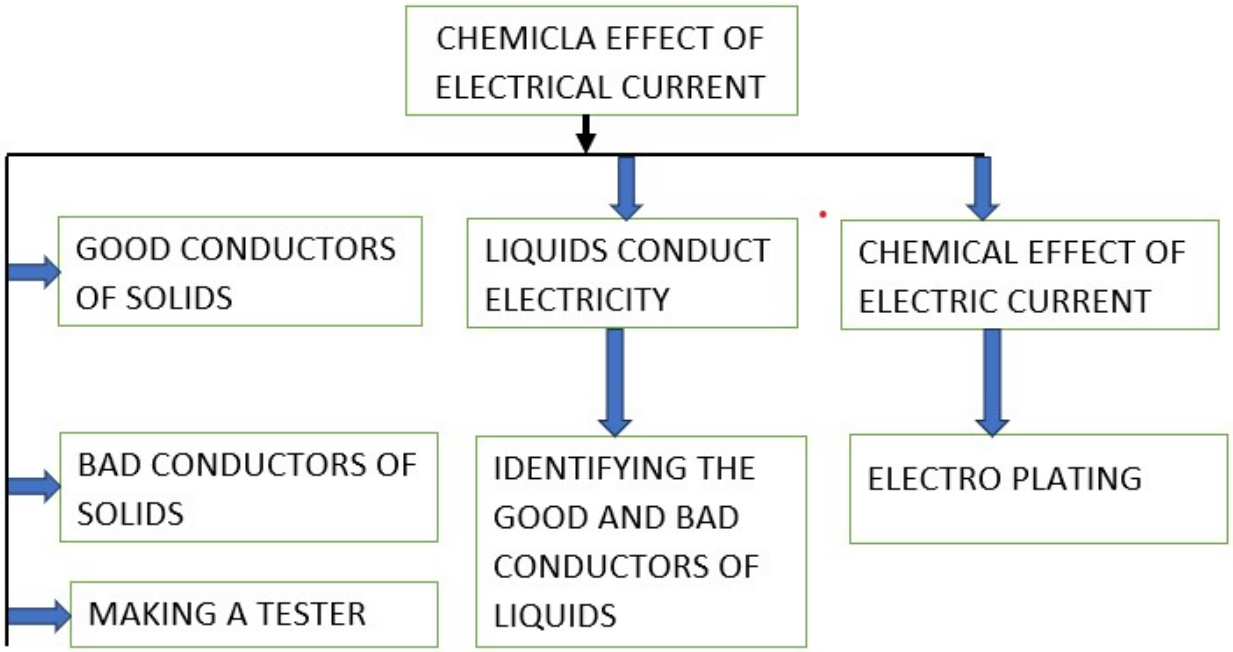
2. Let students predict for each liquid.
3. Pour different liquids into the beaker one by one and complete the circuit.
4. Observe whether the bulb lights up or not for each liquid.

Discussion Questions:

- Why did the bulb light up in some cases but not others?
- What could be present in those liquids that allows electricity to pass?
- Does pure water light the bulb?

Learning Link:
This activity hooks students directly into the lesson by showing them real-world evidence of **chemical effects of electric current**.

Concept Map



Synopsis

- Materials in which it is very difficult for the current to flow through them are called poor conductors of electricity. They have a very low value for electrical conductivity. In general, nonmetals are **poor conductors** of electricity.
- Materials in which current can flow easily are said to be **good conductors**.
- Like solids, some liquids can also conduct electricity. Pure or distilled water is a poor conductor but when some impurities like salts, bases or acids are dissolved in water then it starts conducting electricity.
- Certain substances like acids, bases and salts when dissolved in water then the water starts conducting electricity. This results in some chemical changes like bubbles may form at the electrodes due to evolution of gases or the colour of a solution may change. This is called the **chemical effects of electric current**.
- **Electroplating** is the process of coating a material with a layer of any desired metal using electric current.
- A sample that has to be coated is placed as the cathode in the electrolytic cell and a desired metal which is to be deposited on the surface is placed as an anode.

- **Uses of electroplating:**
  a. Chromium metal is electroplated on car parts, bath taps, kitchen gas burners etc.
  b. Electroplating of gold and silver is done on some less expensive metals.
  c. Electroplating of tin is done on iron as tin is less reactive.
  d. Electroplating of zinc on iron prevents it from corrosion.
  e. Some other phenomenon of current:
- When the current is passed through the filament of the bulb it becomes hot and starts glowing due to the **heating effect** of the electric current.
- When a compass is brought near a wire conducting electricity, the needle of the compass is deflected which indicates the presence of current in the wire. This is called the **magnetic effect** of electric current

Period-wise Topics

Period No.	Concepts/Topic	Learning Outcomes	QR Code	Any Specific Information
4.1	Introduction 4.1 is liquids conduct electricity activity 4.1	understand that liquids also conduct electricity		
4.2	4.1 Do liquids conduct electricity activity 4.1	make his own tester		
4.3	activity 4.2	understand the different situation of electrical conduction		
4.4	Activity 4.3	suggest different methods of preparing tester		
4.5	Activity 4.4	concludes that solution which contain ions will conduct electricity		
4.6	Activity 4.5	do experiments with different solutions like acids, bases and some salts		
4.7	4.2 chemical effects of electric current. Activity 4.6	understand that chemical changes occur during electrical conduction		
4.8	Conduction of electricity through vegetables testing	understand that some vegetables also conduct electricity		
4.9	4.3 Electroplating Activity 4.7	explain the necessity of electroplating		
4.10	Applications of electroplating	apply the principle of electroplating in daily life		

Period 4.1	**TOPIC: Introduction 4.1 is liquids conduct electricity activity 4.1**	**Activity**
Learning Outcome: Students will be able to understand that liquids also conduct electricity		
Behavioural Expectations: a. We will listen attentively when others are speaking b. We will actively participate in classroom discussions and activities		

c. We will bring the required material to class every day
d. We will complete our tasks (classroom and homework) on time

Hook

Imagine a world without electricity! It's hard to picture, right? Electricity powers our homes, gadgets, and the world around us. But have you ever wondered if electricity itself can conduct electricity

Think about the last time you encountered a thunderstorm. Did you ever wonder why it's not safe to be in a swimming pool or near water bodies during a lightning storm?

Explicit Teaching/Teacher Modelling (I Do)	Group Work (We Do)	Independent Work (You Do)
Teacher to introduce the chapter Chemical effects of electric current. Teacher to begin the lesson by discussing **good and bad conductors of electricity.** Ask few questions regarding the same to build the foundation for the upcoming concepts. 1.Think about the materials around you. 2.Can you classify them as good conductors or bad conductors of electricity based on your previous knowledge? 3.What properties of these materials might make them good or bad conductors?	**Activity: Debate** **Topic:** Should liquids be allowed near electrical devices and circuits? Should there be stricter rules about using liquids in areas where electricity is present? **Instructions:** 1. Divide the students into two groups: "For" and "Against. "Assign each group a designated area in the classroom for their discussions 2. Provide time for both groups to research and gather information to support their respective positions. Encourage them to use textbooks, class notes, and any available resources. 3. Give both groups time to prepare their arguments. 4. Conduct the debate, with each group taking turns presenting their arguments and responding to the opposing side. After the debate summarize the key points made by each side and discuss any common ground or areas of agreement.	Ask students to write a list of good conductors and bad conductors of electricity in their notebook.

Summary	Factual Question	CFU	Home Work
Some Liquids are good conductors of electricity and some are poor conductors	1. Write the names of few good conductors? 2. what is a bad conductor?	1. How can you distinguish the good and bad conductors? 2. What is closed circuit? How it is used in a tester?	Is it safe for the electrician to carry out electrical repairs outdoors during heavy downpour? Explain.

Period 4.2	TOPIC: 4.1 Do liquids conduct electricity activity 4.1	Activity
Learning Outcome:		

Students will be able to Make his own tester

Explicit Teaching/Teacher Modelling (I Do)	Group Work (We Do)	Independent Work (You Do)
Introduce the concept of **liquid conducting electricity**. - Do you think liquids can conduct electricity? Why or why not? - Think about examples of liquids you observed daily which conduct electricity?	**Activity:** Testing the electrical conductivity of liquid **Materials** a. Cells b. Bulb c. Connecting wire d. Liquid **Instructions** **1.** Divide the students into two groups. 2. Provide time for the both groups to observe 3. Join free ends of a tester together for a movement 4. This completes the circuit and bulb will glow 5. If the bulb does not glow, replace the bulb. Now the bulb will glow Now tester is ready to test the conductivity of liquid	Ask the students to write materials required for the activity

Summary	Factual Question	CFU	Home Work
Tester, it works with simple circuit concept. When circuit is closed that allows electric current through a bulb and it glows.	1. What is a tester? 2. What is function of battery in the tester**?**	1. Is the cell necessary for working of a tester?	Daw a neat diagram of testIng the conductivity of liquid?

Period 4.3	TOPIC: Activity 4.2	Activity

Learning Outcome:
Students will be able to understand the different situation of electrical conduction

Explicit Teaching/Teacher Modelling (I Do)	Group Work (We Do)	Independent Work (You Do)
Recall the concept of liquid conducting electricity. Perform the activity 4.2, which tests the "conduction of electricity in lemon juice or vinegar". Explain why in certain cases the bulb will not glow	**Activity:** The activity is to test the conductivity of various liquids and determine if they can complete an electrical circuit. **Materials Needed:** a. Small plastic or glass containers (e.g., cups or beakers) b. Various liquids (e.g., water, vinegar, lemon juice, saltwater, sugar water)	**Activity: Reflection** Reflect on what you learned from this experiment about the electrical properties of liquids. Write a

	c. Two metal wires with exposed ends d. Small light-emitting diode (LED) e. 9-volt battery f. Electrical tape **Instructions:** 1. Divide the class into groups. 2. Start by creating a simple circuit using the LED, battery, and wires. Connect one wire to the positive terminal of the battery and the other to the negative terminal. Attach the free end. 3. Pour water into one of the containers. Dip the ends of the wires into the water, making sure they don't touch each other. If the LED lights up when the wires are in the water, it means that water is a conductor of electricity. 4. Repeat the process with various liquids, such as vinegar, lemon juice, salt water, and sugar water. Observe whether the LED lights up when the wires are placed in each liquid 5. Instruct each group to record observations. 6. On the basis of student's observation, do a discussion.	note in your notebook

Summary	Factual Question	CFU	Home Work
All the liquids do not allow the electricity through them. Ex: Sugar, oils etc. Some liquids good conductors of electricity. Ex: All salt solutions, Acids and Bases	1. What is a tester 2. What is a good and bad conductor of electricity?	1. Pure water does not conduct electricity? If we add the salt that conduct electricity? What is the reason?	1. Write a step-by-step description of the experimental procedure you followed to test the conductivity of various liquids.

Period 4.4	TOPIC: Activity 4.3	Activity
Learning Outcome: Students will be able to suggest different methods of preparing tester		
Explicit Teaching/Teacher Modelling (I Do)	**Group Work (We Do)**	**Independent Work (You Do)**

When electric current pass through a liquid that glows a bulb and it is an indication of conducting electricity through liquids. We can make a tester with the help of magnetic effects of current with the help of the activity 4.3. Explain the concept of good and bad conductor of electricity. (Liquid)	**Activity:** Testing Electrical Conductivity of Different Solutions. **Materials required**: • a. A small plastic or glass container for each liquid (e.g. Cups or beakers) b. Various liquids (e.g., lemon juice, water, vegetable oil, milk, honey) c. Copper wire (about 1-foot or30 cm) d. Small compass needle e. A small battery (e.g., 1.5V AA battery) f. Electrical tape g. Insulated wire with exposed ends h. Small LED (optional) **Instructions:** 1. Divide the class into groups. 2. Take the small compass needle and wrap the copper wire around the container (similar to the matchbox tray in your previous activity). 3. Connect one end of the copper wire to the positive terminal of the battery. 4. Leave the other end of the wire free. 5. Connect the insulated wire to the negative terminal of the battery. 6. Ensure that the compass needle is inside the coil of wire but not touching it. 7. Dip the free ends of the wires (copper and insulated) into the first liquid, such as lemon juice. 8. Observe whether the compass needle shows deflection (moves) when the wires are in the liquid. 9. After testing one liquid, remove the wires from the liquid and rinse them with water. 10. Wipe them dry with a cloth or paper towel. 11. Repeat the testing process for each of the other liquids (e.g., water, vegetable oil, milk, honey). 12. Discuss your findings with your class. 13. Explain that the compass needle deflects when the liquid allows the flow of electricity (is conductive).	Students, individuall y fill table 4.1 which shows that some liquids are good conductors of electricity and some are poor conductors

Summary	Factual Question	CFU	Home Work
During the testing of liquids with a tester containing magnetic compass showing deflections are indication of conducting electricity through liquids. If deflections are not fine which sign of bad	1. How does magnetic needle deflect? 2. What is the difference between magnetic compass and electric bulb used in a tester?	1. Did you notice any relationship between the type of liquid and its ability to conduct electricity? 2. For the liquids that did not conduct electricity, can you speculate on the reasons behind their lack of conductivity?	1. Draw a neat diagram of a tester containing magnetic compass. 2.Write your observatio

conductors of liquids.			ns with oil and other liquids

Period 4.5	**TOPIC: Activity 4.4**	**Activity**

Learning Outcome:
Students will be able to Conclude that solution which contain ions will conduct electricity

Explicit Teaching/Teacher Modelling (I Do)	**Group Work (We Do)**	**Independent Work (You Do)**
Teacher explains 1. The testing whether the distilled water conducts electricity or not with the help of activity 4.4 2. Explain the concept of distilled water do not conduct electricity, but salt water conducts electricity.	**Activity:** Testing of conductivity of distilled water and common salt water solution. Materials required: 1. Plastic container 2. Distilled water 3. Common salt 4. Battery 5. Connecting wires 6. Small LED bulb **Instructions** **1.** Divide class into two groups **2.** Take a plastic container with distilled water **3.** Test the conductivity of the distilled water with tester. **4.** Ensure that distilled water does not conduct electricity. **5.** Now dissolve a pinch of salt in distilled water. **6.** Again, test with tester. **7.** Now we observe that the salt solution conducts electricity We conclude that solution which contain ions conducts electricity.	Students can conduct activity individually and identify the solutions which contain ions can conduct electricity. Students can write their observations in their notebook

Summary	**Factual Question**	**CFU**	**Home Work**
Generally, distil water does not conduct the electricity, because no free ions available in it. While adding the salt that provides the free ions and the liquid allows the electricity.	1. What is distil water? 2. How distil water act as a bad conductor?	1. Why distilled water does not conduct electricity? 2. Why common salt solution conducts electricity?	Right brief note about the activity that does not allow the electricity through a distil water.

Period 4.6	TOPIC: Activity 4.5	Activity

Learning Outcome:
Students will be able to Do experiments with different solutions like acids, bases and some salts

Explicit Teaching/Teacher Modelling (I Do)	Group Work (We Do)	Independent Work (You Do)
Teacher demonstrates the activity that shows different solutions like acidic Alkali and salt solutions conduct electricity	**Activity 4.5** **Material** - Three plastic bottles - Distilled water - Few drops of lemon juice - Caustic soda - Sugar - Tester **Procedure** **1.** Take three plastic bottles **2.** Fill them with distilled water **3.** Add few drops of lemon juice to the first bottle **4.** Add caustic soda to second bottle **5.** Add some amount of sugar to the third bottle **6.** Test which solution conducts electricity Lemon solution and caustic soda solution conducts electricity but not sugar solution.	Students are able to conduct the activity with different solutions like HCl, NaOH, Glucose etc. They can identify the solutions a. conducts the electricity and b. which do not conduct electricity

Summary	Factual Question	CFU	Home Work
Acidic, alkaline and salt solutions conduct electricity	1. Does the urea solution conduct electricity? Why?	1.Does the glucose solution conduct electricity? Why? 2.Why NaCl solution and HCl solution conducts electricity?	Write the few names of acidic solutions

Period 4.7	TOPIC: Activity 4.6	Activity

Learning Outcome:
Students will be able to understand that chemical changes occur during electrical conduction

Hook Ask the students

- Have you ever wondered how electricity can do more than just light up a bulb or power your gadgets? Today, we're embarking on a thrilling journey into the world of 'Chemical Effects of Electric Current.
- Have you ever thought about the hidden powers of a simple electric current?
- Have you ever observed any everyday objects changing when exposed to an electric current? If so, what were they, and what changes did you notice?

Explicit Teaching/Teacher Modelling (I Do)	Group Work (We Do)	Independent Work (You Do)
- Give a brief revision of the previous class. - Introduce the concept of **chemical effects of electric current**. - Perform activity 4.6	**Activity:** Discussion on activity 4.6 **Instructions:** 1. Divide the class into groups. 2. Ask each group to observe the activity performed by the teacher in the classroom. 3. Ask groups about their observation in the activity.4. Discuss the important points of the activity with the students	Students to answer the following question. - Explain the role of copper wires in this experiment. - What is the purpose of immersing the electrodes in a solution of water with added salt or lemon juice? - During the experiment, where should the metal caps of the carbon rods be positioned in relation to the water solution? Why?

Summary	Factual Question	CFU	Home Work
Summarize **a.** Chemical effects of electric current **b.** electrode	1. What are the chemical effects of electric current?	1. What is an electrode? 2. Which side is the positive terminal of a battery?	Write and research about William Nicholas in your notebook

Period 4.8	TOPIC: Conduction of electricity through vegetables testing	Activity

Learning Outcome:
Students will be able to
- understand that some vegetables also conduct electricity

Explicit Teaching/Teacher Modelling (I Do)	Group Work (We Do)	Independent Work (You Do)
Explain the concept of chemical effects of electricity with connecting to the previous class activity. Teacher will be covering the below points during this class: 1. The passage of an electric current through a conducting solution causes chemical reactions. As a result, bubbles of a gas may be formed on the electrodes.	**Activity:** Electricity conduction with fruits and vegetables **Instructions:** 1. Divide the class into groups. 2. Cut the potato in two halves. 3. Insert copper wire	Students can write their observations and draw the diagram of the activity. **Note:** Students can refer to fig. 4.6 for the same

2. Deposits of metal may be seen on electrodes. 3. Changes of colour of solutions may occur	tester into it. 4. Leave it for some time Write your observation.	

Summary	Factual Question	CFU	Home Work
1.The passage of an electric current through vegetables also conduct electricity 2. The passage of an electric current through a conducting liquid causes chemical reactions. The resulting effects are called chemical effects of current	1. The passage of an electric current through a solution causes ____effects	1. Why bubbles are observed at electrodes? 2. Did you observe any changes in the colour of solution?	Explain how electric current is involved in chemical reactions

Period 4.9	TOPIC: 4.3 Electroplating Activity 4.7	Activity

Learning Outcome:
Students will be able to

- explain the necessity of electroplating

Hook
Imagine a world where we have the power to transform ordinary objects into dazzling treasures, giving them a shiny and protective coat of metal. This mysterious process is called electroplating, and today, we're diving into the world of shimmering possibilities!

- Can you think of any everyday objects that have a shiny, metallic surface? What are they made of, and why might they have that coating
- Have you ever seen tarnished or corroded metal objects? Can you describe what they looked like and what might have caused the tarnishing or corrosion?

Explicit Teaching/Teacher Modelling (I Do)	Group Work (We Do)	Independent Work (You Do)
- Give a brief revision of the previous class. - Introduce electroplating.to the students by performing Activity 4.7. Ask students for their observation and discuss the observation.	**Activity: Case study** Electroplating is a remarkable process that uses electricity to coat objects with a thin layer of metal, providing them with enhanced properties and aesthetics. In this brief case study, we'll follow the story of Ram, an eighth-grader who encountered a fascinating application of electroplating. Ram loves collecting old and rare coins, and one day, he comes across a worn and tarnished silver coin. Determined to restore its beauty, he decides to explore electroplating as a way to give the coin a fresh look. Ram's challenge is to electroplate the silver coin with a thin layer of silver to restore its shine and	For each of the three objects, provide the following information: 1. Name of the object or product. 2. Description of the object's use or purpose. 3. Details about the metal coating used in electroplating (e.g., gold, silver, chromium). 4. Explain how

	lustre. He seeks guidance from his science teacher, Mr. Shyam. Mr. Shyam explains the basics of electroplating to Ram. He describes how the process involves using electricity to deposit a layer of metal onto the surface of another object. To electroplate the coin, Ram gathers the following materials: a. The tarnished silver coin piece of clean copper wire small plastic container b. 9-volt battery c. Silver plating solution d. Clips **Setting Up the Electroplating Process:** With Mr. Shyam's guidance, Ram sets up the electroplating process. He suspends the coin from the copper wire using a clip. Ram fills the plastic container with the silver-plating solution. He connects the copper wire to the positive terminal of the battery and the coin to the negative terminal using clips. Ram carefully observes the coin as the electroplating process begins. Over time, he notices that the coin's surface starts to change, becoming shinier. After a few hours, Ram removes the coin from the solution. He is thrilled to see that the coin now has a shiny and uniform layer of silver plating, restoring its original beauty. **Discussion Questions for Students:** 1. How did Ram use electroplating to restore his silver coin? 2. What materials did Ram need for the electroplating process? 3. Why is it necessary to connect the coin and the copper wire to the battery in electroplating? 4. Can you think of other real-world situations where electroplating could be valuable?	the electroplating process enhances the object's appearance, functionality, or durability.

Summary	Factual Question	CFU	Home Work
Electroplating is a process that uses electric current to coat a conductive object	1. Have you seen any products in your surroundings that are being electroplated?	1. Have you seen any products in your surroundings that are being electroplated?	For the above task. For one of the objects you've chosen, delve deeper into the industry

(usually a metal) with a thin layer of metal.		2. Can you think of innovative applications of electroplating in various industries? 3. The process that you saw in Activity 4.7 is used for purification of copper. A thin plate of pure copper and a thick rod of impure copper are used as electrodes. Copper from impure rod is sought to be transferred to the thin copper plate. Which electrode should be attached to the positive terminal of the battery and why? **4. True or False:** Electroplating is a process used to remove the metal coating from an object's surface	where it is commonly used. Discuss how electroplating is vital in that industry and the benefits it offers. Include any environmental or sustainability considerations if applicable. Write a short paragraph reflecting on how your understanding of electroplating has evolved through this research. Consider how electroplating impacts your daily life and the world around you.

Period 4.10	**TOPIC: Applications of electroplating**	**Activity**
Learning Outcome: Students will be able to apply the principle of electroplating in daily life		
Explicit Teaching/Teacher Modelling (I Do)	**Group Work (We Do)**	**Independent Work (You Do)**
- Give a brief revision of the previous class. - Explain the uses and importance of electroplating. - Explain Chromium and iron uses.	**Activity: Discussion** **Discussion questions:** 1. Can you think of any objects or products that you encounter regularly that might have been electroplated? 2. Have you ever noticed any changes in the appearance or properties of objects that you suspect were electroplated? 3. What are some industries where electroplating is commonly used, and why is it essential in those industries? 4. Can you name any specific items or materials that benefit from electroplating? 5. Are there any environmental or	Ask students to write what possible environmental hazards electroplating factories might be causing. Draw a poster.

	sustainability considerations associated with electroplating processes? 6. Encourage students to reflect on what they've learned about the uses of electroplating. 7. Ask them to think about how electroplating impacts their daily lives and the world around them. 8. Invite a few students to share their thoughts and insights with the class.	

Summary	Factual Question	CFU	Home Work
The process of depositing a layer of any desired metal another material, by means of electricity id called Electroplating	**1.what is Electroplating?** **2. Prepare a list of objects around you that are Electroplated?**	**1.** Explain the process of electroplating on key? **2.** Name the process used in 1 gram gold ornaments?	Complete question 11 and 12 from exercise questions given in the chapter ending.

Teacher's Diary

Name of the Teacher:		Name of the Month:	
Name of the Lesson:		Class:	

Period No	Name of the Concept to be taught	Date	Activities Conducted during the teaching	TLM Used	Remarks
4.1	Introduction 4.1 is liquids conduct electricity activity 4.1				
4.2	4.1 Do liquids conduct electricity activity 4.1				
4.3	activity 4.2				
4.4	Activity 4.3				
4.5	Activity 4.4				
4.6	Activity 4.5				
4.7	4.2 chemical effects of electric current. activity				

	4.6				
4.8	Conduction of electricity through vegetables testing				
4.9	4.3 Electroplating Activity 4.7				
4.10	Applications of electroplating				

1.	What were some of the specific strategies that I used to encourage participation? How effective were they? What will I do differently next time?
2.	Were there any concepts or activities that students found particularly difficult? How will I adapt my approach to address these difficulties in the next lesson?
3.	What additional resources or modifications could improve the effectiveness of this lesson in future implementations?
4.	How well did I adjust my teaching based on student reactions or unforeseen challenges?

Teacher's Signature Head Teacher's Signature

Head Teacher's Suggestions:

Teacher Notes:

October 2025

Sunday	Monday	Tuesday	Wednesday	Thursday	Friday	Saturday
E8W4U1 No bag Day	H6H1Q4 Cluster Complex	K5M8W3 Teacher Resources	1 Dussera Holiday	2 Mahatma Gandhi Jayanti Vijayadashami	3 NI	4 No Bag Day Sakunthala Devi Jayanti
5 Sunday International Teacher's Day	6 NI	7 NI	8 NI	9 NI	10 NI	11 Second Saturday
12 Sunday	13 FA - 2	14	15	16	17 NI	18 No Bag Day Cluster meeting
19 Sunday	20 Deepavali	21 NI	22 NI	23 NI	24 NI	25 No Bag Day SMC Meeting
26 Sunday	27 NI	28 NI	29 NI	30 NI	31 National Unity Day NI	

Week1:

Week 2:

Week 3:

Week 4:

Week 5:

↓ TEACHER'S NOTES ↓

November 2025

Sunday	Monday	Tuesday	Wednesday	Thursday	Friday	Saturday
						1 No Bag Day Andhra Pradesh Avatarana
2 Sunday	3 5.1	4 5.2	5 5.3	6 5.4	7 NI	8 Second Saturday
9 Sunday	10 SA - 1	11 Day	12	13	14 Day	15 No Bag Day
16 Sunday	17 SA - 1	18	19	20 5.5	21 5.6	22 No Bag Day Cluster meeting Fibonacci Day
23 Sunday	24 5.7	25 5.8	26 National Constituti 5.9	27 Sanjeevaraya 5.10	28 5.11	29 No Bag Day SMC Meeting
30 Sunday						

14 - 20 National Library Week

Week1: **5.1 – 5.4**

Week 2:

Week 3: **5.5 – 5.6**

Week 4: **5.7 – 5.11**

TEACHER'S NOTES

Chapter 5: Sound

Learning Outcomes:

1. Understand the process of production and propagation of sound.
2. Classify audible and inaudible sounds based on frequency.
3. Draw the conclusion that sound needs a medium for propagation.
4. Explain the process of the working of human ear.
5. Give examples of sounds with high frequency and low frequency.

Prior Concepts

1. Sound as means of communication between persons
2. Hearing as a sense
3. Identification of variety of sounds in our surroundings like animals and birds
4. Comfortable and uncomfortable sounds

Teaching Learning Materials

1. Sound is produced by a vibrating body- Activity 5.1,5.2,5.3,5.4,5.5
   (Pan, rubber, steel dish, water, ektara, glass tumblers, pencil.)
2. Sound produced by humans – activity 5.6 (Rubber strips, paper)
3. Sound needs a medium of propagation – Activity 5.7, 5.8, 5.9
   (Glass tumbler, cell phone, bucket, water, bell, stick, toy telephone.)
4. We hear sound through our ears – Activity 5.10
   (Plastic can, thin rubber balloon)

Teacher references

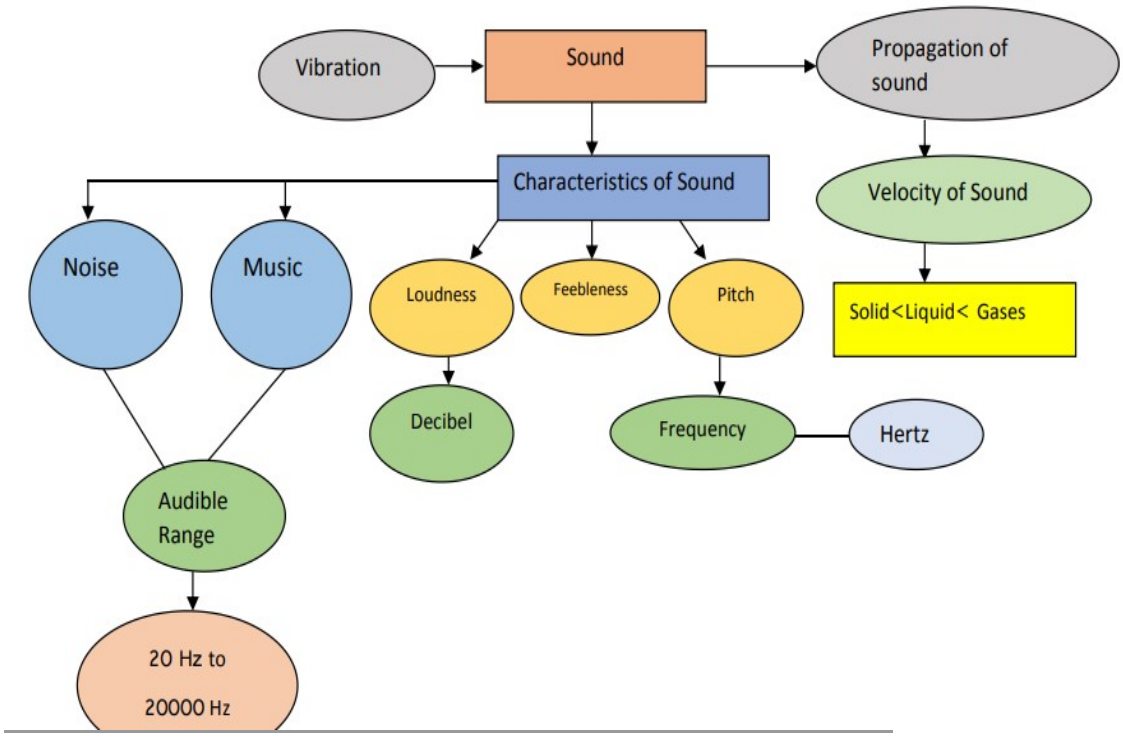
Lakhmir Singh & Manjit Kaur – Science for class 8th

- S Chand publishing

Together wit science --- 8th

- Rachana sagar-

Concept Map



Synopsis

- Sound is produced by a vibrating object/ body.
- If a tightly stretched string or band is plucked, it vibrates and produces sound. When it stops vibrating, it does not produce any sound.
- Different musical instruments have different vibrating parts.
- In humans, the sound is produced by the voice box or the larynx. It is found at the upper end of the windpipe.
- There is a slit between two vocal cords across the voice box or larynx. The slit allows the air to pass between them. The lungs force air through the slit and cause the vocal cords to vibrate and produce sound.
- Vocal cords may be tight and thin. The type or quality of voice depends on the conditions of vocal cords whether they are tight and thin or loose and thick.
- Sound needs a medium for propagation. It can propagate in solid, liquid, and gas.
- The eardrum in human beings is like a stretched rubber sheet. Sound vibrations make the eardrum vibrate and transfer the sound signal to the inner ear and then brain.
- The number of oscillations per second is called the frequency of oscillation.
- Frequency is expressed in Hertz (Hz). A frequency of 1 Hz is one oscillation per second.
- Amplitude and frequency are two important properties of any sound.
- Loudness of sound is proportional to the square of the amplitude of the vibration producing the sound.
- The loudness of sound depends on its amplitude. The loudness is expressed in a unit called decibel (dB).
- The frequency determines the shrillness or pitch of a sound.
- Sounds of frequencies below (20 Hz) and higher than 20 kHz cannot be detected by the human ear. Such sounds are called inaudible.
- For the human ear, the range of audible frequencies is roughly from 20 to 20,000 Hz.
- The unpleasant sounds are called noise and the presence of excessive or unwanted sounds in the environment is called noise pollution.
- Noise pollution in the surroundings may cause many health-related problems. A person who is exposed to a loud sound continuously may get temporary or even permanent impairment of hearing.

Period wise Topics

Period No.	Concepts/Topic	Learning Outcomes		Any Specific Information
5.1	Introduction and 5.1 Sound is produced by vibrating bodies activity 5.1,	concludes that sound is produced by vibrating bodies		
5.2	Activity 5.2 and Activity 5.3	understand that vibration in the body creates sound		
5.3	Activity 5.4, Activity 5.5,	different musical instruments create sounds differently		
5.4	Sound produced by humans	understand that vocal cards in human produce sounds		
5.5	Activity 5.6	concludes that in humans sound is produced by vibrating the vocal cords		
5.6	5.3 Sound needs a medium for	do the experiment that sound needs a medium for its propagation		

	propagation, Activity 5.7			
5.7	Activity 5.8 Activity 5.9	understand that sound can also travels through liquids and solids		
5.8	5.4 We hear sounds through our ears Activity 5.10	understand the need of ears to hear the sound		
5.9	5.5 Amplitude, time period and frequency of vibration	define amplitude, time period and frequency clearly		
5.10	Loudness and pitch Activity 5.11	explain the factors in which loudness and pitch depends		
5.11	5.6 Audible and inaudible sounds. 5.7 Noice and music 5. 8 Noice pollution	differentiate audible and inaudible sounds		

Period 5.1	**TOPIC: Introduction of Sound, Sound is produced by vibrating bodies**	**Activity**

Learning Outcome:
Students will be able to Conclude that sound is produced by vibrant bodies

Behavioural Expectations:

1. We will listen attentively when others are speaking
2. We will actively participate in classroom discussions and activities
3. We will bring the required material to class every day
4. We will complete our tasks (classroom and homework) on time

Hook
Teacher conducts an activity and gives these instructions to the students - Close your eyes. Sit quietly for 2 mins and observe what all sounds you can hear in the surrounding. After the activity, teacher asks the following questions -

- What sounds did you hear? (Expected Responses - Birds, traffic, fan, students in other classes)
- Can anyone tell me how you hear the chirping of birds or barking of dogs? or the meow sound of your lovely pet cat?
- Is there any difference in the sweetness / or hardness of sounds of these animals?
- How are we able to hear different types of sounds from our surroundings such as our favourite beautiful song, car horns, or someone playing flute at his home?

Explicit Teaching/Teacher Modelling (I Do)	**Group Work (We Do)**	**Independent Work (You Do)**
• Begin the lesson by engaging students in discussion about the different types of sound. • Give examples of some soft and loud sounds. • Examples of some soft sound are ◦ The ticking of a clock.	• Divide the class in groups of three where students discuss and write in their notebook about sound and its different sources such as ◦ animals, ◦ wind, ◦ flowing streams	***Activity:*** Production of sound by tapping various objects such as pencil on desk, spoon on desk/tiffin **Material required:** • A pencil • A spoon • Tiffin

• A creature's heartbeat. • Sound formed by a musical instrument such as Guitar or Sitar. • The slow drenched water from a tap. • playing of a piano • The unique sound of wind blowing. • Examples of some loud sounds are • Banging of a hammer • Car's horn • Sound of home appliances such as mixer, • Lion's roar. • Show images of different types of sound producing objects such as musical instruments, metallic utensils, etc. **Closing** Students will be given 1 minute to remember and write the name of various instruments that vibrate and produce sound.	avalanches, • Volcanoes. • aero planes, • helicopters, • road vehicles, • trains, • explosions, • factories, • Home appliances such as vacuum cleaners and fans. • Make a poster on A3 sheet about different types of sources of sound and show it in the classroom.	**Method:** • Tap the desk with the pencil /spoon, or the tiffin box with the spoon • Listen and watch carefully • Record what you see and hear. • Touch the object and feel if it is vibrating? **Observation:** • Every time you tap the object; a sound is produced. • Classify it as a soft or loud sound. • Vibration will be felt whenever you touch the object.

Summary	Factual Question	CFU	Home Work
Sound is produced by a vibrating object/ body.	1. List some musical instruments that produce sound by vibrations 2.In Veena the ……string produce sound	1. How is Sound produced? 2. How does vibration of an object produce sound?	• Write 5 examples of the following as they find in their surroundings: (a) Soft sound producing objects (b) Loud sound producing objects

Period 5.2	TOPIC: Activity 5.2, Activity 5.3	Activity

Learning Outcome:
Students will be able to understand that vibration in the body creates sound

Explicit Teaching/Teacher Modelling (I Do)	Group Work (We Do)	Independent Work (You Do)
• Recap of the topic/ concepts discussed in the last class. • Give a brief introduction about various types of sounds. • Present the concept of sound production by showing different	• Divide the class in groups of three **and p**erform the **Activity 5.2** from the textbook given on **Page No 104** • Divide the class in groups of three where students will	**Activity:** **Materials required:** • Pictures of various sound producing objects/ animals / musical instruments etc

objects that vibrate and produce sound.

- Explain different types of sounds produced by musical instruments by vibrating objects.
    - Table
    - Guitar
    - Sitar
    - Harmonium
    - Mridangam
    - Ektara
    - Manjira

- Show video to develop a better understanding in the class
  Link 1

Closing
Students will share the key points from the topic studied in class.

write different vibrating objects and check whether, during vibration, they produce sound or not.

- A chart paper
- Fevicol/ glue

Method:

- Collect and bring the pictures of various sound producing objects/ animals/ appliances in class.
- Paste it into the chart paper.
- Show it in the class
- Compare it with your friend's chart paper.
- Make a list of objects that produce sound.
- How many different types of sound-producing objects are common in both the chart papers?
- Can you think of any other objects that produce sound? make a list of it.
- Is there any similarity in how these objects produce sound?
- Note it in your notebook.

Key points:

- To and from or back and forth motion of an object is called vibration
- The objects in vibrations produce sound.
- When the object stops vibrating, it does not produce any sound.

In some cases, the vibrations in objects are easily visible to us. However, in some cases we cannot see vibrations.

Summary	Factual Question	CFU	Home Work
• Different musical instruments have different vibrating parts.	1. A ……. object produces sound 2. Vibration of small …… are not visible to us	1.How do we hear sound? 2 Can a wood produce sound by vibrating? 3. To and fro or back and forth motion of an object is	- Identify the part which vibrates to produce sound in the following instruments. a. Dholak b. Sitar c. Flute

		called…… 4. A ……. object produces sound 5. Vibration of small …… are not visible to us 6. Which of the following is true about sound? ● Vibrations are visible to us ● The vibration with small movement can be felt by touching the vibratory body ● Both are correct ● None of the above	- What is vibration? - Name a musical instrument which produces sound by blowing air into it. - What are the different objects that vibrate and produce sound? - How is sound produced? - How does sound help us?

Period 5.3	**TOPIC: Activity 5.4, Activity 5.5**	**Activity**

Learning Outcome:
Students will be able to understand how different musical instruments creates sounds differently

Explicit Teaching/Teacher Modelling (I Do)	**Group Work (We Do)**	**Independent Work (You Do)**
● Recap of the topic/ concepts discussed in the last class. ● Give a brief introduction about various types of sounds produced by a vibrating object. ● **Activity:** Do a Textbook **Activity 5.5,** given on **Page No 108,** to show the sound produced by jaltarang ● Show video to develop a better understanding in the class Link 2	**Activity: To show that vibration of objects produce sound** ● Divide class in groups of 4 ● Each group will perform its own activity and record their observations in the notebook. **Materials required:** ● A steel plate/ bowl ● A spoon **Activity:** ● Ask a student to hold the spoon in his/her hand and strike the side of the plate or the bowl gently. ● Now ask another student to touch the plate / bowl with his/her fingers and feel the vibration of the plate/bowl. ***Observation:*** ● Students who are touching the plate/ bowl will feel the	**Activity:** Perform the **Activity 5.3** from the textbook given on **Page No 104 and answer the following questions:** 1. Do you hear a sound? 2. Again, strike the dish and then touch it. 3. Can you feel the dish vibrating? 4. Strike the dish again. 5. Look at the surface of the water. Do you see any waves there? 6. Now hold the dish. What change do you observe on the surface of water? Can you explain the change? Is there a hint to connect sound

	vibration of it. ● Motivate them to have a group discussion, and ask them to write all the steps discussed in a sequence in their notebook. ● Move around to observe student's discussion and ask questions to clarify their misunderstanding. **instructions for the activity:** ● We will work in our learning groups ● Discuss the observations. ● Make notes from discussion ● We have 10 minutes for Discussion	**Key points:** ● Striking the edge of the bowl or tumblers produces a vibration in it ● These vibrations travel through the water and also make it vibrate. ● This vibration when transferred to surrounding air produces a sweet sound. ● Therefore, the experiment proves that vibration of objects/ things produces sound.

Summary	Factual Question	CFU	Home Work
Musical Sound: The sound which produce a pleasing effect on the ear. **Noise:** The sounds which produce a jarring or unpleasant effect.	1. What is jaltarang? 2. How does a jaltarang produce sound? 3. Which part of Jal-tarang vibrates to produce sound? 4. What is the cause of sound produced by jaltarang?	1.What are the different types of musical instruments that vibrate and produce sound? 2. Do all bodies produce sound? 3.How does vibration produce sound? 4 Do all vibrating materials make sound?	Make a list of familiar musical instruments and identify their vibrating parts.

Period 5.4	TOPIC: Sound produced by humans	Activity

Learning Outcome:
Students will be able to understand that vocal cords in human produce sounds

Explicit Teaching/Teacher Modelling (I Do)	Group Work (We Do)	Independent Work (You Do)
● Recap of the concepts discussed in the last class. ● Show the diagram of larynx to students to make them understand the sound producing organ of the human body. ● Students should note that - Larynx is the voice box in the human body. - It is situated at the upper	**Activity:** ● Divide the class in groups of three where students can make a box which can show the functioning of the vocal cords. **Materials Required:** ● An empty tiffin box, ● One straw ● 4-5 rubber bands. ● Arrange all the rubber bands	**Activity:** Engage students in a hands-on ***Activity 5.6*** **Procedure:** ● Take two rubber strips of the same size. ● Place these two pieces one above the other and stretch them tight. ● Now blow air through the gap between them

end of the trachea (windpipe) in the throat, and is responsible for the production of sound. - As we know, sound can only be produced due to vibration, the larynx is provided with two vocal cords. These vocal cords are separated by a narrow slit for the passage of air. - When the lungs force air through the slit, the vocal cords vibrate, producing sound. - Muscles attached to the vocal cords can make the cords tight or loose. ● Show the following video to students for proper understanding of sound produced by vocal cords in humans. Link 3	on the tiffin box as shown in the given figure. ● Image 7 ● Ask a student to blow the air through the straw on the rubber bands. ● What did you observe? **Observations:** ● Blowing air through the straw on the rubber bands causes them to vibrate. ● The vibration of the rubber band creates sound. ● This is how the vocal cords in our body produce sound. ● Students can watch the following video for activity Link 4	● As the air blows through the stretched rubber strips, a sound is produced. ● You can also take a piece of paper with a narrow slit and hold it between your fingers. ● Now blow through the slit and listen to the sound. ● Our vocal cords produce sound in a similar manner.

Summary	Factual Question	CFU	Home Work
In humans, the sound is produced by the voice box or the larynx. It is found at the upper end of the windpipe	1. What is the sound producing organ of the body? 2. What is larynx? 3. Where is the larynx located in the human body?	- How does larynx produce sound? - What is the effect of a thick vocal cord on the voice of a person?	1. Sketch the larynx and explain its function in your own words. 2. What is the larynx in the human body? 3. How is sound produced by vocal cords in humans?

Period 5.5	TOPIC: Activity 5.6	Activity

Learning Outcome:
Students will be able to conclude that in humans sound is produced by vibrating the vocal cords

Explicit Teaching/Teacher Modelling (I Do)	Group Work (We Do)	Independent Work (You Do)
Students should note that a) The larynx is the voice box in the human body. b) Position of vocal cords c) Vibration of vocal cords d) How do vocal cords produce sound	**Activity:** Sound is produced by vibrating bodies **Material:** Two rubber stripes, paper **Instructions:** 1. Take two rubber stripes of the same size 2. Place them one above the other and stretch them tightly	Students can conduct the activity individually and the functioning of vocal cords

	3. Now blow out through the gap 4. Sound is produced 5. Take a paper with narrow slit 6. Hold it between your fingers 7. Now blow the air through slit 8. Sound is produced We conclude that our vocal cords produce sound in similar manner.	

Summary	Factual Question	CFU	Home Work
• There is a slit between two vocal cords across the voice box or larynx. The slit allows the air to pass between them. The lungs force air through the slit and cause the vocal cords to vibrate and produce sound. • Vocal cords may be tight and thin. The type or quality of voice depends on the conditions of vocal cords whether they are tight and thin or loose and thick.	What role do vocal cords play in sound production? Where are the vocal cords located?	1) Why is sound produced by stretching rubber strips? 2) How is sound produced by vocal cords?	1. How sound is produced?

Period 5.6	TOPIC: Sound needs a medium for propagation (Activity 5.7)	Activity

Learning Outcome:
Students will be able to

- do the experiment that sound needs a medium for its propagation

Explicit Teaching/Teacher Modelling (I Do)	Group Work (We Do)	Independent Work (You Do)
Recap of the previous class 1) Give a brief introduction about movement of sound requires a medium. 2) The movement of sound is different in various mediums such as solid (wood), liquid (water) and gas (air). 3) Explain to the students that the speed of sound in all the mediums is not the same. It differs in different mediums.	Perform the Activity ***5.7*** **Procedure:** • Take a metal or glass tumbler • Make sure that it is dry. Place a cell phone in it. (Remember that the cell phone must not be kept in water.) • Ask any student to give a ring on this cell phone from another cell phone. • Listen to the ring carefully. • Now, surround the rim of the tumbler with your hands. Put your mouth on the opening between your hands. • Indicate any student to give a ring again. Listen to the ring	**Activity:** You can perform the activity by placing your ear at one end of a long wooden or metallic table and asking your friend to gently scratch the other end of the table Note your response in the notebook. - Can sound travel through wood or metal. - In which object the sound was very clear? Wood or metal? Perform the same activity with other solids and note your observation in the

	while sucking air from the tumbler. Ask the following questions from students; 1) Does the sound become fainter as you suck air? 2) Remove the tumbler from your mouth. Does the sound become loud again?	notebook.

Summary	Factual Question	CFU	Home Work
Sound needs a medium for propagation . It can propagate in solid, liquid, and gas.	• In which medium is the movement of sound faster? • Does sound travel faster in liquids than in gases? • Does sound travel slower in liquids than in gases	• In the context of sound, what is a medium? • What could be the reason if you cannot hear the sound of a musical instrument on the moon? • Is there any other liquid in which sound can be heard?	1) Choose the correct answer. Sound can travel through *(a)* gases only *(b)* solids only *(c)* liquids only *(d)* Solids, liquids and gases. 2) Can sound be heard in vacuum? 3) How do humans communicate on the moon?

Period 5.7	TOPIC: Activity 5.8, Activity 5.9	Activity

Learning Outcome:
Students will be able to understand that sound can also travel through liquids and solids

Explicit Teaching/Teacher Modelling (I Do)	Group Work (We Do)	Independent Work (You Do)
• Introduce the topic of sound and how it travels in different mediums. Discuss the following questions with students? • Can you hear the sound of the bell? • Does it indicate that sound can travel through liquids?		

Summary	Factual Question	CFU	Home Work
Sound is produced by vibrating objects. These vibrations disturb the particles of the surrounding medium (like air), which carry the sound.	1. Is there any other liquid in which sound can be heard? 2. Is there any other liquid in which sound can be heard?	1. In the context of sound, what is a medium? 2. What are the different mediums in which sound can travel? 3. Why is a medium	• How do whales and dolphins communicate under water? • Perform the same activity at home with other solids and note your

	3. In which medium is the movement of sound faster? 4. Does sound travel faster in liquids than in gases? 5. Does sound travel slower in liquids than in gases?	important for movement of sound?	observation in the notebook. **Objects** / **Sound heard or not?** Steel utensil Wooden Amirah

Period 5.8	**TOPIC: We hear sounds through our ears**	**Activity** 5.10

Learning Outcome:
Students will be able to understand the need of ears to hear the sound

Explicit Teaching/Teacher Modelling (I Do)	**Group Work (We Do)**	**Independent Work (You Do)**
● Recap of the previous class ● Introduce the structure of the human ear by showing and/or drawing its figure. ● Explain the structure of human ears. **Closing** - What did we discuss today? - Recollect the name of the internal structure of the human ear, and write in your notebook	**Activity:** Engage students in a hands-on ***Activity 5.10*** **Procedure:** ● Take a plastic or tin-can. Cut its ends. ● Stretch a piece of rubber balloon across one end of the can and fasten it with a rubber band. ● Put four or five grains of dry cereal on the stretched rubber. ● Now ask your friend to speak "Hurray, Hurray" from the open end. ● Observe what happens to the grain. ● Why do the grains jump up and down?	**Activity:** Engage students in a hands-on activity of formation of a sound collecting instrument like ear to hear sound more clearly. **Materials required:** ● A4 size Paper ● Make a funnel from the given paper and hold it near your ear as shown in figure. **Observation:** ● Students will find that when they hold the paper funnel near their ear, they hear the sound very clearly. ● This experiment suggests that our ears are shaped and structured in such a way that they can collect more sound waves.

Summary	**Factual Question**	**CFU**	**Home Work**
The human ear helps us hear sounds. It collects sound waves and converts them into signals that the brain can	1. What is the name of the passage in the outer ear which carries sound waves to the eardrum? 2. Name the nerve	1. Name the organ of hearing in human beings? 2. The inner part of the ear has a coiled tube called	1. The sound waves are collected by the pinna of the …… part of the ear. 2. When the sound waves fall on the ……..it starts vibrating back and

understand. The eardrum vibrates with sound. Three bones in the middle ear help pass on the vibrations. The cochlea changes vibrations into nerve signals. The brain makes sense of these signals as sound.	which carries electrical impulses from the cochlea of the ear to the brain?	…… 3. What is the function of eardrum in the human ear?	forth rapidly. 3. Sound vibrations make the ……. vibrate 4. The eardrum sends vibrations to the ……. ear. **5.** Make a poster presentation of Human ear. (Use Chart Paper) and show in your classroom.

Period :5.9	**TOPIC: Amplitude, time period and frequency**	**Activity**

Learning Outcome:
Students will be able to define amplitude, time period, and frequency

Explicit Teaching/Teacher Modelling (I Do)	**Group Work (We Do)**	**Independent Work (You Do)**
● Do a brief recap of the last class. ● Discuss the various properties of sound. ● For example: Amplitude, Time Period, Frequency, Loudness, Pitch ● Let's watch a video to understand these terms.	**Objective:** To engage students in an interactive **Activity 5.11**that will help to understand the relationship between vibrations and amplitude of sound. **Materials Needed:** ● Metallic tumbler ● Tablespoon. **Instructions:** 1. Divide the class into small groups of 3-4 students. 2. Provide each group with materials and instruct them to follow the steps. 3. Each student records the observations. 4. Encourage students to share their observations and insights about the experiment.	***Activity:*** To show the relationship between loudness and the amplitude of the sound **Materials required:** ● A drum ● A stick **Method:** ● Struck the drum with the stick slowly. ● Observe the sound produced. ● Now again hit the drum with the stick by applying more force. ● Again, observe the sound produced by the drum. **Observation:** ● When you hit the drum with less force, it produces less sound. This is due to less vibration of the membrane of a drum that vibrates to produce sound. ● The harder you strike, the louder is the sound. ● This is because of the increase in amplitude and when amplitude increases loudness increases as it sets more

		air into vibration for which the sound becomes loud.

Summary	Factual Question	CFU	Home Work
• The number of oscillations or vibrations per second is called the frequency of oscillation. • The frequency is expressed in hertz (Hz) • Larger the amplitude of vibration, louder is the sound. • Higher the frequency of vibration, the higher is the pitch, and shriller is the sound.	- If an object oscillates 20 times in one second, what would be its frequency? - What is the loudness of a sound?	- What are examples of loud sounds? - How do we identify a low sound? - What is pitch? - How are frequency and oscillations associated with each other? Explain	• Define I. Amplitude II. Time Period III. Frequency IV. Loudness V. Pitch • A pendulum oscillates 40 times in 4 seconds. Find its time period and frequency. • The sound from a mosquito is produced when it vibrates its wings at an average rate of 500 vibrations per second. What is the time period of the vibration?

Period 5.10	TOPIC: Loudness and Pitch (activity 5.11)	Activity

Learning Outcome:
Students will be able to explain the factors in which loudness and pitch depends

Explicit Teaching/Teacher Modelling (I Do)	Group Work (We Do)	Independent Work (You Do)
• Do a brief recap of the last class. • Discuss the following properties of sound 1. Loudness 2. Pitch	**Objective:** • to engage students in an interactive **Activity as shown in Fig. 5.18** **Materials Needed:** • A small drum / a whistle **Instructions:** • Begin the activity by gathering the students around a table or designated area. • Explain to the students that they will be observing a demonstration and actively participate in the process. • Struck the drum and observe the vibrations of the flexible skin of the drum. • Encourage students to ask questions and actively engage in	***Activity:*** Engage students in an interactive **activity** to show that the frequency or pitch of a wire depends on its length. **Materials Required:** • A cardboard box • Rubber band • 2 pencils/ sketch pens **Activity:** • Take a box and stretch a rubber band around it as shown in the given image • Put two pencils beneath the band and listen to the sound produced on plucking the band. • Keep increasing the distance between the pens and listen to the sound produced each time. **Observation:** • You will notice that at the shortest

	the demonstration. ● Prompt the students with questions related to the activity such as: 1. What did you observe during the drumming? 2. Is the drum a loud sound? 3. How loud is playing drums? 4. Do drums have a low or high sound? 5. Is the sound of a drum a force? Summarize the key points discussed during the activity and emphasize the significance of striking drum with small or great force.	length the sound is sharp and as the length increases the pitch becomes lower. ● In musical instruments, we either have strings of different lengths (piano, harp) or use our hands to change the length of the strings (guitar, violin) and thus produce sounds of different pitches.

Summary	Factual Question	CFU	Home Work
● The loudness of sound depends on its amplitude. The loudness is expressed in a unit called decibel (dB). ● The frequency determines the shrillness or pitch of a sound.	I. The unit of frequency is ……. II. The loudness is expressed in a unit called …… III. What is the loudness of a sound?	1. What are examples of loud sounds? 2. How do we identify a low sound? 3. What is pitch?	1. Define loudness 2. How does a low-pitched sound produce?

Period 5.11	TOPIC: Audible & Non-audible Sounds, Noise & Music, Noise pollution	Activity

Learning Outcome:
Students will be able to differentiate audible and inaudible sounds

Explicit Teaching/Teacher Modelling (I Do)	Group Work (We Do)	Independent Work (You Do)
● Do a brief recap of the last class. ● Give an overview of the chapter, topics that will be covered throughout the chapter. ● Begin the lesson by engaging students in discussion about the different types of sound produced by various animals such as humming bees, whales, octopuses, squid, cuttlefish, cod, Guinea fowl, etc. ● The sound produced by these animals are not audible. These are called inaudible sounds. ● Sounds of frequencies less than	● Divide the class in groups of three where students can discuss and write about different types and range of sound in their notebook. ● Make a list of sounds that we cannot hear. For example, **Infrasonic sound:** ● The frequency range of these types of waves is below 20Hz. ● Humans cannot detect it. ● **Example:** - Sound produced by earthquake, - Volcanic eruption and ocean waves,	**Activity:** Making a collage of images of different natural or manmade things that produce inaudible sounds. **Materials required** ● Images of different objects / animals / things that produce inaudible sound. ● A fevicol/ glue ● Chart paper **Method:** ● Make collages of

about 20 vibrations per second (20 Hz) cannot be detected by the human ear. Such sounds are called inaudible. ● Sounds of frequencies higher than about 20,000 vibrations per second (20 kHz) are also not audible to the human ear. ● Therefore, for the human ear, the range of audible frequencies for humans is nearly from 20 to 20,000 Hz. ● Encourage the students to ask questions about various other types of sounds produced which are inaudible to humans. ● ***Activity: The teacher can show the following videos to students:*** ***Link 6*** ***Link 7***	- Weather, - Avalanche, - Waterfalls, - Meteors, - Lightening, etc. **Ultrasonic sound:** ● Ultrasonic wave is defined as "inaudible sound with high frequency for human" ● The frequency of which generally exceeds 20 kHz (20000 Hz). ● Most likely humans cannot hear the ultrasonic sound but many animals such as dogs, bats, monkeys, etc. can hear ultrasound. Divide the class in groups of three or four where students can discuss different types of noise pollution in their surroundings. ● ***Activity:*** Play different types of sounds through speakers such as Mridangam, Tabla, Veena dogs barking, traffic, cats purring, alarm clocks, a group of children shouting and playing together, loud music, DJs, rocket launching, and ask students to identify each sound and categorise them as soft and loud sound. ● Ask students which sounds they think would cause hearing problems and why? ● Write about different sources of noise pollution in your notebook. ● What can be done to minimise noise pollution? ● What are the different effects of noise pollutants on humans?	different images that you have collected and brought with you in school. ● paste it in the chart paper under two sections: infrasonic and ultrasonic sounds ● if possible, search frequencies of sounds produced by these things/ animals and write below their images. ● Show the chart paper to your teacher.

Summary	Factual Question	CFU	Home Work
Sounds of frequencies below (20 Hz) and higher than 20 kHz cannot be detected by the human ear. Such sounds are called inaudible. For the human ear, the	Which of the following sound waves can we hear? 10 Hz, 500 Hz, 1500 Hz, 12000 Hz, 25000 Hz.	1. How can you say whether a sound is audible or not? 2. Do we hear all types of sounds in the environment? 3. Can you define the term loudness?	Fill in the blanks with suitable words. 1.Time taken by an object to complete one oscillation is called_____. 2.Loudness is determined by the

range of audible frequencies is roughly from 20 to 20,000 Hz. The unpleasant sounds are called noise and the presence of excessive or unwanted sounds in the environment is called noise pollution. Noise pollution in the surroundings may cause many health-related problems. A person who is exposed to a loud sound continuously may get temporary or even permanent impairment of hearing.		4. What do you understand by the term pitch? 5. How does pitch define the voice of male and female	_____ of vibration. 3.The unit of frequency is _____ 4.Unwanted sound is called _____. 5.Shrillness of a sound is determined by the _____ of vibration. **5.Define** a) Audible sound b) Inaudible sound **6. Give examples:** (a) The sound which cannot be heard. (b) The sound range that we can hear

Teacher's Diary

Name of the Teacher:		Name of the Month:	
Name of the Lesson:		Class:	

Period No	**Name of the Concept to be taught**	**Date**	**Activities Conducted during the teaching**	**TLM Used**	Teacher Signature
5.1	Introduction and 5.1 Sound is produced by vibrating bodies activity 5.1,				
5.2	Activity 5.2 and Activity 5.3				
5.3	Activity 5.4, Activity 5.5,				
5.4	Sound produced by humans				
5.5	Activity 5.6				

5.6	5.3 Sound needs a medium for propagation, Activity 5.7				
5.7	Activity 5.8 Activity 5.9				
5.8	5.4 We hear sounds through our ears Activity 5.10				
5.9	5.5 Amplitude, time period and frequency of vibration				
5.10	Loudness and pitch Activity 5.11				
5.11	5.6 Audible and inaudible sounds. 5.7 Noise and music 5. 8 Noise pollution				

1.	What were some of the specific strategies that I used to encourage participation? How effective were they? What will I do differently next time?
2.	Were there any concepts or activities that students found particularly difficult? How will I adapt my approach to address these difficulties in the next lesson?
3.	What additional resources or modifications could improve the effectiveness of this lesson in future implementations?
4.	How well did I adjust my teaching based on student reactions or unforeseen challenges?

Teacher's Signature Head Teacher's Signature

Head Teacher's Suggestions:

Teacher Notes:

December 2025

Sunday	Monday	Tuesday	Wednesday	Thursday	Friday	Saturday
	1 World AIDS 6.1	2 6.2	3 6.3	4 6.4	5 NI	6 No Bag Day Mega PTM/SMC
7 Sunday	8 6.5	9 6.6	10 Human 6.7	11 6.8	12 NI	13 Second Saturday
14 Sunday	15 6.9	16 6.10	17 6.11	18 NI	19 NI	20 No Bag Day Cluster Meeting
21 Sunday	22 Mathematics Day	23	24 National	25 Christmas	26	27 No Bag
28 Sunday	29 NI	30 NI	31 NI			

Christmas Holiday

E8W4U1 H6H1Q4 K5M8W3

No bag Day | Cluster Complex | Teacher Resources

Christmas Holidays for Christian Minority Institutions
21.12.25 to 28.12.25

TEACHER'S NOTES

Week1: **6.1 – 6.4**

Week 2: **6.5 – 6.8**

Week 3: **6.9 – 6.11**

Week 4:

Week 5:

Chapter 6: Combustion and Flame

Learning Outcomes:

1. Classify materials as combustible and non-combustible substances based on chemical properties.
2. Prove experimentally that air is essential for burning.
3. Draw a labelled diagram showing different zones of candle flame.
4. Make efforts to protect the environment from evil effects formed by the burning of fuels.
5. Explain about different types of combustion.

Prior Concepts/Skills:

1. Basic understanding of combustion and flame.
2. Understanding the combustible and non-combustible substances based on chemical properties.
3. Understand the process of BURNING of fuels.

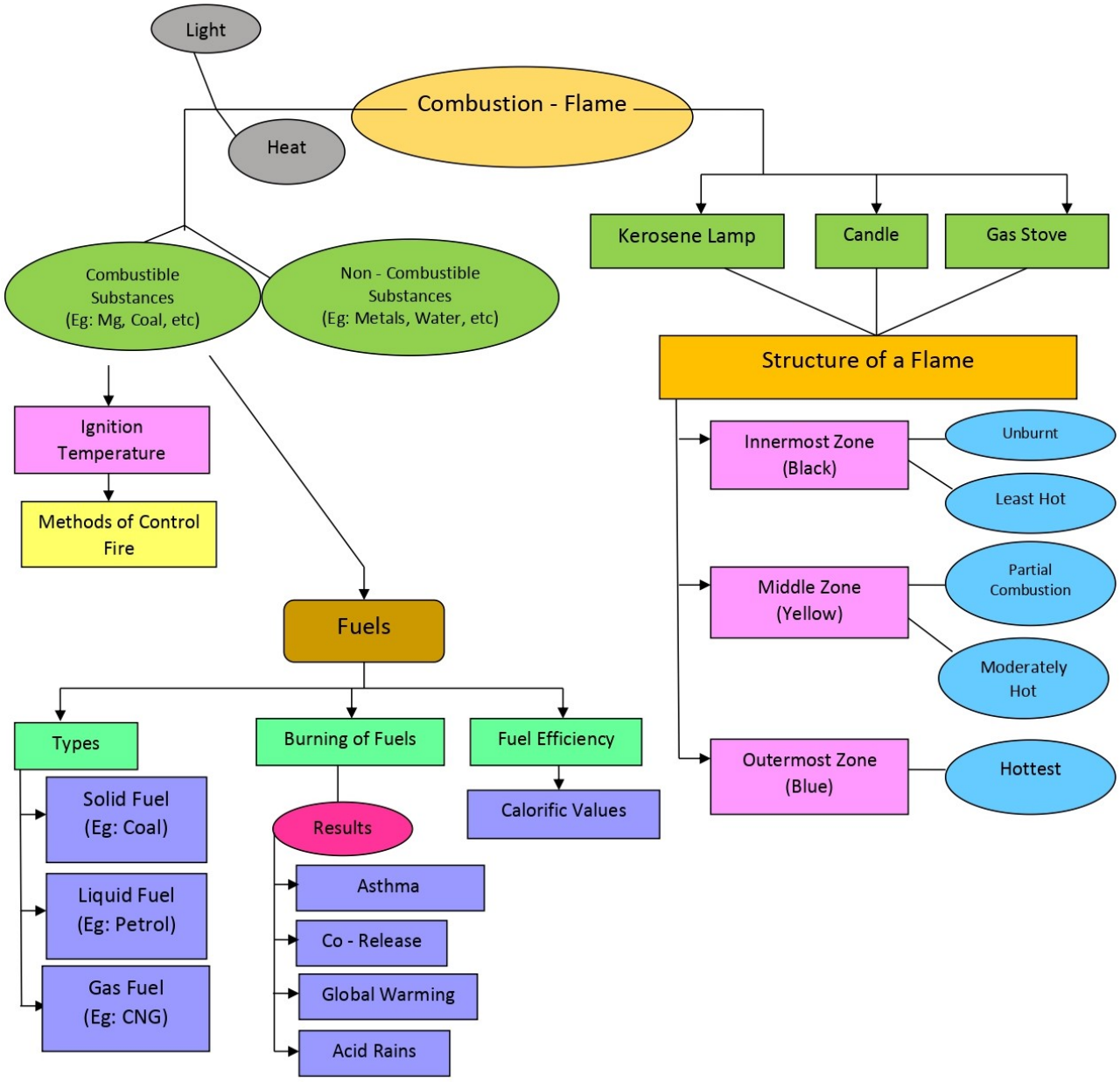
Teaching Learning Materials

Magnesium ribbon, candle, glass,
paper funnel, fire extinguisher,
chimney, metal string, metal
plate, tongs, forceps,
Bunsen burner,
Images explaining combustion and flame.

Teacher Resources

https://images.app.goo.gl/71zhBNfyd7mZRQ2M7
https://images.app.goo.gl/xzW1CZFjEPnzv3tYA
https://images.app.goo.gl/snS9CMbEpF5fYaE99
https://images.app.goo.gl/TVe6i1jgzCsDZgPS8

Concept Map



Synopsis

- **Combustion:** The process in which a substance undergoes a chemical reaction in the presence of air (oxygen) to produce heat and light is called combustion.
- **Fuel:** A substance that is burned to provide energy, heat or power.
- **Combustible substances:** The substances which can burn are called combustible substances. For example: cloth, straw, cooking gas, kerosene oil, coal, charcoal, wood, leaves, paper, wax, hydrogen gas, ethanol, methane, propane, propene.
- **Non-combustible:** A non-combustible substance is that which is not capable of igniting and burning. Examples of non - combustible materials include Portland cement concrete.
- **Conditions under which combustion takes place**: (a) Presence of combustible substance. (b) Presence of a supporter of combustion i.e., oxygen. (c) Attainment of ignition temperature.
- **Ignition temperature:** The ignition temperature of a substance is the minimum temperature that is required to start or cause combustion.
- **Inflammable substances:** Inflammable substances are those substances which catch fire very easily and their ignition temperature is very low.

- Essential requirements for producing fire are fuel, air (to supply oxygen), and heat (to raise the temperature of the fuel beyond the ignition temperature).
- **Rapid combustion:** The gas burns rapidly and produces heat and light. Such combustion is known as rapid combustion
- **Spontaneous combustion:** The type of combustion in which a material suddenly bursts into flames, without the application of any apparent cause is called spontaneous combustion.
- **Explosion:** When a cracker is ignited, a sudden reaction takes place with the evolution of heat, light, and sound. A large amount of gas formed in the reaction is liberated. Such a reaction is called an explosion.
- **Flame:** The substances which vapourise during burning give flames.
- The sources of heat energy for domestic and industrial purposes are mainly wood, charcoal, petrol, kerosene etc. These substances are called **fuels.**
- The amount of heat energy produced on complete combustion of 1 kg of a fuel is called its calorific value.
- The calorific value of a fuel is expressed in a unit called kilojoule per kg (kJ/kg).
- Increased concentration of carbon dioxide in the air is believed to cause global warming.
- Oxides of Sulphur and nitrogen dissolve in rainwater and form acids. Such rain is called acid rain.
- **Deforestation:** Cutting down of trees leads to deforestation which is quite harmful to the environment.
- The use of diesel and petrol as fuels in automobiles is being replaced by CNG (Compressed Natural Gas) because CNG produces harmful products in very small amounts as compared to diesel and petrol.

Period wise Topics

Period No.	Concepts/Topic	Learning Outcomes	Any Specific Information
6.1	Introduction 6.1 What is combustion?	understand the process of combustion	
6.2	Activity 6.1	differentiate combustible and non-combustible substances	
6.3	Activity 6.2	concludes that oxygen is required for combustion	
6.4	Activity 6.3 Ignition temperature	understand the concept of ignition temperature	
6.5	Activity 6.4	do experiments to test the ignition temperature of paper	
6.6	6.2 how do we control fire	suggest some methods to control fire	
6.7	6.3 Types of combustion	understand the difference between different types of combustion	
6.8	6. 4 Flame	explain about flames formed by burning different objects	
6.9	6.5 structure of flame Activity 6.5	draw the structure of flame and label its parts	
6.10	6.6 What is a fuel 6.7 Fuel efficiency	explain about fuel efficiency of different fuels	
6.11	Burning fuels leads to harmful products	suggest remedies from harmful effects by burning fuels	

Period 6.1	**TOPIC: Introduction, what is Combustion?**	Activity

Learning Outcome:
Students will be able to understand the process of combustion

Hook Activity: "Guess the Flame — Fuel Mystery!"
Objective:
To ignite curiosity about how different fuels burn, the type of flame they produce, and why fuels differ.

Setup:
Arrange 3-4 safe, small burning samples (done by the teacher or in a controlled video demonstration):

- A candle (wax)
- Kerosene lamp
- Alcohol burner
- Small wood chip or charcoal piece (if safe, or shown via image/video)

(Always ensure safety — real flames should be handled by the teacher or done via video in class!)

Steps:

1. Blind Observation Game:
   Cover the setup or use images or videos of flames. Ask students:
   - Can you guess which fuel produces this flame?
   - Why does one flame look yellow and another blue?
   - Why do some fuels give more smoke than others?
2. Discussion Points:

- Why do fuels burn differently?
- Which fuels give more heat and less smoke?
- How does this connect to their calorific value and their environmental impact?

✅Key Concept Link:
This activity makes students curious about:

- Types of combustion: rapid, spontaneous, explosion.
- Flame colours.
- Fuels' roles in daily life and industries

Explicit Teaching/Teacher Modelling (I Do)	**Group Work (We Do)**	**Independent Work (You Do)**
● Begin the lesson by engaging students in a discussion about fuels and fire. ● Ask the following questions to the students. For example: ● How do we get fire? ● Can you name a few fuels used in our homes? ● Name a few fuels used in trade and industry. ● Name various things used for burning, fuels in daily life, etc. ● Have you ever wondered how a candle flame or a bonfire is created? ● Can you name any materials or substances that can burn? ● What do you think is needed for	***Activity 1*** ● Teacher and students together do an activity where the teacher will hold a piece of charcoal by using a pair of tongs and bring it near the candle or Bunsen burner. ● They observe how the charcoal is burning. ● Teacher explicitly explains the process of combustion through this activity. ***Activity 2*** ● The teacher divides the students into 3-4 groups and	Students write in their notes. 1. Where do they observe the process of combustion in their daily lives? 2. What are fuels? 3. What are the different fuels used in combustion?

something to catch fire or burn?

- Have you heard of the term "combustion" before? If so, what do you think it means?

- After the activity, the teacher explains the concept of combustion and fuels. (referring to the textbook)

A chemical process in which a substance reacts with oxygen to give off heat is called combustion. The substance that undergoes combustion is said to be combustible. It is also called a fuel. The fuel may be solid, liquid or gas. Sometimes, light is also given off during combustion, either as a flame or as a glow.

provides different materials like paper, wood, cow dung, candle to burn and discuss the different changes that they observed in different items.

Ask the following questions to the students:

- What did you observe while burning the different things?
- Did they burn in the same way?
- Did you observe any changes?

Summary	**Factual Question**	**CFU**	**Home Work**
Students can understand the process of combustion.	Is food being a fuel for our body?	• What is combustion? • State some examples where we can observe combustion.	The teacher asks the students to write the answers of the following questions in their notebook- Write names of different types of fuels 1. Solid 2. Liquid 3. Gas

Period 6.2	**TOPIC: Activity 6.1**	Activity

Learning Outcome:
Students will be able to differentiate between combustible & non-combustible substances

Explicit Teaching/Teacher Modelling (I Do)	**Group Work (We Do)**	**Independent Work (You Do)**
• Recap of the topic/concepts discussed in the last class. • Ask the following questions from students: • What did we learn yesterday? • Where can you apply the learnings in your daily life? • Teacher shares the objective of the class and both students and teacher perform the activity given in group work • From the activity, the teacher	**Activity 10.1** • Divide the class in groups of three and perform the Activity **6.1** from the textbook given on **page number 90.** (Teacher to keep the students under close observation when using fire) • Teacher asks students to note their observations from the activity. **Observe and discuss the materials' reactions:** whether	***Activity:*** Based on your observations from activity 6.1 complete the following Table 6.1

Material	Combustible	Non-combustible
Wood		
Paper		
Iron		
Nails		
Kerosene oil		
Stone piece		
Straw		

explains about the combustible and non-combustible items. (by referring to the textbook).	they burn, how they burn, the colour of the flames, and if they leave any residue after burning	Charcoal Match sticks Glass

Summary	Factual Question	CFU	Home Work
Students can differentiate combustible and non-combustible substances.	1. Write some examples of inflammable substances from your daily lives.	2. Does a matchstick burn by itself? 3. How does a matchstick burn? 4. Why do you have to use paper or kerosene oil to start a fire in wood or coal?	1. Give two examples of the combustible substances. 2. Classify the following as combustible and non-combustible substances- Paper, iron nails, cloth, glass.

Period 6.3	**TOPIC: Activity 6.2**	Activity

Learning Outcome:
Students will be able to conclude that oxygen is required for combustion

Explicit Teaching/Teacher Modelling (I Do)	**Group Work (We Do)**	**Independent Work (You Do)**
• Teacher recalls the concepts discussed in previous class by asking the following questions - • What did we learn yesterday? • Where can you apply the learnings in your daily life? • The teacher introduces the various conditions under which combustion takes place. • Explain the conditions under which combustion takes place like- air • Ignition temperature: The lowest temperature at which a substance catches fire Ignition temperature of common fuels: • Kerosene - 290°C • Gasoline - 400°C • Alcohol - 365°C • Coal - 580°C	***Activity*** • Divide the class in groups of three/ four and perform the *Activity **6.2** from* the textbook (*Teacher will be with the students by continuously observing them and giving instructions to use the fire and participating in the activity)* • Observe and discuss the conditions which are required for the combustion to take place. • Teacher will ask the following questions: • What happens in the three cases? • Does the flame flicker off? • Does it flicker and give smoke? • Does it burn unaffected? • Can you infer anything at all about the role played by air in the process of burning? The teacher explains the three scenarios - 1. The candle burns freely in case (a) when air can enter the chimney from below.	***Activity:*** The teacher asks students to think and answer. Imagine if a **person's clothes catch fire.** • What will people do first? • Why do people cover him with a blanket? What happens in this situation? • What will happen if the person is wearing wet clothes? • Students should write their answers in their notebook and show it to the teacher.

		2. In case (b), when air does not enter the chimney from below, the flame flickers and produces smoke. 3. In case (c), the flame finally goes off because the air is not available.	
Summary	**Factual Question**	**CFU**	**Home Work**
Students will understand that OXYGEN gas is essential for combustion .	We have read that THE SUN produces its own HEAT and LIGHT, how? Is it a type of COMBUSTION?	1. What are fuels?	- List conditions under which combustion can take place. - Explain the role of oxygen in the process of combustion. - What is the ignition temperature? - Why is ignition temperature necessary for combustion? - What is necessary for combustion?

Period 6.4	**TOPIC: Ignition temperature**	Activity
Learning Outcome: Students will be able to understand the concept of ignition temperature		
Explicit Teaching/Teacher Modelling (I Do)	**Group Work (We Do)**	**Independent Work (You Do)**
The teacher begins the class by asking some open-ended questions like ● Does a matchstick burn by itself? How does it burn? ● You have had an experience of burning a piece of paper. Does it burn when a burning matchstick is brought near it? ● Can you burn a piece of wood by bringing a lighted matchstick near it? ● Why do you have to use paper or kerosene oil to start a fire in wood or coal? ● Have you heard of forest fires? (Students share their thoughts by raising their hands. The teacher appreciates their responses). ● After the discussion, the teacher explains why the above things won't happen on their own. ● The teacher explains about the ignition temperature by referring to the textbook. The teacher explains the history of	Divide the students in groups of two and give this resource (section : Fiery Fire, pages 9 to 14) as a passage for students to read in pairs. After reading the article students are asked to discuss in groups the following questions - ● In India, religious fires in the form of 'havens' are created. How do you think that started? ● In a forest fire what are the things that act as fuels? ● What is the role of heat to maintain the fire? ● What is the reason why fires in the open are more difficult to control?	Give this resource (section: Wildfire, pages 15 to 24) as a passage for students to read individually. The teacher asks students to think and answer. ● What is good fire and bad fire? ● What are some of the countries where wildfires are very common? ● What are the different types of wildfires? ● What are the natural catalysts/causes of wildfires? ● What are some of the devastating effects of wildfires?

matchsticks by showing the video (Link 2_History of Matchstick) to students for better understanding.			
Summary	**Factual Question**	**CFU**	**Home Work**
Students will understand that no body immediately catches fire. Every substance requires minimum initial temperature- IGNITION TEMPARATURE.	If we put on our hand on candle fire, is it immediately burns?	We have read that the sun produces its own heat and light. Is it an example of combustion?	• Do these experiences tell you that different substances catch fire at different temperatures? • Can you tell now why a matchstick does not catch fire on its own at room temperature? • Why does the matchstick start burning on rubbing it on the side of the matchbox?

Period 6.5	**TOPIC: Activity 6.4**	Activity

Learning Outcome:
Students will be able to

- experiments to test the ignition temperature

Explicit Teaching/Teacher Modelling (I Do)	**Group Work (We Do)**	**Independent Work (You Do)**
• The teacher starts the lesson by recalling the concepts from the previous class and explains about the inflammable items from their daily lives. • The teacher conducts the activity in the group work section (Activity 10.4 from the Textbook Page Number 94) and then proceeds with the following questions. • What did you observe? • What happened to the empty paper cup? • What happened to the paper cup with water? • Did the water in this cup become hot? • The teacher then explains the reason that a combustible substance cannot catch fire or burn as long as its temperature is lower than its ignition temperature. • The heat supplied to the paper cup is transferred to water by conduction. So, in the presence of water, the ignition temperature of paper is not reached. Hence, it does not burn. The substances which have very low	***Activity:*** Perform the ***activity 5.4*** from the ***textbook page*** • (**Caution : Be** careful while handling burning candles). • The teacher will have a positive attitude towards students' challenges and be there with children to support them in every step.	Students are asked to write down the answers to the following questions in their notebooks - 1. In the previous experiment, if we continue heating the cup, we can even boil water in the paper cup. Explain why? 2. Define inflammable substances. 3. Give names of some inflammable substances

ignition temperature and can easily catch fire with a flame are called inflammable substances. Examples of inflammable substances are petrol, alcohol, Liquified Petroleum Gas (LPG) etc.		

Summary	Factual Question	CFU	Home Work
Students will understand that different materials have different IGNITION TEMPARATURES.	**How do BABAS swallow fired balls of camphor?**	**1.Prepare a list of materials having different ignition temperatures.** **2draw a graphical representation of different materials ignition temperatures.**	1.Survey 20 houses in your area. Find the percentage of households using LPG, kerosene, wood and cattle dung as fuel. 2.Observe the cooking process at your home and write the answers in your notebook. 3.Have you ever seen cooking oil catching fire when a frying pan is kept for long on a burning stove? 4.Does it mean that the ignition temperature of kerosene oil is lower? Does it mean that we need to take special care in storing kerosene oil?

Period 6.6	**TOPIC: How do we control Fire**	Activity

Learning Outcome:
Students will be able to

- suggest some methods to control fire

Explicit Teaching/Teacher Modelling (I Do)	**Group Work (We Do)**	**Independent Work (You Do)**
• The teacher starts the lesson by recalling the previous class and re-explains about the inflammable items from their daily lives. • The teacher asks some open-ended questions- ◦ What will you do, if you see some home or shop on fire? ◦ To whom will you inform? • Teacher explicitly explains about the fire extinguishers and the process involved in it.	• Students and the teacher watch the video in the classroom using ICT Link 3_Fire Extinguisher • Ask the following questions about fire and fire extinguishers after the video- • What are the essential requirements for producing Fire? • What is the fire extinguisher? • Which gas is used in a fire extinguisher? • What is the basic principle of functioning of any fire extinguisher?	*Activity:* • Students will be divided into 3-4 groups and asked to prepare posters about combustion, ignition temperature, giving examples from their daily lives and safety measures taken while handling the fire.

• Teacher also explains about the various types of fire extinguishers based on the different types of inflammable material. Such as, fires involving • electrical equipment • wood and paper • petrol • Water cools the combustible material so that its temperature is brought below its ignition temperature. This prevents the fire from spreading. Water vapours also surround the combustible material, helping in cutting off the supply of air. So, the fire is extinguished.	Or, Find out the number, type and location of fire extinguishers available in your school, nearby shops and factories. Write a brief report about the preparedness of these establishments to fight fire.	

Summary	**Factual Question**	**CFU**	**Home Work**
Students can understand - How fire is extinguished.	1. What is the most common fire extinguisher? 2. What is used in fire extinguishers?	1.Why do we need fire extinguishers? 2.What is the basic principle of any fire extinguisher? 3. How can fire be controlled in case of fire produced by oil or gas? 4. Why is it dangerous to use water to extinguish an oil or electrical fire?	• Explain how CO_2 is able to control fires. • Find out the telephone number of the fire service in your area. If a fire breaks out in your house or in your neighbourhood, the first thing to do is to call the fire service. • What are the different types of fire extinguishers?

Period 6.7	**TOPIC: Types of Combustion**	Activity
Learning Outcome: Students will be able to understand the difference between different types of combustion		
Explicit Teaching/Teacher Modelling (I Do)	**Group Work (We Do)**	**Independent Work (You Do)**
• Teacher starts the lesson by recalling the previous class and builds up on that. • The teacher will mention the lesson objective- "types of combustion". • The teacher explains the types of combustion: • Rapid combustion, • Spontaneous combustion • Explosion	• Teacher shows a video on types of combustion. • The teacher and students will be watching the video using ICT. Link 4_Types of Combustion • The teacher will ask a few questions related to video and students will write their answers in the notebook: • In which type of combustion, gas burns	Students will solve the following worksheet in their notebooks. *Worksheet 3*

		very fast? • What did you not understand about the video? • Can anyone explain what was shown in the video?	
Summary	**Factual Question**	**CFU**	**Home Work**
Students will understand the -Types of Combustion.	1. What factors affect explosion? 2. How does pressure cause combustion? 3. Name different types of combustion.	*1.* Which substance burns in air at room temperature? 2. What will you observe when you ignite crackers? 3. Differentiate between spontaneous and rapid combustion	• Which substance burns in air at room temperature? • What will you observe when you ignite crackers? • Differentiate between spontaneous and rapid combustion.

Period 6.8	**TOPIC: Flame**	Activity

Learning Outcome:
Students will be able to explain about flames formed by burning different objects

Explicit Teaching/Teacher Modelling (I Do)	**Group Work (We Do)**	**Independent Work (You Do)**
- The teacher starts the lesson by recalling the previous class and builds up on that. - The teacher explains the types of substances which produce flame.	The teacher divides the students in group and does textbook activity (table 6.2) **Table 10.2 Materials forming Flame on Burning** S.No. | Material | Forms flame | Does not form flame 1. Candle 2. Magnesium 3. Camphor 4. Kerosene Stove 5. Charcoal • Which substance burns and produces flame? • Fill the table in your notebooks.	• Make a poster presentation of different objects producing different coloured flames.

Summary	**Factual Question**	**CFU**	**Home Work**
Students can understand How FLAME is formed by Burning.	Differentiate between the flame of a cooking gas and the flame of a candle?	- Define flame - What is the colour of a candle flame? - What is the colour of a flame of magnesium ribbon? - Why do different substances produce flames of different colours?	Why is the colour of candle flame and the flame of the kitchen stove different? **Activity:** With the help of your parents/ elders at home, burn a kerosene lamp, light a candle and a Bunsen burner. Observe the flame of the colour and note it down in your notebook. Discuss your observation with your teacher in the next class

Period 6.9	**TOPIC: Structure of Flame (Activity 6.5)**	Activity

Learning Outcome:

Students will be able to

- to draw the structure of flame & label it's parts

Explicit Teaching/Teacher Modelling (I Do)	Group Work (We Do)	Independent Work (You Do)
• Teacher demonstrates the activity and students will observe-Structure of flame- • Light a candle (Caution: Be careful). • Hold a 4-5 cm long thin glass tube with a pair of tongs and introduce its one end in the dark zone of a non-flickering candle flame. • Bring a lighted matchstick near the other end of the glass tube. • Do you see a flame caught at this end of the glass tube after a while? If so, what is it that produces a flame? • Notice that the wax near the heated wick melts quickly Teacher explains the three zones of flame and luminous and non-luminous zones of flame.	• The teacher and students discuss the procedure of the activity and the reason behind forming the flame on the other side of the glass tube. • Show the video to students for proper understanding of the structure of a flame. Link 6_Structre of a flame • Teacher will ask the following question from the students: • How many zones are there in the flame? • What are the different zones in the flame? • Is there anything that you find difficult to understand? • Can anyone explain the structure of a flame?	• Children will note their observations and write the reason in their notes. • Draw the diagram of different zones of a flame in your notebook.

Summary	Factual Question	CFU	Home Work
Students can DRAW the STRUCTURE OF FLAME and label its parts.	1. What are the different zones of a flame? 2. Which zone of the flame is the hottest part?	1. Why do goldsmiths use the outermost zone of the flame? 2. Can you think of situations in your daily life where flames play a crucial role? How does fire impact our lives?	1. Which zone of a flame does a goldsmith use for melting gold and silver and why? 2. Make a labelled diagram of a candle flame. 3. Abida and Ramesh were doing an experiment in which water was to be heated in a beaker. Abida kept the beaker near the wick in the yellow part of the candle flame. Ramesh kept the beaker in the outermost part of the flame. Whose water will get heated in a shorter time?

Period 6.10	**TOPIC: What is fuel, fuel efficiency**	Activity

Learning Outcome:
Students will be able to explain about fuel efficiency of different fuels

Explicit Teaching/Teacher Modelling (I Do)	Group Work (We Do)	Independent Work (You Do)
• Teacher starts the lesson by recalling the previous class and builds up on that. • The teacher will mention the	• Give the following passage to students to read in pairs. In India, households use more than one fuel	• Make a list of fuels familiar to you. Group them as solid,

lesson objective- “fuels and its different types.” • Sources of heat energy for domestic and industrial purposes are mainly wood, charcoal, petrol, kerosene etc. These substances are called fuels. • A good fuel is one which is readily available. It is cheap. It burns easily in air at a moderate rate. ◦ It produces a large amount of heat. ◦ It does not leave behind any undesirable substances. Teachers and students discuss the calorific values of different fuels. Suppose you were asked to boil a given quantity of water using cow dung, coal and LPG as fuel. Which fuel would you prefer? Give your reason. You may take the help of your parents. Do these three fuels produce the same amount of heat? The amount of heat energy produced on complete combustion of 1 kg of a fuel is called its calorific value. The calorific value of a fuel is expressed in unit kilojoule per kilogram (kJ/ kg).	for the fulfilment of energy requirements. So, they have to decide not only how much of a particular fuel to use, but also which fuel to choose. Energy is a necessity for a household. The demand for energy for cooking and lighting is increasing rapidly in India. Presently, India faces a dual challenge of providing clean fuel for the vast majority of households and also taking measures to reduce greenhouse gas emissions. If India adopts a carbon tax to reduce the carbon emissions so as to affect the price of fossil fuels, then this carbon tax may create considerable effect on fuel choice of households. (A carbon tax is a tax levied on the carbon emissions required to produce goods and services. Carbon taxes are intended to make visible the "hidden" social costs of carbon emissions, which are otherwise felt only in indirect ways like more severe weather events. Under a carbon tax, the government sets a price that emitters must pay for each ton of greenhouse gas emissions they emit. Businesses and consumers will take steps, such as switching fuels or adopting new technologies, to reduce their emissions to avoid paying the tax.) This study examines the effect of adopting a carbon tax on fuel choice of rural households in India. It uses 2009-10 National Sample Survey data covering over 59,119 households of rural areas. The study found that with the increase in income of households, the expenditure on modern fuels is increased. On the other hand, with the increase in price of modern fuels due to imposing a carbon tax, households are less likely to choose modern fuels than traditional fuels. <Link for study> Ask the students to discuss the following points - • What is the main argument of the study? • Why does the government apply a carbon tax? What is the main finding of the study?	liquid and gaseous fuels as in Table 10.3., *Ask students to write the answer of the following in their notebooks* - If LPG has more calorific value when compared to diesel, petrol, and kerosene, what does it mean in terms of cooking time?

Summary	**Factual Question**	**CFU**	**Home Work**
Students will understand and	1. What is global warming?	1. What is fuel?	Ask your parents for the LPG bill. Try to find out the quantity specified in the

apply in their life, How the different fuels have different calorific values.	2. What is acid rain? 3. What is CNG made of?	2. Define a good fuel. 3. Give some examples of fuels from our daily lives. 4. Define the calorific value of fuel	bill in a cylinder.

Period 11	**TOPIC: Burning fuels leads to harmful products**	Activity

Learning Outcome:
Students will be able to suggest remedies from harmful effects by burning fuels

Explicit Teaching/Teacher Modelling (I Do)	**Group Work (We Do)**	**Independent Work (You Do)**
• Recap- Teacher recalls the previous class. • The teacher shares the natural scenarios where fuel consumption as well as its effect is increasing. • Teacher explains about causal reasons of global warming, environmental pollution, etc. - Global warming is the long-term warming of the planet's overall temperature. - Though this warming trend has been going on for a long time, its pace has significantly increased in the last hundred years due to the burning of fossil fuels. - As the human population has increased, so has the volume of fossil fuels burned.	• Show the video to students on global warming and its consequences on the earth and its organisms. Link 7_Global Warming • The teacher will ask a few questions related to video and students will write their answers in the notebook: ◦ What is global warming? ◦ What is greenhouse gas? • The teacher will ask a few questions related to video and students will write their answers in the notebook: ◦ What is acid rain? ◦ What is the cause of acid rain? ◦ What is the effect of acid rain?	• Students will make a poster presentation on global warming. • Show it in the class.

Summary	**Factual Question**	**CFU**	**Home Work**
Students can suggest remedies from harmful effects of burning fuels.	What are the steps do you take when your hand is burnt accidentally?	1.What is global warming? 2.What is the cause of acid rain on plants and animals?	1. Ask your parents and grandparents about the temperature and climate in their times. Are the temperature and rain the same as before or have they changed. Ask them to describe how the climate and environment changed from those times. 2. What can you do to reduce global warming at your school/ home level? 3. What are different greenhouse gases? 4. How are greenhouse gases

			produced? 5. Write a short note on acid rain and its consequences. 6. What is CNG? Why CNG is considered a cleaner fuel?

Teacher's Diary

Name of the Teacher:				Name of the Month:	
Name of the Lesson:				Class:	
Period No	**Name of the Concept to be taught**	**Date**	**Activities Conducted during the teaching**	**TLM Used**	Remarks
6.1	Introduction 6.1 What is combustion?				
6.2	Activity 6.1				
6.3	Activity 6.2				
6.4	Activity 6.3 Ignition temperature				
6.5	Activity 6.4				
6.6	6.2 how do we control fire				
6.7	6.3 Types of combustion				
6.8	6. 4 Flame				
6.9	6.5 structure of flame Activity 6.5				
6.10	6.6 What is a fuel 6.7 Fuel efficiency				
6.11	Burning fuels leads to harmful products				

1.	What were some of the specific strategies that I used to encourage participation? How effective were they? What will I do differently next time?

2.	Were there any concepts or activities that students found particularly difficult? How will I adapt my approach to address these difficulties in the next lesson?
3.	What additional resources or modifications could improve the effectiveness of this lesson in future implementations?
4.	How well did I adjust my teaching based on student reactions or unforeseen challenges?

Teacher's Signature Head Teacher's Signature

Head Teacher's Suggestions:

Teacher Notes:

January 2026

Sunday	Monday	Tuesday	Wednesday	Thursday	Friday	Saturday
				1 New year 7.1	2 7.2	3 No Bag Day
4 Sunday	5	6	7	8	9 7.3	10 Second Saturday
11 Sunday National Youth Day	12	13	14	15	16	17 No Bag Day
18 Sunday D.R.Kaprekar Jayamti	19 7.4	20 7.5	21 7.6	22 7.7	23 7.8	24 No Bag Day Cluster,PTM, SMCmeeting
25 Sunday Republic day	26 7.9	27 7.10 Annual Day	28 7.11	29 7.12	30 NI Gandhi Vardhanti	31

No bag Day

Cluster Complex

Additional Resources

FA - 3

Pongal Holidays

Pongal Holidays 10-01-2026 TO 18-01-2026,
Pongal Holidays for Christian Minority Schools 10-01-2026 TO 15-01-2026

Week1: **7.1 – 7.2**

Week 2: **7.3**

Week 3:

Week 4: **7.4 – 7.8**

Week 5: **7.9 – 7.12**

TEACHER'S NOTES

Chapter 7: Light



Learning Outcomes:

1. Draw the experimental setup to verify laws of reflection.
2. Differentiate regular and diffused reflections.
3. Explain the process and phenomena of formation of multiple images.
4. Construct a kaleidoscope using materials from surroundings and explain its working.
5. Understand the structure and working of the human eye.

Prior Concepts/Skills:

1. Different types of mirrors
2. Image of an object and its features formed by a plane mirror.
3. Knowledge about light
4. Basic knowledge about vision and taking care of eyes

Teaching Learning Materials

Plane mirrors, transparent plastic sheet, black paper, card temperature/mobile phone, tape, thread, Braille system.

Teacher Resources

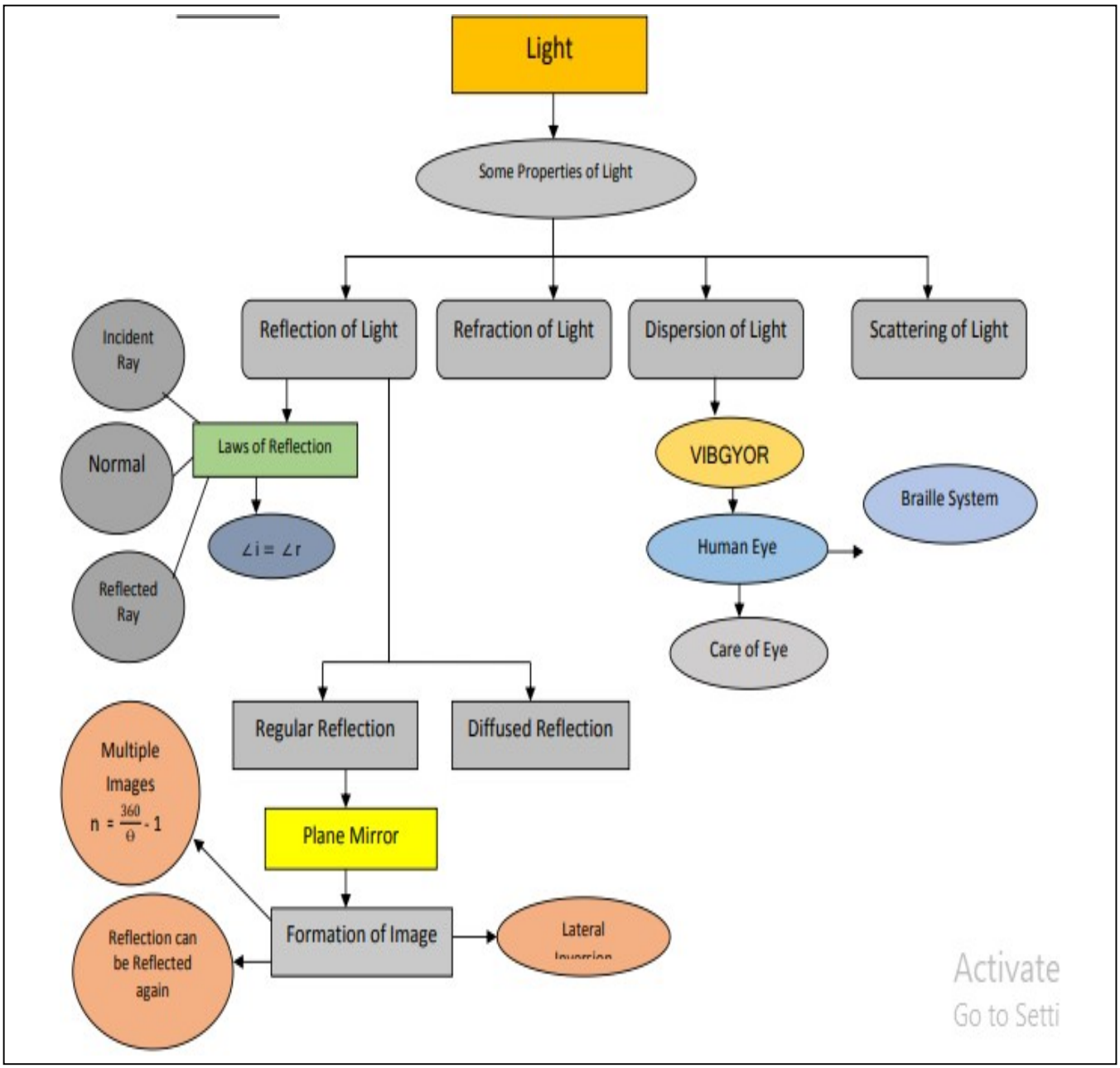
1. A detailed lesson plan, 2. Biju's notes for light lesson, 3. ekshiksha.org.in

4. vedantu.com, 5. physics catalyst.com

Hook Activity:

Teacher Notes :

Concept of Map



Synopsis

- We can see the object when light from an object enters our eyes. The light may have been either emitted or reflected by the object.
- **Incident ray:** After striking the mirror the ray of light is reflected in another direction The light ray that strikes on any surface is called incident ray.
- **Reflected ray:** The ray that comes back from the surface after reflection is known as the reflected ray.
    - ➔ **Normal:** The ray perpendicular to the surface, at the point of incidence is called normal.
    - ➔ **Angle of incidence:** The angle between the normal and the incident ray is called the angle of incidence.
    - ➔ **Angle of reflection:** The angle between the normal and the reflected ray is called the angle of reflection.

- **The laws of reflection:**
  - ➔ The angle of incidence is equal to the angle of reflection.
  - ➔ The incident ray, the normal and the reflected ray, all lie in the same plane.
- **Lateral Inversion:** In an image formed by a mirror, the left side of the object appears on the right and the right appears on the left. This is known as lateral inversion.
- **Diffused or irregular reflection:** When all the rays reflected from a plane surface are not parallel the reflection is known as diffused or irregular reflection.
- **Regular reflection:** Reflection from a smooth surface like that of a mirror is called regular reflection.
- **Multiple Image:** Large number of images are formed by mirrors placed at an angle to one another. e.g., kaleidoscope.
- **Dispersion of light:** Splitting of light into its colours is known as dispersion of light. Rainbow is a natural phenomenon showing dispersion.
- **Human eye:** It is roughly spherical in shape.
  - ➔ **Cornea:** Transparent front part of eye.
  - ➔ **Pupil:** A small opening in the iris
  - ➔ **Iris:** Coloured part of eyes, it controls the size of pupil.
  - ➔ **Lens:** Focuses light on retina.
  - ➔ **Retina**: Site of image formation.
  - ➔ **Cones and Rods:** There are two kinds of cells present in the retina.
    - I. Cones, that are sensitive to bright light
    - II. Rods, which are sensitive to dim light.
- **Blind spot:** At the junction of the optic nerve and the retina there are no sensory cells so no vision is possible at this spot. This is called the blind spot.
- **Yellow spot:** It is located in the centre of the retina. It has a maximum concentration of light sensitive cells.
- **Least distance of distinct vision:** The most comfortable distance at which one can read with a normal eye is about 25 cm. It is also called the least distance of distinct vision.
- **Care of Eyes:** Use suitable spectacles. Too little or too much light is bad. Do not look at the sun directly. Wash your eyes frequently.
- **Periscope: Periscopes** are used in submarines, lakes and also by soldiers in bunkers to see things which are not in the line of sight.

Braille system: **Visually challenged persons can read and write using the Braille system. It has 63 dot patterns or characters. Each character represents a letter, a combination of letters, a common word or a grammatical sign.**

Period wise Topics

Period No.	Concepts/Topic	Learning Outcomes	QR Code	Any Specific Information
7.1	Introduction 7.1 What makes things visible 7.2 laws of reflection	explain about the laws of reflection		
7.2	Activity 7.2,	verify the laws of reflection of light		
7.3	Activity 7.3	concludes that image		

		formed by plane mirror is laterally inverted		
7.4	7.3 Regular and diffused reflection, activity 7.47.4 Reflected light can be reflected again	differentiate regular and diffused reflections		
7.5	7.4 Reflected light can be reflected again 7.5 Multiple images	Explain the formation of multiple images.		
7.6	Activity 7.5 Kaleidoscope Activity 7.6	prepare kaleidoscope of his own from the local available materials		
7.7	7.6 Sunlight white or coloured. Activity 7.7	explain the dispersion of light from white colour		
7.8	7.7 What is inside our eyes? Activity 7.8.	understand the structure of human eye		
7.9	Activity 7.9	explain the persistence of vision in our eye		
7.10	Activity 7.10.	understand persistence of vision in play way method		
7.11	7.8 care of eyes 7.9 Visually impaired persons can read and write	suggest some precautions of care of eyes		
7.12	7.9 Visually impaired persons can read and write 7.10 What is the Braille system	explain Braille system and its usefulness to the visually impaired		

Period 7.1	**TOPIC: Introduction 7.1 What makes things visible 7.2 laws of reflection**	**Activity**

Learning Outcome:
Students will be able to

- explain about the laws of reflection

Behavioural Expectations:

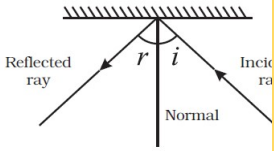
1. We will listen attentively when others are speaking
2. We will actively participate in classroom discussions and activities
3. We will bring the required material to class every day
4. We will complete our tasks (classroom and homework) on time

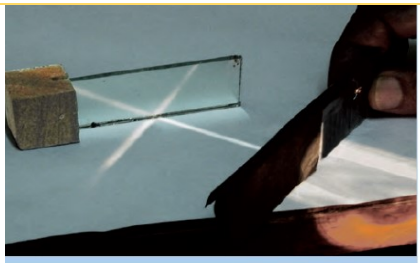
Hook
Can anyone tell me what makes things visible?
Every day we see different types of things around us. You know very well that we cannot see things in the dark. We need light to see things in our surroundings. In this chapter we will study about light.

- Teachers can start with a conversation on what is light and how it helps us to see various things around us.

- Brief students about the topics that will be covered in this chapter by going through the lesson outcomes.
- Do you know how we are able to see things around us?

What are the properties of light that help us to see things?

Explicit Teaching/Teacher Modelling (I Do)	Group Work (We Do)	Independent Work (You Do)
- Begin the lesson by engaging students in discussion about the light, and how it helps in seeing things around us. - Give an overview of the chapter, topics that will be covered throughout the chapter. - Encourage the students to ask questions or discuss prior knowledge of light and its various properties, such as - Does light travel in straight lines - How your image will be in a plane mirror? - Show video of the same to develop a better understanding in the class Link 1 ***Laws of reflection:*** - *Angle of incidence is always equal to the* ***angle of reflection.*** - Show a video of the same for an experimental verification of the law of reflection in the class. Link 1 or Link 2	**Activity 1** ● Divide the class in groups of three where students perform the activity given below. ● ***Activity:*** *Perform the* ***Activity 7.1*** *from the textbook* ***Procedure:*** - Fix a white sheet of paper on a drawing board or a table. - Take a comb and close all its openings with the help of a strip of black paper except one in the middle. - Hold the comb perpendicular to the sheet of paper. - Throw light from a torch through - the opening of the comb from one side. - With slight adjustment of the torch and the comb you will see a ray of light along the paper on the other side of the comb. Keep the comb and the torch steady. - Place a strip of plane mirror in the path of the light ray. - What do you observe?	● Solve the following worksheet in your notebook. Worksheet 1 ● ***Activity:*** Draw the following rays and angles in an A4 sheet and show it in the class. 



Activity 2

- Discuss the law deduced from the ***activity 7.1 with*** students and ask them to write in their notebook.
- *Complete* ***Table 7.1***

S. No	***Angle of Incidence (∠i)***	***Angle of Reflection (∠r)***
1		
2		

Summary	**Factual Question**	**CFU**	**Home Work**
1. Light is reflected from all surfaces 2. The light ray, which strikes any surface is called incident ray. The ray that comes back from the surface after refraction is known as the refracted ray.	Define 1. Angle of incidence 2. Angle of reflection 3. Incident Ray 4. Reflected ray	1.Can you see an object in the dark? 2.What is reflection? 3.How do we see various objects? 4.Does any object emit light?	• Write your observation of **Activity 7.1** in your own words in detail. • Write about different features of light in your notebook. • Suppose you are in a dark room. Can you see objects in the room? Can you see objects outside the room. Explain. • Explain your answer in your own words and discuss with your teacher and friends in the classroom. • *Angle of incidence is equal to the angle of reflection.* a. *Always* b. *Sometimes* c. *Under special conditions* d. *Never*

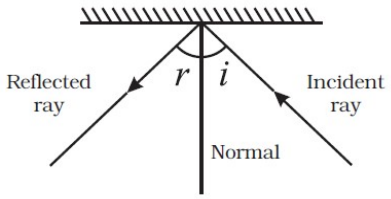
Period 7.2	**TOPIC: Activity 7.2**	**Activity**
Learning Outcome: Students will be able to verify the laws of reflection of light		
Explicit Teaching/Teacher Modelling (I Do)	**Group Work (We Do)**	**Independent Work (You Do)**

- Recap of the topic/ concepts discussed in the last class.

- Discuss the law of reflection with students.

Laws of reflection:

- *Angle of incidence is always equal to the angle of reflection.*
- *The incident ray, the normal at the point of incidence and the reflected ray all lie in the same plane.*



- Show the following videos of the law of reflection to students for better understanding.

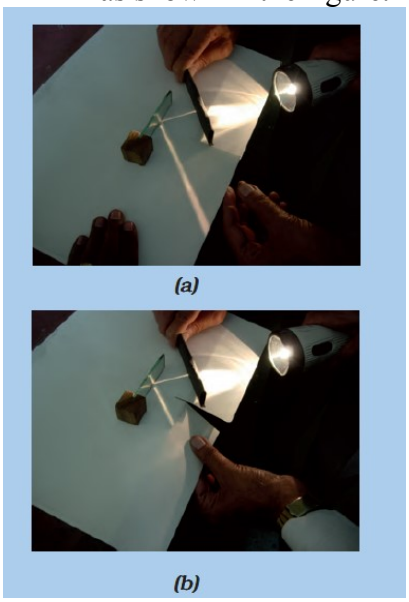
Link 3

- Divide the class in groups of three or four and ask them to perform the activity given below.

- ***Activity:*** *Perform the* ***Activity 7.2 again*** *from the textbook given on* ***Page No. 4***

Procedures: Activity 7.2:

- ***Perform Activity 7.1 again.***
- This time use a sheet of stiff paper or a chart paper. Let the sheet project a little beyond the edge of the Table as shown in the figure.



Ask the following questions from the students:

- Can you still see the reflected rays?
- Bring the paper back to the original position.

or, can perform the following activity to show that the incident ray, the reflected ray and the normal at the point of incidence lie in the same plane.

Materials Required:

- *Plane mirror, holder, ray box, etc.*

Procedure:

- Follow the procedure given in the textbook.

Solve the following question in your notebook:

- *Two mirrors meet at right angles. A ray of light is incident on one at an angle of 30° as shown in Fig. 7.19. Draw the reflected ray from the second mirror.*

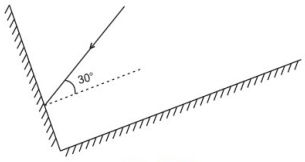
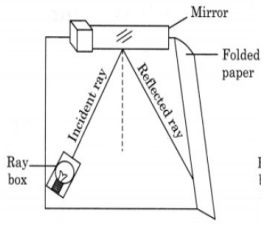
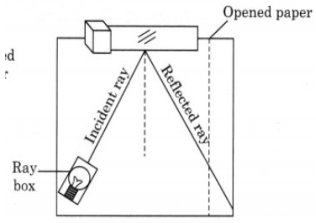


Fig. 7.19

Observation:

- You will observe that the reflected ray is not seen in the folded portion of the chart paper.
- Now bring the folded portion back to its original position.
- The reflected ray of light is again *seen on the page.*





Conclusion:

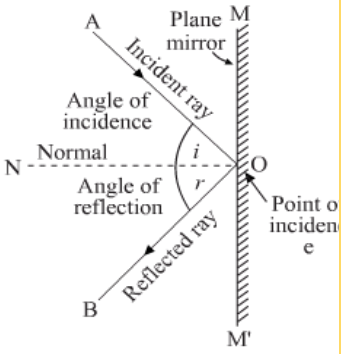
The sheet on the board can be considered as a plane. The incident ray, the reflected ray, the normal at the point of incidence lie in the same plane.

Summary	Factual Question	CFU	Home Work
- Angle of incidence is always equal to the angle of reflection. - The incident ray, the normal at the point of incidence and the reflected ray all lie in the same plane.	1. What are the laws of reflection? 2. Angle of incidence is always equal to the angle of refraction? explain.	- What are the different laws of reflection of light? - What is 'normal' in the reflection of light from a plane mirror? - What happens if you change the angle of incidence? - List the characteristics of an image formed in a plane mirror. - What is meant by lateral inversion?	• Describe an activity to show that the incident ray, the reflected ray and the normal at the point of incidence lies in the same plane. • What is the angle of incidence of a ray if the reflected ray is at an angle of 90° to the incident ray? • Boojho stands at A just on the side of a plane mirror as shown in Fig. 7.20. Can he see himself in the mirror? Also, can he see the image of objects situated at P, Q and R? A(Boojho) .P .Q .R

Period 7.3	TOPIC: Activity 7.3	Activity

Learning Outcome:
Students will be able to

- concludes that image formed by plane mirror is laterally inverted

Explicit Teaching/Teacher Modelling (I Do)	Group Work (We Do)	Independent Work (You Do)
- Recap of the topic concepts discussed in the last class. - Ask the students about Laws of reflection taught in previous classes. - Show the following video to students for better understanding of images formed by a plane mirror. **Link 4**	***Activity:*** *Perform the* ***Activity 7.3*** *from the textbook* ***Activity: Formation of an image by a plane mirror*** • A source of light O is placed in front of a plane mirror PQ. • Two rays OA and OC are incident on it ***Can you find out the direction of the reflected rays?*** • Draw normal to the surface of the mirror PQ, at the points A and C. • Then draw the reflected rays at the points A and C. • How would you draw these rays? Call the reflected rays AB and CD, respectively. • Extend them further. Do they meet? • Extend them backwards. Do they meet now? If they meet, mark this point as I. • For a viewer's eye at E (Fig. 7.5), do the reflected rays appear to come from the point I. • Since the reflected rays do not actually meet at I, but only appear to do so, we say that a virtual image of the point O is formed at I. • As you have learnt already in Class VII, such an image cannot be obtained on a screen.	• ***Activity:*** Make a poster presentation of the given diagram on A3 sheet and show it in the classroom 

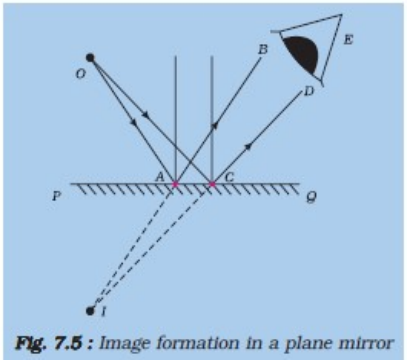
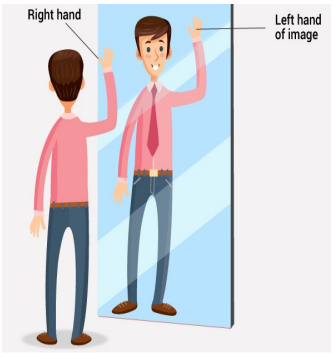


Fig. 7.5 : Image formation in a plane mirror

- An image formed by a mirror the left of the object appears on the right and the right appears on the left. This is known as **lateral inversion.**



Summary	Factual Question	CFU	Home Work
The reflections of the objects in the mirror will be lipped/mirr ored.	**1. What is lateral inversion?** **2. W**here do you feel lateral inversion of images in your daily life?	1.What is the angle of reflection when a ray of light is incident normally on a plane mirror? 2.If the angle between the mirror and the incident ray is 30°, the angle of reflection is (a) 30°. (b) 60°. (c) 15°. (d) 90°. 3.Draw the image of the letter 'P' as formed by a plane mirror placed on the right side of 'P'.	• Image formed by a plane mirror is (a). virtual, behind the mirror and enlarged. (b) virtual, behind the mirror and of the same size as the object. (c) real at the surface of the mirror and enlarged. (d) real, behind the mirror and of the same size as the object. • Differentiate between real and virtual images? • Define lateral inversion.

Period 7.4	TOPIC: 7.3 Regular and diffused reflection, activity 7.47.4 Reflected light can be reflected again	Activity
Learning Outcome: Students will be able to differentiate regular and diffused reflections		

Hook

- Imagine you're walking in a park, and you come across a shiny, smooth mirror hanging on a tree. When you look into it, what will you see?
- Now, picture a different scenario. You're in the same park, but this time, you notice a rough, matte, and non-shiny wall. When you look at your reflection on this surface, what will you see? What will your reflection look like on that surface? How is this different from first experience?
- Teacher will introduce topics '' regular and diffused reflections'' through this hook. The discussions will be built by asking given questions above.
- Everyday appliances like stainless steel appliances or polished metal objects show regular reflections.
- Diffused reflections happen on rough or irregular surfaces like walls. When light hits these surfaces, it scatters in various directions, creating a softer, less defined reflection.

Understanding the differences between regular and diffused reflections helps us appreciate the aesthetics and practical aspects of the materials and surfaces we encounter daily

Explicit Teaching/Teacher Modelling (I Do)	Group Work (We Do)	Independent Work (You Do)
- Recap of the concepts discussed in the last class. - Introduce the concept of regular and (Types of reflection). - Reflection from a smooth surface like that of a mirror is called **regular reflection.** Show the diagram of regular reflection	● Divide the class in groups of three where students discuss the regular reflection, and write their answers in their notebook. ● Ask students to draw the figure of regular reflection in their notebook. Ask the following questions from students: ● What did students keep in mind while drawing the figure of regular reflection in their notebook? Do you find any object in the class in which regular reflection can be expected? Name them.	● Solve the given worksheet in your notebook. <u>Worksheet 2</u>

Summary	Factual Question	CFU	Home Work
Reflection from a smooth surface like that of a mirror is called **regular reflection.**	1. Difference between regular and diffused reflection? 2. Does diffused reflection mean failure of laws of reflection?	- Which reflection is known as diffuse reflection? - Does a paper reflect light? - Does diffused reflection mean the failure of the laws of reflection?	- What is regular reflection? - Find out some surfaces in the home in which regular reflections can be expected. Note it down in your notebook and show it to the teacher the next day.

Period 7.5	TOPIC: 7.4 Reflected light can be reflected again 7.5 Multiple images	Activity

Learning Outcome:
Students will be able to explain the formation of multiple images.

Explicit Teaching/Teacher Modelling (I Do)	Group Work (We Do)	Independent Work (You Do)

- Recap of the previous class - Ask some questions about regular reflection from students. - Introduce the concept of diffused reflection. - When all the parallel rays reflected from a rough or irregular surface are not parallel, the reflection is known as **diffused or irregular reflection.** - Show the diagram of diffused reflection. - Show the following video to students for proper understanding of the topic. Link 5	**Activity:** Engage students in a hands-on ***Activity7.4*** **Procedure:** ● Imagine that parallel rays are incident on an irregular surface as shown in Fig. 7.6. ● Remember that the laws of reflection are valid at each point of the surface. ● Use these laws to construct reflected rays at various points. **Are they parallel to one another?** **Observation:** ● All the parallel rays reflected from a rough or irregular surface are not parallel ● It is caused by the irregularities in the reflecting surface for example, cardboard.	**Activity:** Consider the diagram at the right of five rays of light approaching a microscopically rough surface. For each incident ray, estimate the normal line and draw the corresponding reflected ray of light. Make a poster presentation of such reflection from the uneven surfaces, and show it in the class.

Summary	Factual Question	CFU	Home Work
When all the parallel rays reflected from a rough or irregular surface are not parallel, the reflection is known as **diffused or irregular reflection.**	**1. Give some examples of material with diffuse reflection** **2. Where diffuse reflection is useful**	- Diffused reflection occurs at the ……. surface. - Which type of reflection of light, regular reflection or diffuse reflection, leads to the formation of images? - How is the reflection of light from a plane mirror different from the reflection of light from a plane sheet of paper? - Why is the image not formed in diffused reflection?	- ……is caused by their irregularities in the reflecting surface. - Images are formed by ……. reflection. - Differentiate between regular and diffused reflection. - Search for the objects at your home that can cause diffused reflection. - Give three examples of materials with diffuse reflection? - Give one example where diffuse reflection is useful. - A microscopic view of a sheet of paper is shown in the diagram given below. Would you expect this sheet of paper to cause light to undergo regular or diffuse reflection? Explain.

Period 7.6	TOPIC: Activity 7.5 Kaleidoscope Activity 7.6	Activity

Learning Outcome:
Students will be able to prepare kaleidoscope of his own from the local available materials

Explicit Teaching/Teacher Modelling (I Do)	Group Work (We Do)	Independent Work (You Do)
- Ask the questions from students from the topic studied in the last period. - Introduce the term kaleidoscope. And explain about its structure, function and its formation in the class. - A kaleidoscope is an optical instrument consisting of two or more reflecting surfaces. - These surfaces are tilted towards each other at a certain angle to obtain symmetrical patterns when viewed from the other end. - The tube with reflecting surfaces consists of coloured pieces of glass. - Show the video to the students for the formation of a kaleidoscope. Link 8	**Activity: Formation of Kaleidoscope** **Materials Required** ● Plastic (See-through and coloured) ● Overhead Transparency Paper ● Pencil ● Masking/Duct tape (One Roll) ● Mirrored Perspex (3 Pieces) **Procedure** ● Use the three pieces of mirrored Perspex and place them in the form of a triangle. Try to ensure that it has a solid structure and is taped on the outside of the triangle. ● Sketch the small triangle located at the edge of the kaleidoscope to the overhead transparency paper (set aside 1 cm extra around the triangle to allow folding of the paper). ● Keep the transparency paper to the kaleidoscope edge and cut the narrow openings at the corners, which would facilitate the folding of the corners easily. ● Attach the paper to a separate place. ● Make another triangle by drawing it, and this time it should be 2 cm larger than the earlier one. ● Choose the kind of plastic colour you would like to put inside the kaleidoscope. Take off small paper cuttings that would sit well on the transparent paper. ● Place the coloured plastic at the end of the kaleidoscope that has a transparency paper and add another transparency paper (triangle) a little bigger than the earlier one. Keep the second triangle upside down so that there would be ample space for plastic to shift between the two transparencies. Once you have finished making the kaleidoscope, design it beautifully with colours, glitters and so on.	**Activity:** ● Ask the students to draw the given image in the chart paper and present it in the class.

Summary	Factual Question	CFU	Home Work
A kaleidoscope is an optical instrument consisting of two or more reflecting surfaces.	1. What is the concept of a kaleidoscope ? 2. Kaleidoscope features how many reflecting surfaced?	1. Reflected light can be reflected again. This principle is used in which of the following instruments? (a) Telescope (b) Periscope (c) Stethoscope (d) Microscope 2. What is reflection? 3. The principle of successive reflections of light at inclined mirrors is used in a. Telescope b. Periscope c. Stethoscope d. Microscope 4. Who invented the kaleidoscope? 5. Kaleidoscope features how many reflecting surfaces?	Answer the questions in your notebook. 1. Describe the construction of a kaleidoscope. 2. Where can a kaleidoscope be used? 3. What is the principle of Kaleidoscope?

Period 7.7	TOPIC: 7.6 Sunlight white or colored. Activity 7.7	Activity

Learning Outcome:
Students will be able to explain the dispersion of light from white colour

Hook

- Can you tell me what the colour of the sunlight is?
- Have you seen a rainbow in the sky during the rainy season?
- How many colours are there in a rainbow?
- Do you know that sunlight is also a mixture of colours like the rainbow?
- Do you know the combination of water and mirror can show the different colours of sunlight?

In this class we will try to find the answer to these questions in this period.

- Understanding the science behind the colours of sunlight and their manifestation in rainbows enriches our appreciation of the natural world, and experiencing a rainbow after the rain is a magical reminder of the wonders of physics and optics in everyday life.

One of the most common experiences with rainbows is seeing them after rain showers when the sun comes out. Sunlight is typically perceived as white light. The sunlight breaks down into its component colours as it passes through raindrops forming a rainbow, due to dispersion

Explicit Teaching/Teacher Modelling (I Do)	Group Work (We Do)	Independent Work (You Do)
- Do a brief recap of the last class. - Discuss about the sunlight and its different component colours - For example: Whether it is white light or coloured? - Let's watch a video to	• Objective: To engage students in an interactive **Activity 7.7 (Text Book page no 14).** **Materials Needed:** • plane mirror • sheet of white paper • Bowl with water.	**Activity:** With the help of a prism, show the splitting of white light into component colours. • Place a piece of white paper on the

understand dispersion of sunlight into different component colours.

Link 9

- Explain the following to the students:
- **Dispersion of light:** Splitting of light into its colours is known as dispersion of light.
- VIBGYOR: VIBGYOR refers to (**Violet–Indigo–Blue–Green–Yellow–Orange–Red).**
- The band of seven colours is called a spectrum of visible light.

Rainbow is a natural phenomenon showing dispersion.

Instructions:

- Divide the class into small groups of 3-4 students.
- Provide each group with materials and instruct them to follow the steps.
- Each student records the observations.
- Encourage students to share their observations and insights about the experiment.

Procedure:

- Get a plane mirror of a suitable size.
- Place it in a bowl (Katori)
- Fill the bowl with water.
- Put this arrangement near a window in such a way that direct sunlight falls on the mirror.
- Adjust the position of the bowl so that the reflected light from the mirror falls on a wall.
- If the wall is not white, fix a sheet of white paper on it.

.

Observations:

- Reflected light will be seen to have many colours.

Ask the question from the students:

- How do you explain this?
- The mirror and water form a prism.

ground under the sunlight.

- Put the prism on or above the paper.
- Rotate and move the prism around until you see rainbow colours on the paper.

Summary	Factual Question	CFU	Home Work
Dispersion of light: Splitting of light into its colours is known as dispersion of light.	- What is dispersion? - What is prism? - What is the significance of the dispersion of light into its spectrum of colours? - How does light travel through prism?	- Explain the dispersion of white light by a glass prism. - Why does light split into its spectrum of colours in a prism but not in a glass slab? - What happens when a ray of light passes through a prism? - How many times does a ray of light bend on passing through a prism? What is the principle of a prism? - What do you mean by dispersion? Write the sequence of colours in the rainbow. - How is a rainbow formed?	1. Define I. Dispersion of light II. VIBGYOR III. Spectrum 2. On the basis of an experiment done in the classroom, explain the formation of a rainbow in the sky during the rainy season in your own words. 3. What is the principle of a prism? 4. In the dispersion experiment by prism, name the light that bends the (a) minimum (b) maximum.

Period 7.8	TOPIC: 7.7 What is inside our eyes? Activity 7.8.	Activity

Learning Outcome:
Students will be able to understand the structure of human eye

Explicit Teaching/Teacher Modelling (I Do)	Group Work (We Do)	Independent Work (You Do)
- Do a brief recap of the last class. - Ask some questions about topics taught in the previous class. - For example: - How is a rainbow formed in the sky? - Expand VIBGYOR - Define dispersion. - Introduce the concept: Internal structure of the human eye. - Explain the internal structure of eye to the students: - The internal components of an eye are: - **Lens:** It is a transparent, biconvex, lens of an eye. The lens is attached to the ciliary body by ligaments. The lens along with the cornea refracts light so that it focuses on the retina. - **Retina:** It is the innermost layer of the eye. It is light sensitive and acts as a film of a camera. Three layers of neural cells are present in them, they are ganglion, bipolar and photoreceptor cells. It converts the image into electrical nerve impulses for the visual perception by the brain. - **Optic nerve:** It is located at the posterior portion of the eyes. The optic nerves carry all the nerve impulses from the retina to the human brain **for** perception. - **Cornea:** The cornea is the clear, outer part of the eye. - **Iris:** Behind the cornea,	● Objective: To engage students in an interactive **Activity 7.8 (Text Book page no 16)** that will help to understand the relationship between size of the pupil in relation to the light. **Materials Needed:** ● Torch **Procedure:** ● Divide the class into small groups of 3-4 students. ● Provide each group with materials and instruct them to follow the steps given in the book. ● Each student records the observations. ● Encourage students to share their observations and insights about the experiment. Ask the following questions from students now: ● Do you notice any change in the size of the pupil? ● In which case was the pupil larger ● Why do you think it was so? ● In which case do you need to allow more light in the eye, when the light is dim or bright? or, students can also perform the following short activity: ● **Activity:** Demonstration of blind spot **Procedure:** ● Divide the class into small groups of 3-4 students. ● Provide each group with materials and instruct them to follow the steps given in the book.	**Activity:** With the help of a prism, show the splitting of white light into component colours. ● Place a piece of white paper on the ground under the sunlight. ● Put the prism on or above the paper. Rotate and move the prism around until you see rainbow colours on the paper.

there is a dark muscular structure called iris. In the iris, there is a small opening called the pupil. It controls the size of the pupil. - **Pupil:** In the iris, there is a small opening called the pupil. The size of the pupil is controlled by the iris. - Let's watch the following videos to understand the internal structure (Link 10) and functioning (Link 11) of human eye: Link 10 Link 11	Ask questions from students: ● Does the round mark disappear at some point? ● Now close your right eye. Look at the round mark now and repeat the activity. Does the cross disappear? **Result:** ● The disappearance of the cross or the round mark shows that there is a point on the retina which cannot send messages to the brain when light falls on it. .	

Summary	Factual Question	CFU	Home Work
Summarize the concept: Internal structure of the human eye.	1. What are a few ways to protect your eyes?	1.Why is eye health important? 2. Which food is good for the eye? 3. Why do eyes need glasses?	1. Explain the terms: A. Retina B. Iris C. Pupil D. Cornea E. Blind spot F. Optic nerve Draw a labelled sketch of the human eye.

Period 7.9	Topic: Activity 7.9	Activity

Learning Outcome:
Students will be able to explain the persistence of vision in our eye

Explicit Teaching/Teacher Modelling (I Do)	Group Work (We Do)	Independent Work (You Do)
- Student should know functioning of cones and rods - About no sensory cells at the junction of optic nerve and retina. - About blind spot	Teacher demonstrates the blind spot activity. **Material:** Sheet of paper with round mark (**0**) and cross (**X**) **Instructions** - The distance between two points may be 8cm. - Hold the sheet of paper at arm's length from eye. - Close your left eye, look continuously at the cross, move the sheet slowly towards you keeping your eye on the cross. - We observe that round mark disappears some	- Students can identify the disappearance of dot or cross for some time. - Students can conduct this activity at any time.

	time. - It means that the image does not vanish immediately from retina. - It persists there for about 1/16th of the second.	

Summary	Factual Question	CFU	Home Work
1. Cones, which are sensitive to bright and rods, which are sensitive to dim light	1. What are the cones, rods in eyes?	- Name the receptors which can identify the colour of light. - How much time is needed to persist clear image on retina.	What is blind spot.

Period 7.10	**TOPIC: Activity 7.10.**	**Activity**

Learning Outcome:
Students will be able to understand persistence of vision in play way method

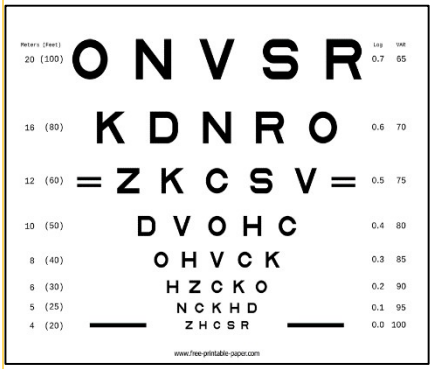
Explicit Teaching/Teacher Modelling (I Do)	Group Work (We Do)	Independent Work (You Do)
- Teacher demonstrates the bird in cage activity. - Students could know about moving pictures in the movie	**Activity 7.10** **Material:** 6-8 cm cardboard thread, cage image and bird in papers. **Procedure** - Paste a cage image on one side of card. - Paste the bird image on another side of cardboard - Put two holes on either side of cardboard - Thread a string through two holes - Twist the strong end and make the cardboard twirl rapidly. - We observe that a bird in a cage.	- Students are able to demonstrate the activity. - They know the concept of moving bodies in the movies.

Summary	Factual Question	CFU	Home Work
1. The comfortable distance at which one can read with normal eye is about 25 cm	1. What is the persistence of vision?	1. How the bird appears? (in a cage or out of cage) 2. Why the bird appears in a cage	- What did you learn about a bird in a cage activity?

2. The impression of an image does not vanish immediately from the retina. Its persistence there for about 1/16 the of a second			

Period 7.11	**TOPIC: 7.8 care of eyes 7.9 Visually impaired persons can read and write**	**Activity**

Learning Outcome:
Students will be able to suggest some precautions of care of eyes

Explicit Teaching/Teacher Modelling (I Do)	Group Work (We Do)	Independent Work (You Do)
- Do a brief recap of the last class. - Ask some questions about topics taught in the previous class. - For example: - How do you see things through your eyes? - Differentiate between retina and cornea. - Introduce the topic: Care of eyes - Explain the students about the importance of taking care of our eyes - Explain the students to various tips to take care of eyes - If advised, use suitable spectacles. - Too little or too much light is bad for the eyes. - Insufficient light causes restrain and headaches. - Too much light, like that of the Sun, a powerful lamp or a laser torch can injure the retina. - Do not look at the Sun or a powerful light directly. - Never rub your eyes. If particles of dust go into your eyes, wash your eyes with clean water. If there is no improvement go to a doctor. - Always read at the normal distance for vision.	**Objective:** To engage students in an interactive way to find out the distance of distinct vision for each student in the class. This activity will also be helpful for students if there is any challenge in vision to any students or in case their eyesight has become weak due to any reason. **Materials Required:** • Eye Chart • Tape to measure the  distance. **Instructions:** • Paste an eye chart on the wall of the class. • Start taking the eye test one by one. • Students can move as per their finest vision form where they can read the chart properly. • Mark the point on the ground, and measure between the ball and the	**Activity:** Make a collage of various nutrients including which are required for the proper vision of eyes. (Include images of fruits/ vegetables/ vitamins).

- Do not read by bringing the book too close to your eyes or keeping it too far.
- If food is deficient in some components, eyes may also suffer.
- Lack of vitamin an in foodstuff is responsible for many eye troubles. Most common amongst them is night blindness.
- Include in the diet components which have vitamin A.
- Raw carrots, broccoli and green vegetables (such as spinach) and cod liver oil are rich in vitamin A.
- Eggs, milk, curd, cheese, butter and fruits such as papaya and mango are also rich in vitamin A.
- Let's watch a video about how to take care of our eyes.

Link 12

student from the tape.

- Note the distance in your notebook from each student and make a list.
- Fill the following table.

Name of student	**Distance of distinct vision.**

Summary	**Factual Question**	**CFU**	**Home Work**
1.Lack of vitamin an in-food stuff is responsible for many eye troubles. 2. Too little or too much light is bad for eyes 3. You take proper care of your eyes.	**1. Define the least distance of distinct vision** **2. Which vitamin is necessary for the healthy eye?**	- Define the term: Least distance of distinct vision. - What are 4 things that protect the eye? - Name some diseases which are related to the eye. - Why is the image not formed at the blind spot?	1. Explain how you can take care of your eyes. 2. Gurmit wanted to perform Activity 7.8 using a laser torch. Her teacher advised her not to do so. Can you explain the basis of the teacher's advice? 3. Define night blindness. How does it happen? 4. Which vitamin is necessary for the healthy eye? 5. Meet an eye specialist. Get your eyesight checked and discuss how to take care of your eyes.

Teacher's Diary

Name of the Teacher:		Name of the Month:	
Name of the Lesson:		Class:	

Period No	**Name of the Concept to be taught**	**Date**	**Activities Conducted during the teaching**	**TLM Used**	Remarks
7.1	Introduction 7.1 What makes things visible 7.2 laws of reflection				
7.2	Activity 7.2,				
7.3	Activity 7.3				
7.4	7.3 Regular and diffused reflection, activity 7.47.4 Reflected light can be reflected again				
7.5	7.4 Reflected light can be reflected again 7.5 Multiple images				
7.6	Activity 7.5 Kaleidoscope Activity 7.6				
7.7	7.6 Sunlight white or coloured. Activity 7.7				
7.8	7.7 What is inside our eyes? Activity 7.8.				
7.9	Activity 7.9				
7.10	Activity 7.10.				
7.11	7.8 care of eyes 7.9 Visually impaired persons can read and write				
7.12	7.9 Visually impaired persons can read and write 7.10 What is the Braille system				

1.	What were some of the specific strategies that I used to encourage participation? How effective were they? What will I do differently next time?

2.	Were there any concepts or activities that students found particularly difficult? How will I adapt my approach to address these difficulties in the next lesson?
3.	What additional resources or modifications could improve the effectiveness of this lesson in future implementations?
4.	How well did I adjust my teaching based on student reactions or unforeseen challenges?

Teacher's Signature Head Teacher's Signature

Head Teacher's Suggestions:

Teacher Notes:

February 2026

Sunday	Monday	Tuesday	Wednesday	Thursday	Friday	Saturday
Sunday 1	2 8.1	3 8.2	4 8.3	5 8.4	6 NI	7 No Bag Day
Sunday 8	9 FA - 4	10	11	12	13 NI	14 Second Saturday
Sunday 15 Maha Sivaratri	16 8.5	17 8.6	18 8.7	19 8.8	20 NI	21 ~~No Bag~~ Day Cluster, PTM, SMC meeting
Sunday 22	23 8.9	24 8.10	25 NI	26 NI	27 NI	28 No Bag Day National Science Day

E8W4U1 — No bag Day

H6H1Q4 — Cluster Complex

K5M8W3 — Teacher Resources

Week1:

Week 2:

Week 3:

Week 4:

Week 5:

↓ TEACHER'S NOTES ↓

Chapter 8: Some Natural Phenomena

B4I8P4

Learning Outcomes:

1. Apply scientific concepts in lightning conductors to protect buildings from the effect of lightning.
2. Explain the process of earthquakes.
3. Draw the labelled diagram of seismograph.
4. Understand the concept of 'transfer of charges' between two bodies.
5. Analyze the causes and precautions of earthquakes

Prior Concepts/Skills:

1. Destructive phenomena such as cyclones.
2. Sparks appear when wires become loose.
3. Attraction and repulsion unlike poles and like poles in magnetism.
4. Conductors and insulators.
5. Mythical stories, if any, on lightning and earthquakes.

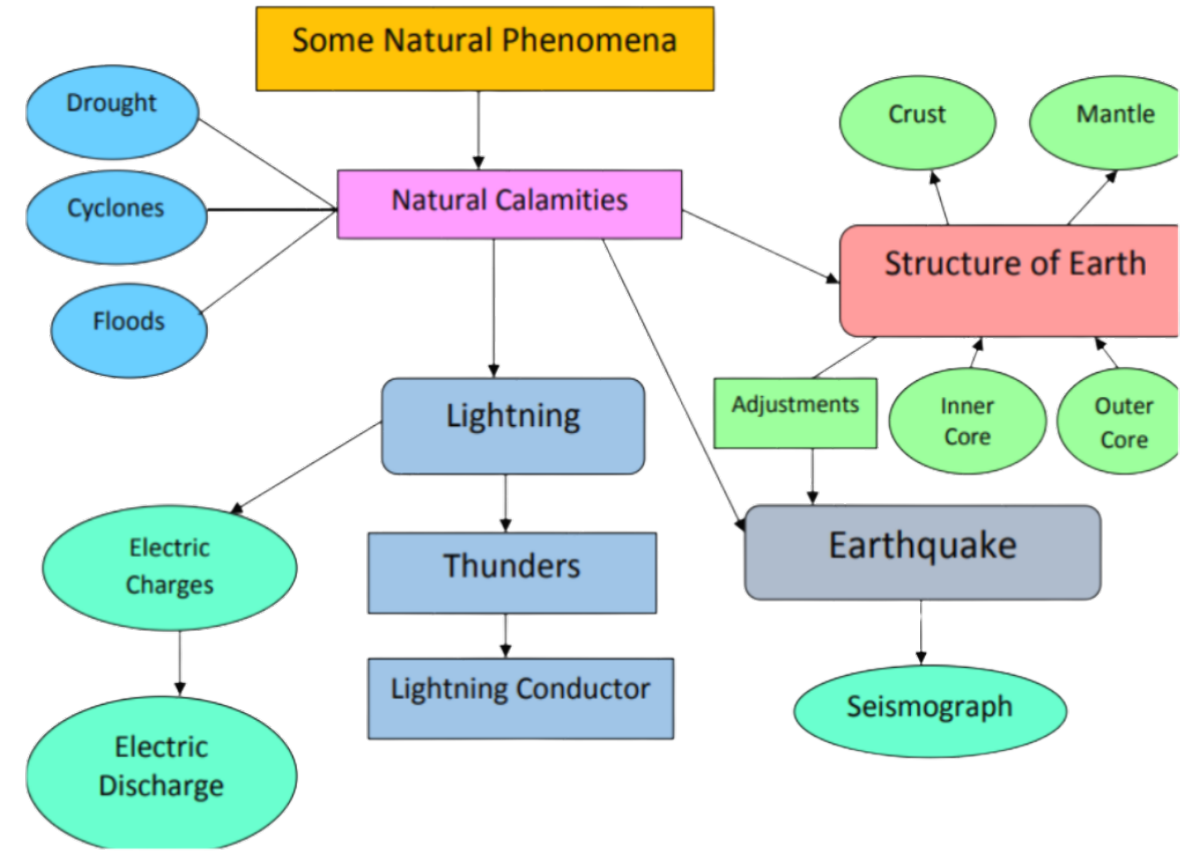
Teaching Learning Materials

Green Board, Videos on Natural phenomena like Lightning, Earthquake through IFP panel, Charts or Images, Models (Seismograph).

Teacher Resources

AP SCERT 8Th Class text book, NCERT 8th class text book, Scientific discoveries related to natural phenomena, Newspaper or Magazine Articles and YOU tube links.

Concept Map



Synopsis

- *Charging by Rubbing*:
  a. When objects like plastic get rubbed with hair, it acquires a charge. Similarly, a glass rod gets charged when rubbed with a silk cloth.
  b. They get charged as they gain or lose electrons, and that is why they can attract or repel the small pieces of paper depending on the type of charge they carry.
- *Static charges*: The electric charges that are typically stationary, meaning they do not flow as an electric current, are called static charges.
- *Charged objects*: Objects that carry a charge by means of rubbing or other processes are called charged objects.
- *Interaction between like and unlike charges*: Like charges repel, while unlike charges attract.
- *Electroscope:* A device that can detect the presence of charge is called an electroscope.
- *Discharged Objects*: When objects lose their charge by transfer, they are known as discharged objects.
- *Earthing:* The process of transferring charges from a charged object to the earth is called earthing.

- *Lightning*: During thunderstorms, air currents move upwards, and water droplets move downwards. This causes the separation of charges between clouds and between clouds and the Earth. When the magnitude of charges increases, air (normally a bad conductor) starts conducting and allows the electric discharge". This is called lightning, as this flow of charge is accompanied by bright streaks of light and sound.
- *Electric Discharge*: The flow of charge through a medium, typically a gas or a vacuum, accompanied by the release of energy.
- *Lightning safety*: Steps to follow:
    a. Outside the house: (a) Find a safe place or stay away from all trees". (b) If inside a car or vehicle, stay inside with doors and windows shut (c) Stay away from metal poles (d) Do not lie on the ground. Instead, crouch with your head in between your hands.
    b. Inside the house: (a) Avoid contact with telephone and electrical wires (b) avoid bathing (c) Unplug electrical appliances.
- *Lightning Conductors*: Lightning conductors help to protect buildings during lightning.
- *Earthquake:* An earthquake is the sudden trembling of the surface of the earth due to a disturbance deep inside the earth's crust. Causes large-scale damage to life and property. It cannot be predicted.
- *Causes of earthquake*: Earthquakes are caused due to movement or collision of tectonic plates deep down inside the uppermost layer of the earth called the crust.
- Movement of plates: Earth's crust is fragmented, and each such fragment is known as a plate. These plates are constantly in motion and sometimes can collapse under one another, causing an earthquake on the surface.
- *Seismic/fault Zones*: Boundaries of the plates on the earth's crust are weak zones where earthquakes are most likely to occur. These are known as seismic or fault zones.
- *Power of earthquake*:
    a. The power of an earthquake is expressed in terms of magnitude on a scale. This scale is called a Richter Scale.
    b. An earthquake with a magnitude > 7 on the Richter scale is considered destructive.
    c. This scale is not linear. i.e. an increase of 2 in magnitude implies 1000 times more destructive energy.
- *Seismic Waves*: Tremors deep inside the earth produce waves, which are called seismic waves.
- *Seismograph:*
    a. An instrument that records seismic waves is called a seismograph.
    b. It consists of a vibrating rod or pendulum that starts vibrating when tremors occur.
- *Protection from an earthquake*: Steps to follow
    a. Outdoors: (a) Find a clear spot, away from overhead power lines and buildings (b) if in a car, then go to a clear spot and stay inside the car till the tremors stop.
    b. Indoors: (a) stay under a table till the tremors stop (b) Avoid staying near tall heavy objects.

c. Structural measures: (a) in highly seismic areas, build mud houses with light roofs in order to minimize damage (b) Fix cupboards and shelves to the walls (c) During an earthquake, some buildings catch fire and therefore must have proper working firefighting equipment

Period wise Topics

Period No.	Concepts/Topic	Learning Outcomes	Any Specific Information
8.1	Introduction, 8.1 Lightning	explain how lightning occurs in the atmosphere	
8.2	8.2 Charging by rubbing Activity 8.1 Activity 8.2	understand how bodies get charged	
8.3	8.3 Types of charges and their interaction Activity 8.3	explain about different charging methods	
8.4	8.4 Transfer of charge ACTIVITY 8.4	Do experiments with simple electroscope	
8.5	8.5 The story of lightning	understand the story of lightning	
8.6	8.6 Lightning safety	suggest some safety measures during lightning	
8.7	8.7 Earth quakes Activity 8.5	explain how earthquakes occur	
8.8	Activity 8.6 What causes an earth quake?	locates the places of earthquake in the globe	
8.9	What causes an earth quake?	understand the main causes of earthquake	
8.10	Protection against an earthquake	give some protection methods during earthquake	

Period 8.1	**TOPIC: Introduction, 8.1 Lightning**	**Activity**

Learning Outcome:

Students will be able to

- explain how lightning occurs in the atmosphere

Behavioural Expectations:

1. We will listen attentively when others are speaking
2. We will actively participate in classroom discussions and activities

3. We will bring the required material to class every day
4. We will complete our tasks (classroom and homework) on time

Hook

Can anyone tell me what is lightning?

Sometimes we see some kind of sparks when a plug is loose in its socket or during the rainy season. In this chapter we will study about lightning.

- Teachers can start with a conversation on what lightning is and how it occurs in the sky?
    a. Have you seen the sparks on an electric pole when wires become loose?
    b. Have you seen the sparks when a plug is loose in its socket?
    c. Do you know how lightning occurs in the sky?
- Brief students about the topics that will be covered in this chapter by going through the lesson outcomes.

Explicit Teaching/Teacher Modelling (I Do)	**Group Work (We Do)**	**Independent Work (You Do)**
• Begin the lesson by engaging students in discussion about lightning, and how does it occur? • Give an overview of the topics that will be covered throughout the chapter. • Lightning is caused by the accumulation of charges in the clouds. • Encourage the students to ask questions or discuss their prior knowledge of lightning	• Show video of the same to develop a better understanding in the class Link 1 - Make a group of 3-4 students. Ask them to write the key points from the video they have watched. - Each group will share and discuss their key points with the teacher and other groups. • Teacher asks questions related to video from students. • How does lightning occur? • What is produced when we rub two objects against each other? • Which type of charges	*Activity:* • What are some natural phenomena? Write 8-10 lines about it.

	attract each other? • Why is lightning dangerous? Students will answer these questions and write their responses in the notebook.	

Summary	Factual Question	CFU	Home Work
Students learnt about the occurrence of Lighting in the atmosphere.	1) Did you observe the Sparks at electric poles in the rainy season? 2) Why lighting occurs in the rainy season?	1. What is lightning? 2. What causes lightning?	1. Who was Benjamin Franklin? What was his discovery? 2. How is lightning caused in the sky? 3. Perform the activity 8.1 given on Page No 60. Students will record their observations in their notebook and show it to the teacher.

Period 8.2	TOPIC: 8.2 Charging by rubbing Activity 8.1 Activity 8.2	Activity

Learning Outcome:

Students will be able to understand how bodies get charged

Explicit Teaching/Teacher Modelling (I Do)	Group Work (We Do)	Independent Work (You Do)
• Recap of the topic discussed in the last class. • Introducing today's topic: how charges are produced by rubbing? • Explain the following to the students: • When a plastic comb is rubbed with dry hair, it acquires a small charge. These objects are called charged objects.	• Show video of the same to develop a better understanding in the class Link 2 • The teacher will make a group of 3-4 students and will ask questions related to video from each group: • What was the topic of the video? • What were the two things in the video which were rubbed together? • Which type (material) of cloth	*Activity 9.2* Answer the following questions in your notebook: 1. Which of the following cannot be charged easily by friction? a) A plastic scale (b) A copper rod (c) An inflated balloon (d) A woollen cloth. 2. The like charges

	was taken for charging? • What happened after rubbing the two objects together? • How many types of charges are gained by rubbing objects • What can you conclude from the video? • The teacher will ask the students to write their key points in their notebook.	a. Repel each other b. Attract each other c. No interaction takes place d. All of these

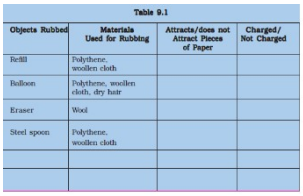
Summary	**Factual Question**	**CFU**	**Home Work**
Students learnt about the Charges produced by rubbing of bodies.	1)What happens when the ball pen refill is rubbed with polythene? 2)Why the plastic comb moved through the dry hair for few minutes is attracting the paper pieces?	• Which kind of material is used to transfer charges from one body to another? • What happens when amber is rubbed with fur? • What kind of electrical charges are generated by rubbing?	• Select a few random things at your home and rub with each other. take the rubbed things near the pieces of paper. Observe carefully and record your observation in the notebook and show it to the teacher in the class. • Collect the objects and the materials listed in Table 9.1. (Textbook Page No 60). Try to charge each by rubbing with the materials mentioned in the Table. Record your findings. You can add more items to the Table. 

Table 9.1

Objects Rubbed	Materials Used for Rubbing	Attracts/does not Attract Pieces of Paper	Charged/ Not Charged
Refill	Polythene, woollen cloth		
Balloon	Polythene, woollen cloth, dry hair		
Eraser	Wool		
Steel spoon	Polythene, woollen cloth		

Period 8.3	**TOPIC: 8.3 Types of charges and their interaction Activity 8.3**	**Activity**

Learning Outcome:

Students will be able to explain about different charging methods

Explicit Teaching/Teacher Modelling (I Do)	**Group Work (We Do)**	**Independent Work (You Do)**
• Ask students questions related to the topic studied in the last period. • Introduce the today's topic tin the class: Types of charges and	Activity: 1. Perform activity 9.3 to understand the interaction between like charges and unlike charges." Instructions:	Activity: 1.Make a poster presentation showing that like charges repel each other and

their interactions ● Explain the following to the students: - There are two kinds of charges. - Charges of the same kind repel each other, while charges of different kind attract each other. - When a charged glass rod is brought near a charged plastic straw and rubbed with polythene there is attraction between the two.	● Be careful not to touch the charged end with your hand. ● Ask the following questions from the students ➢ Is there any effect on the refill in the tumbler? ➢ Do the two attract each other, or repel each other? ● In this activity we have brought close together the charged objects that were made of the same material. ➢ What happens if two charged objects made of different materials are brought close to each other? ➢ Does it indicate that the charge on the balloon is of a different kind from the charge on the refill?	unlike charges attract each other.

Summary	**Factual Question**	**CFU**	**Home Work**
1)Students learnt about the Two kinds of charges – Namely Positive charge and Negative charge. 2)The electric charges produced by rubbing bodies are called static charges.	1)How many charges are there? What are they? 2)Do you know Like charges repel each other?	● What do you mean by electric discharge? How does it occur?	Answer the following questions in your notebook: 1.Like charges attract while unlike charges repel each other. (True/ false) 2.When a glass rod is rubbed with a piece of silk cloth the rod a) the cloth both acquire positive charge. b) becomes positively charged while the cloth has a negative charge. c) the cloth both acquire negative charge becomes negatively charged while the cloth has a positive charge.

Period 8.4	**TOPIC: 8.4 Transfer of charge**	**Activity 8.4**
Learning Outcome:		

Students will be able to do experiments with simple electroscope

Explicit Teaching/Teacher Modelling (I Do)	Group Work (We Do)	Independent Work (You Do)
Introduce the topic to the students: Transfer of charges Explain the following to the students: - The electrical charges generated by rubbing are static. They do not move by themselves. When charges move, they constitute an electric current. - The current in a circuit which makes a bulb glow, or the current that makes a wire hot, is nothing but a motion of charges. - Electrical charge can be transferred from a charged object to another through a metal conductor. - The process of transferring the charge from a charged object to the earth is called earthing. Earthing is provided in buildings to protect us from electrical shocks due to any leakage of electrical current.	Activity: - Each group performs the activity 8.4 illustrating the construction and working of "a simple electroscope" ● After the activity ask the following questions from students: ● Is there any effect on the foil strips? Do they repel each other or attract each other? Now, touch other charged bodies with the end of the paper clip. ● Do foil strips behave in the same way in all cases? ● Can this apparatus be used to detect whether a body is charged or not? ● Can you explain why the foil strips repel each other?	● Solve the following worksheet in your notebook. *Worksheet 1* - *Answer the following questions in your notebook:* a) The strips carrying ……. charges repel each other and they become ……. Open b) Electrical charge can be transferred from a charged object to another through a ……. conductor.

Summary	Factual Question	CFU	Home Work
Students know about the 1) working of Electroscope. 2)An electroscope is used to detect the body is charged or not.	1)What is the use of Electroscope? 2)What is meant by earthing?	- What is the main purpose of providing earthing in buildings?	1. How will you show that like charges repel and unlike charges attract each other? 2. Why do charges repel each other? Why don't they attract each other? 3. Explain why a charged balloon is repelled by another charged balloon whereas an uncharged balloon is attracted by another charged balloon? 4. Explain why a charged body loses its charge if we touch it with our hand. 5. Describe with the help of a

			diagram an instrument which can be used to detect a charged body.

Period 8.5	**TOPIC: 8.5 The story of lightning**	**Activity**

Learning Outcome:

Students will be able to understand the story of lightning

Explicit Teaching/Teacher Modelling (I Do)	**Group Work (We Do)**	**Independent Work (You Do)**
- Ask students questions related to the topic studied in the last period." • Quick recap of the topics studied in the previous class. • Explain the following to the students: • Explain lightning in terms of the charges produced by rubbing. • During the development of a thunderstorm, the air currents move upward while the water droplets move downward. • The positive charges collect near the upper edges of the clouds and the negative charges accumulate near the lower edges. • There is accumulation of positive charges near the ground also. • When the magnitude of the accumulated charges becomes very large, the air which is normally a poor conductor of electricity, is no longer able to resist their flow. • Negative and positive charges meet, producing streaks of bright light and sound. We see streaks as lightning. The process is called an electric discharge.	• Show the following video to the students for proper understanding of lightning. Link 4 • Ask students to write key points about what they have learnt from the video in their notebook.	• Solve the following worksheet in your notebook. *Worksheet 2*

Summary	**Factual Question**	**CFU**	**Home Work**
Students explain the story of lighting.	1)How the charges are deposited on clouds?	a) Why does lightning strike tall buildings so easily?	- How does lightning occur? Write in your own words

Period 8.6	**TOPIC: 8.6 Lightning safety**	**Activity**

Learning Outcome:

Students will be able to suggest some safety measures during lightning

Explicit Teaching/Teacher Modelling (I Do)	**Group Work (We Do)**	**Independent Work (You Do)**
• Do a quick recap of the topic studies in the previous class. • Ask students questions related to the topic studied in the last period. • Describe the Lightning Safety to students. • Explain the safety measures to follow during lightning when outside in the park: - Open vehicles, like motorbikes, tractors, construction machinery, open cars are not safe - Open fields, tall trees, shelters in parks, elevated places do not protect us from lightning strokes. - Carrying an umbrella is not at all a good idea during thunderstorms. If in a forest, take shelter under shorter trees. - If no shelter is available and you are in an open field, stay far away from all trees. - Stay away from poles or other metal objects. Do not lie on the ground. Instead, squat low on the ground. - Place your hands on your knees with your head between the hands. This position will make you the smallest target to be struck.	• Show the following video to the students for safety outside in the park during lightning. Link 5 • Ask the questions from students related to video: • What do you learn from the video? • What are the things that should be avoided during lightning?	*Activity:* • Ask students to make a poster presentation of dos and Don'ts during lightning while outside in a park or any open area.

Summary	**Factual Question**	**CFU**	**Home Work**
Students Utilizes the lightning safety measures in	1)Which is the safety place during a thunder storm?	1.What is a lightning conductor ?	1. Suggest three measures to protect ourselves from lightning. 2. The weather department has predicted that a thunderstorm is likely to occur on a certain day.

their day-to-day life.	A house or Open place.		Suppose you have to go out on that day. Would you carry an umbrella? Explain 3. How does a lightning conductor protect a building? explain

Period 8.7	**TOPIC: 8.7 Earth quakes Activity 8.5**	**Activity**

Learning Outcome:

Students will be able to explain how earthquakes occur

Hook

Imagine you're sleeping on your bed in your home. Suddenly you feel that the bed is shaking. Everything in your home is shaking. Sometimes it happens mildly, whereas sometimes it shakes so much that it becomes scary. What do you do in that case?

Teacher will introduce the topic ‘’ Earthquake’ through this hook. The discussions will be built by asking given questions above.

Explicit Teaching/Teacher Modelling (I Do)	**Group Work (We Do)**	**Independent Work (You Do)**
• Recap of the concepts discussed in the last class. • Introducing the topic of earthquakes. • Ask students some questions and their experience, if any, about earthquakes or if they have watched or read any news about earthquakes recently in India or any part of the world. • What is an earthquake? • Have you ever experienced an earthquake so far? • Have you heard from your parents/ teachers or read anywhere about the Earthquake on 26th January 2001 in Bhuj, Gujarat, that was disastrous and caused massive destruction of property and lives? • What do people feel when an earthquake occurs? • How do people protect themselves during earthquakes?	• Show the following video about earthquakes to students: Link 9 • Divide the class in groups of three/four where students discuss the earthquake and write their answers in their notebook. • Ask students to recall some news about past events when earthquakes happened in India. • Can scientists predict accurately about earthquakes?	Activity: Do the activity 9.6 for locating various places on a world map which suffered devastation due to tsunami.

Summary	Factual Question	CFU	Home Work
Students understand the reasons for earthquake.	1)what is an earthquake? 2)What happens when an earthquake occurs?	1. What is the epicentre? 2. The point where movement occurred which triggered the earthquake is the _____. (a) Dip (b) Epicentre (c) Focus (d) Strike 3. Which of the following observations may indicate a forthcoming destructive earthquake/ a) An increase in the frequency of smaller earthquakes in the region b) Rapid tilting of the ground c) Rapid changes in water levels in wells d) All of these	1. Do *Activity 9.5L:* Collect a few pictures from newspapers and magazines showing the damage caused by earthquakes. Paste the images in the chart paper and show them in the classroom. Prepare a short report on the suffering of the people during and after the earthquakes List three states in India where earthquakes are more likely to strike. With the help of the internet, make a list of places where earthquakes are very frequent. Note down the year of earthquakes in these areas.

Period 8.8	**TOPIC: Activity 8.6 What causes an earth quake?**	**Activity**
Learning Outcome: **Students will be able to locates the places of earthquake in the globe**		
Explicit Teaching/Teacher Modelling (I Do)	**Group Work (We Do)**	**Independent Work (You Do)**
• Recap of the previous class. • Introducing the topic: What Causes an Earthquake? • Explain the following to the students about cause of earthquake - Tremors are caused by the	• Show the following video to the students for proper understanding of causes of earthquakes. Link 10 • Ask some questions from students after watching	Activity: • Draw the structure of the earth in an A4 sheet and show it in the class.

disturbance deep down inside the uppermost layer of the earth called the crust. - The outermost layer of the earth is not in one piece. It is fragmented. - Each fragment is called a plate. These plates are in continual motion. - When they brush past one another, or a plate goes under another due to collision, they cause disturbance in the earth's crust. - It is this disturbance that shows up as an earthquake on the surface of the earth.		the video • What is a tectonic plate? • How does an earthquake occur? • What is crust? • What is a mantle? • Ask students to write their answers in the notebook.	
Summary	**Factual Question**	**CFU**	**Home Work**
Students recognize places of earthquake in the globe. Students know the reasons for earthquake.	1) What is a tsunami? 2)When major earthquake occurred in Bhuj district of Gujarat?	1.What are seismic or fault zones? 2.Which of the following can be triggered by an earthquake? (a) Tsunami (b) Intense ground shaking (c) A landslide (d All of these	1. Draw a diagram of the seismograph in your notebook.

Period 8.9	**TOPIC: What causes an earth quake?**	**Activity**
Learning Outcome: Students will be able to understand the main causes of earthquake		
Explicit Teaching/Teacher Modelling (I Do)	**Group Work (We Do)**	**Independent Work (You Do)**
The students should be explained 1. The seismic zones or fault lines 2. That the tremors are caused by the disturbance 3. The continual motion of	- The teacher shows the video to students about causes of earthquake and ask some questions - Teacher demonstrates the working of seismograph - Students observe and know the movement of rotating drum and pen of	Students are able to - Draw the diagram of seismograph - Identify the seismic zones - Know the movement of earth's plate - Know the Richter

fragmented plates in the earth crust 4. The working of seismograph	seismograph - Students can estimate the power of seismograph to cause destruction.	scale

Summary	Factual Question	CFU	Home Work
Students understand the causes of earthquake	1)What is the uppermost layer of earth?	1. What is the uppermost layer of the earth? 2. List out three states in India where earthquakes are more likely to strike.	Write down seismic zones

Period 8.**10**	**TOPIC: Protection against an earthquake**	**Activity**

Learning Outcome:

Students will be able to Give some protection methods during earthquakes

Explicit Teaching/Teacher Modelling (I Do)	Group Work (We Do)	Independent Work (You Do)
- Quick recap of the topic studied in the previous class. - Ask some questions about the topic taught in the previous class. - Introduce the topic: Protection against Earthquakes. - Explain the measures to be taken during an earthquake. - Take shelter under a table and stay there till the shaking stops. - Stay away from tall and heavy objects that may fall on you. - If you are in bed, do not get up, protect your head with a pillow.	• Show the video to the students about safety measures to be taken during an earthquake Link 11/or Link 12 • Ask some questions about videos from students. • What do they understand from the video? What are the other things they know that can be followed during an earthquake?	Make a list of Do's and Don'ts during an earthquake on an A3 sheet. Show them to your teacher and paste it in the class.

Summary	Factual Question	CFU	Home Work

Students utilize the protection methods during earthquake	1)What are places in Andhra Pradesh, Where earthquake have occurred?	1.What instrument is used to measure the magnitude and duration of an earthquake? 2.How can you protect yourself during earthquakes?	1. Suppose you are outside your home and an earthquake strikes. What precaution would you take to protect yourself? 2. How will you protect yourself from an earthquake if you are inside any building?

Teacher's Diary

Name of the Teacher:		Name of the Month:	
Name of the Lesson:		Class:	

Period No	Name of the Concept to be taught	Date	Activities Conducted during the teaching	TLM Used	Remarks
8.1	Introduction, 8.1 Lightning				
8.2	8.2 Charging by rubbing Activity 8.1 Activity 8.2				
8.3	8.3 Types of charges and their interaction Activity 8.3				
8.4	8.4 Transfer of charge ACTIVITY 8.4				
8.5	8.5 The story of lightning				
8.6	8.6 Lightning safety				
8.7	8.7 Earth quakes Activity 8.5				
8.8	Activity 8.6 What causes an earth quake?				
8.9	What causes an earth quake?				
8.10	Protection against an earthquake				
1.	What were some of the specific strategies that I used to encourage participation? How effective were they? What will I do differently next time?				

2.	Were there any concepts or activities that students found particularly difficult? How will I adapt my approach to address these difficulties in the next lesson?
3.	What additional resources or modifications could improve the effectiveness of this lesson in future implementations?
4.	How well did I adjust my teaching based on student reactions or unforeseen challenges?

Teacher's Signature Head Teacher's Signature

Head Teacher's Suggestions:

Teacher Notes:

March 2026

No bag Day

Cluster Complex

Teacher Resources

Sunday	Monday	Tuesday	Wednesday	Thursday	Friday	Saturday
1 Sunday	2	3	4 Holi	5	6	7 No Bag Day
8 Sunday International Women's Day	9	10	11	12	13	14 Second Saturday International π day
15 Sunday	16 Srirama Navami Potti Sriramulu Jayanti	17	18	19 Ugadi	20	21 No Bag Day Ramadan
22 Sunday	23	24	25	26 Srirama Navami	27	28 No Bag Day
29 Sunday	30	31				

TEACHER'S NOTES

Week1:

Week 2:

Week 3:

Week 4:

Week 5:

Preparation Strategies and material for National Merit Scholarship Exam (NMMS)



Tentative NMMS Exam Date (for reference): Sunday, November 16, 2025 (This is a lik on previous years, please confirm with your state's official notification).

Preparation Strategy: This plan assumes you can dedicate approximately 2-3 hours daily to your NMMS preparation. Adjust the time allocation based on your individual learning pace and other commitments.

Key Areas to Focus On:

- **NCERT Class 7 and 8 Science Textbooks:** These are the primary resources. Ensure thorough understanding of all concepts.
- **Conceptual Clarity:** Focus on understanding the 'why' behind the concepts, not just memorizing facts.
- **Problem-Solving Skills:** Practice numerical problems and application-based questions.
- **Time Management:** Learn to solve questions accurately and within the stipulated time.
- **Regular Revision:** Consistent revision is crucial to retain information.

60-Day Action Plan:

- **Phase 1: Concept Building and Initial Practice (Days 1-30)**

Week	Days	Class 7 Science Chapters	Class 8 Science Chapters	Activities
1	1-3	Nutrition in Plants	Crop Production and Management	Read the chapters thoroughly. Understand key terms and definitions. Make short notes of important points and formulas.
	4-5	Nutrition in Animals	Microorganisms: Friend and Foe	Review solved examples in the textbook. Try to solve similar questions on your own.
	6-7	Fibre to Fabric	Coal and Petroleum	Identify and understand the differences between related concepts (e.g., autotrophs and heterotrophs, natural and synthetic fibres).
2	8-10	Heat	Combustion and Flame	Draw diagrams and flowcharts to visualize processes (e.g., the water cycle, the process of digestion).
	11-12	Acids, Bases and	Conservation of	Attempt the exercises at the end of each

		Salts	Plants and Animals	chapter in your textbook.
	13-14	Physical and Chemical Changes	Reproduction in Animals	Look for real-life applications of the concepts you are learning. Discuss the concepts with a friend or family member.
3	15-17	Weather, Climate and Adaptations of Animals to Climate	Reaching the Age of Adolescence	Solve objective-type questions (MCQs, fill in the blanks, true/false) related to the topics covered. You can find these in supplementary books or online resources.
	18-19	Winds, Storms and Cyclones	Force and Pressure	Focus on understanding the units of measurement and their conversions (e.g., Celsius to Fahrenheit, Newton to Pascal).
	20-21	Soil	Friction	Relate the concepts to your everyday experiences (e.g., why do we wear woollen clothes in winter? How does friction help us walk?).
4	22-24	Respiration in Organisms	Sound	Practice drawing well-labelled diagrams of important biological processes and structures (e.g., the human respiratory system, the life cycle of a silkworm).
	25-26	Transportation in Animals and Plants	Chemical Effects of Electric Current	Solve numerical problems related to topics like heat transfer, force, and pressure. Pay attention to the formulas and their applications.
	27-28	Reproduction in Plants	Some Natural Phenomena	Review all the notes and key points you have made so far.
5	29-30	Motion and Time	Light	Take a short quiz on the topics covered in the first four weeks to assess your understanding. Identify areas where you need more attention.

- **Phase 2: Focused Practice and Revision (Days 31-50)**
- During this phase, you will focus on practicing different types of questions and revising the concepts you have already learned.

Week	**Days**	**Class 7 & 8 Science (Mixed Topics)**	**Activities**
6	31-33	Revision of Nutrition in Plants & Animals, Crop Production and Management, Microorganisms	Solve a variety of MCQs from different sources. Focus on questions that test your understanding of the concepts and their applications.

	34-35	Practice questions on Fibre to Fabric, Coal and Petroleum, Heat, Combustion and Flame	Attempt previous years' NMMS question papers for your state (if available). This will give you an idea of the exam pattern, difficulty level, and types of questions asked.
	36-37	Revision of Acids, Bases and Salts, Physical and Chemical Changes, Conservation of Plants and Animals, Reproduction in Animals	Solve assertion-reason type questions. These questions test your analytical and reasoning skills.
7	38-40	Practice questions on Weather, Climate and Adaptations, Winds, Storms and Cyclones, Soil, Friction	Focus on time management while solving practice papers. Set a timer and try to complete the questions within the allocated time. Analyse your performance after each practice session. Identify your strengths and weaknesses.
	41-42	Revision of Respiration, Transportation, Reproduction in Plants, Reaching the Age of Adolescence	Pay special attention to diagrams and graphical representations. Practice drawing them accurately and quickly.
	43-44	Practice questions on Motion and Time, Force and Pressure, Sound, Chemical Effects of Electric Current	Solve numerical problems with varying difficulty levels. Ensure you understand the underlying formulas and their applications.
8	45-47	Revision of Light, Some Natural Phenomena, and remaining topics	Focus on understanding the interrelationships between different concepts. Try to connect ideas from different chapters.
	48-50	Full-length mock tests based on the NMMS exam pattern	Take at least two full-length mock tests under exam conditions. This will help you simulate the actual exam environment and refine your time management skills. Evaluate your performance and identify areas that still need improvement.

- **Phase 3: Final Revision and Exam Readiness (Days 51-60)**
- This is the final phase dedicated to thorough revision and building confidence for the exam.

Week	Days	Activities
9	51-53	Review all your notes, key points, and formulas. Focus on the areas where you made mistakes in the mock tests.
	54-55	Solve a few more practice questions, focusing on accuracy rather than speed.
	56-57	Revise important diagrams, definitions, and scientific terms.
10	58-59	Take one final mock test to assess your overall preparedness. Analyse your performance and identify any remaining weak areas. Avoid learning any new concepts at this stage. Focus on consolidating what you already know. Ensure you get adequate rest.
	60	Relax and stay calm. Do a light review of important formulas and concepts if you feel the need. Ensure you have all the necessary materials for the exam (admit card, pen, etc.). Get a good night's sleep before the exam.

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TLM Required

LIST OF MATERIAL FOR PERFORMING THE ACTIVITIES			Class -VIII
Activity Number	**Activity Name**	**Required Material**	**Required Chemicals**
CHAPTER- 1 : MOTION			
1.1	Identify actions as push or pull	Some objects available in class room	
1.2	Observe the resultant force when apply in the same direction and opposite direction.	Heavy object / table	
1.3	Observe the position of an object when force applied on it	Rubber ball	
1.4	Observe the direction of the moving object when force apply on it	Scale / rubber ball	
1.5	Observe effect of action of force on objects	Clay , Spring , Rubber band	

1.6	Observing attraction and repulsion between two magnets	Bar magnets -2 Round shaped Pencils – 10 / wooden rollers	
1.7	understand the electrostatic force	Plastic straw , Blade / cutter Thread White paper	
1.8	Obtain the relation between liquid pressure and height of the water column	Transparent glass tube / pipe (length 25 cm, Diameter 5-7cm) Thick rubber sheet (thick ballon) Thread	Water
1.9	observe the liquid pressure in a container	Plastic bottle / Discarded water bottle Cylindrical glass tube Thick ballon Candle Match box	Water
1.10	Obtain relation between liquid pressure and depth	Plastic water bottle Rounder Plastic bucket	Water
1.11	How large the atmosphere is?	Rubber sucker (Looks like rubber cap)	
Extend activity -1	Observe the atmospheric pressure	Glass/ beaker Cardboard With cloth	Water
Extend activity-2	Observe liquid pressure vs shape and volume of the container	Plastic bottles having different shapes and volume Glass tubes -3	Colored water
CHAPTER -2: FRICTION			

2.1	Observe the motion of an object	Book	
2.2	Observe the friction depends on the smoothness of the surface or not?	Spring balance Nylon thread Brick	
2.3	Observe the friction depends on the smoothness of the surface or not?	Plank Different objects (small size)	
2.4	Observe the rolling friction	Book Round pencils – 6	
CHAPTER -3: COAL AND PETROLEUM			
3.1		Collect various material and classify into natural and man made	
3.2		Any containers -4 Pop corn	
CHAPTER – 5: SOUND			
5.1	Observe how sound be produced?	Metal plate Nylon thread	
5.2	Vibrations produce sound	Compass box / pencil box Rubber band Pencils – 2	
5.3	Observe How vibrating dish, produces sound	Metal dish / steel plate Spoon	Water
5.4	identify the vibrating parts of musical instruments	Hallow coconut	
5.5	Make pleasant sound with ‘altering’	Glass tumblers/ pingani dishes -8	Water
5.6	Observe the working of our vocal cards	Rubber strips Piece of paper	
5.7	Sound needs a medium for propagation	Glass tumbler	

		Cell phone -2	
5.8	Test whether sound travelling through water or not?	Plastic bucket / glass jar Bell	Water
5.9	Test whether sound travel through solids or not?	Meter scale / long iron road	
5.10	Sound has energy	Plastic tin Thick ballon Thread Thermocouple pieces	
5.11	Obtain relation between loudness and amplitude of the vibration producing sound	Metal tumbler Table spoon	
Extend activity	Multiple toy telephone -observe propagation of sound in strings	Long thread, Paper tea cups - 10	
CHAPTER- 7: LIGHT			
7.1	Verification of first law of reflection	White paper, Comb Black paper, Torch/ mobile phone Mirror	
7.2	Verification of second law of reflection	White paper, Comb Black paper, Torch/ mobile phone, Mirror	
7.3	Image formation in a plane mirror	White paper, Comb Black paper, Torch/ mobile phone, Mirror	
7.4	Observe the regular and irregular reflection	White paper, Comb Black paper, Torch/ mobile phone, Mirror	
7.5	Observe number of images formed when inclined plane mirrors	Plane mirrors – 2, Tape Any object (small)	
7.6	Making a kaleidoscope and observe images formed	Rectangular mirror strips Thick card board	

		Transparent plastic sheet	
7.7	Observe the dispersion of the light	Plane mirror, Metal plate Screen	Water
7.8	Observe the functioning of pupil of human eye	Torch / mobile flash	
7.9	Demonstration of blind spot in human eye	White paper, Blak marker	
7.10	Bird in cage activity	Card board, Thread	
CHAPTER- 8: CHEMICAL EFFECTS OF ELECTRIC CURRENT			
8.1	Make a tester	Tester	
8.2	Testing the conduction of electricity in lemon juice	Rubber caps, Tea spoon caps, Tea	Lemon juice Vinegar
8.3	Testing the conductivity of electricity of liquids	Empty match box Magnetic compass Battery, Connecting wires, Copper coil	
8.4	Testing electric conductivity of distilled water	Rubber caps, Tea spoon Tester, Bulb, Battery	Distilled water
8.5	Testing the electric conductivity of acids, bases and sugar solutions	Plastic rubber caps – 3 Battery, Graphite rods Battery	Distilled water Lemon juice Dilute HCl Caustic soda
8.6	Observe effect of electric conductivity of solutions	Copper wire, Battery Connecting wires LED	Potato
8.7	Observe the electroplating effect	Copper plates, Battery, Beaker, Sand paper Connecting wires	Copper sulphate Distilled water
Extend activity	To test whether the fruits and vegetables conduct electricity or not?	LED, Connecting wires Battery	Vegetables

CHAPTER – 9: SOME NATURAL PHENOMENA			
9.1	Observe how objects charging by rubbing	Used ball pen Refill Polyethene	
9.2	Collect the objects and the material listed in below table.	Used ball pen Refill Polyethene, Ballon, Eraser, Steel spoon	
9.3	Observe types of charges and their interaction	Ballons -2, Cello tape Glass / beaker, Pen refill	
9.4	Making a simple electroscope	Empty glass bottle Aluminum strips (food cover), Card board Pen refill	
9.5		Discussion on earth quake	
9.6		India map, Globe	
CHATER – 10: COMBUSTION AND FLAME			
10.1	Identify combustible and non-combustible material	Wood, Paper, Iron nails Kerosene, Stone piece, Straw, Charcoal Match stick, Glass	
10.2	To prove air is essential for burning	Glasses – 3, Card board Candles – 3, Wooden pieces – 2, Match box	
10.3	Testing the ignition temperature of the materials	Iron plate Burning wood / charcoal Glass jar / tumbler	
10.4	Ignition temperature of water	Paper cups, Candles – 2 Match box	Water
10.5	Observe why substances gives flames	Glass tube, Tongs – 2 Glass plate, Candle, Match box	

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C3R3J4	D9N9Z1	U2B8A2	E6J6E7
Force and Pressure	Coal and Petroleum	Friction	Chemical effects of Electric Current
B8Y9I4	T5Y3Y6	V8R6T9	B4I8P4
Sound	Combustion and Flame	Light	Some Natural Phenomena
E8W4U1	H6H1Q4	K5M8W3	L2I5B9
NO Bag Day	Cluster Complex	Teacher References	Additional Resources